

72nd BPSC INTEGRATED PRELIMS · CHEMISTRY

CHEMISTRY CONSOLIDATED NOTES

The complete “Eduteria Chem Notes Class Wise” (Chemistry by Sakshi Ma’am) re-arranged & de-duplicated into 10 clean units — every original fact, table, equation, chart and visual explanation kept, duplicate topics merged. Print-ready A4 edition.

Unit	Topic	Consolidated from
01	Matter & Its States	Basic Chemistry Part-I + gas laws of Part-II
02	Classification of Matter	Basic Chemistry Part-II
03	Atomic Structure	Atomic Structure lecture
04	Periodic Classification & Trends	Periodic Table lecture + revision parts of units 03 & 09
05	Chemical Bonding & Molecular Structure	Chemical Bonding + Chemical Bond & Hydrogen Bond
06	Chemical Reactions	Chemical Reaction lecture
07	Acids, Bases & Salts	Acids, Bases and Salts lecture
08	Metals, Non-metals & Metallurgy	Metal & Nonmetal + element classes of Part-II
09	s, p, d & f Block Elements	Important S,P,D,F Elements lecture
10	Organic Chemistry & Everyday Chemistry	Organic Chemistry lectures + organic part of unit 07

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This edition only re-organises the original notes; apart from layout, captions and rapid-revision capsules no new teaching content is added.

What changed in this edition

The source PDF has 157 pages across 12 lecture files. Several topics repeat because the batch revised them in later lectures. This edition keeps all original material but gives every topic exactly one home, merging the repeated versions (the most complete wording is kept). Lecture cover pages, page watermarks and repeated contact headers were removed.

Original file (12 files · 157 pp.)	Now in	Duplicates resolved
01 Basic Chemistry Part-I	Unit 1	—
02 Basic Chemistry Part-II	Units 2 · 1 · 8	States of matter re-explained → merged with Part-I; metals list → Unit 8
03 Atomic Structure	Units 3 · 4	Periodic history (Dobereiner → Modern) duplicated with ch-04 → kept once, in Unit 4; electron rules merged
04 Periodic Table & Classification	Unit 4	Condensed revision of ch-03 → consolidated
05 Chemical Bonding & Molecular Structure	Unit 5	—
06 Chemical Bond & Hydrogen Bond	Unit 5	Bond-type tables repeated → one combined set
07 Chemical Reaction	Unit 6	—
08 Acids, Bases and Salts	Unit 7 · 10	Carbon/covalent basics moved beside organic chemistry
09 Revision (Acids/Bases/Salts + Periodic Table)	Unit 4	Modern PT + trends repeat ch-04 → one trends table + detailed trend notes
10 Important S,P,D,F Elements	Unit 9	Noble-gas uses repeated in ch-11 → single table
11 Metal & Nonmetal	Unit 8	Metal / non-metal properties from ch-02 merged here
12 Organic Chemistry	Unit 10	Organic intro repeated from ch-02 & ch-08 → merged

How to use

- Each UNIT opens with a banner naming the original lectures it consolidates.
- Rapid Revision boxes at the end of each unit list the high-yield one-liners.
- Original diagrams and charts are reproduced with captions; comparison tables were re-typeset for A4.

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UNIT 01

Matter & Its States

Source: Basic Chemistry Part-I (+ gas laws from Part-II)

1.1 What is Matter?

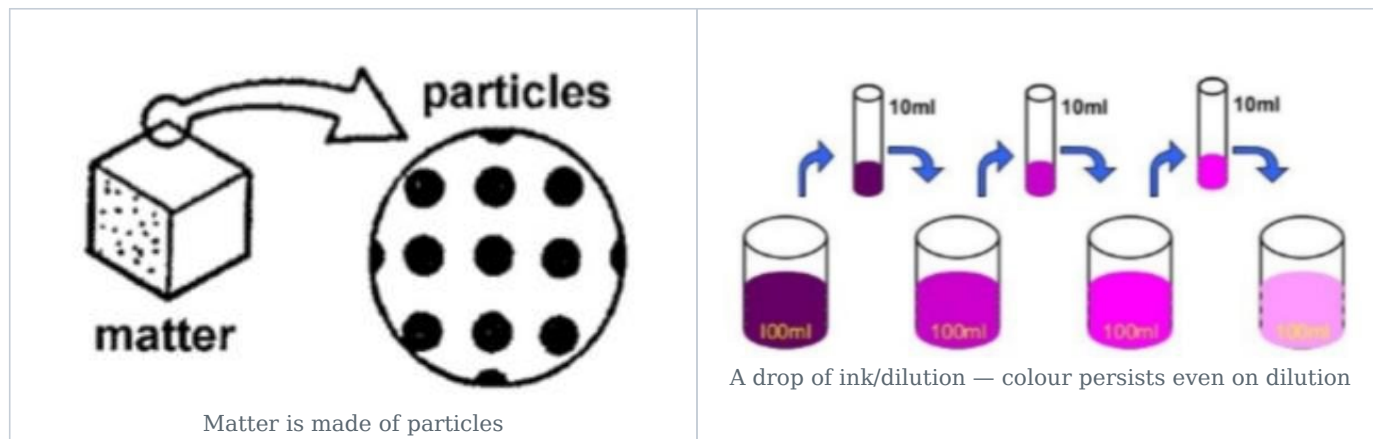
- Air, water, stones, sand, clouds, pencils, books — everything is made up of matter. Matter is everything in this universe that occupies space and has mass (it offers resistance and can be perceived directly by our senses — e.g. book, pen, pencil, water, air, all living beings).

Constituents of Matter

- According to the early Indian philosophers, every living and non-living thing is made of five basic elements called the Panchtatva — Air, Water, Earth, Sky and Fire. Therefore, matter is a composition of these five constituents.

Physical Nature of Matter — continuous or particulate?

- Matter is particulate in nature — it consists of particles, as seen in the microscopic image of a cube.
- Example: if we put a drop of red colour in water, the water turns red because the particles of red colour mix with the particles of water.



What is the size of these particles?

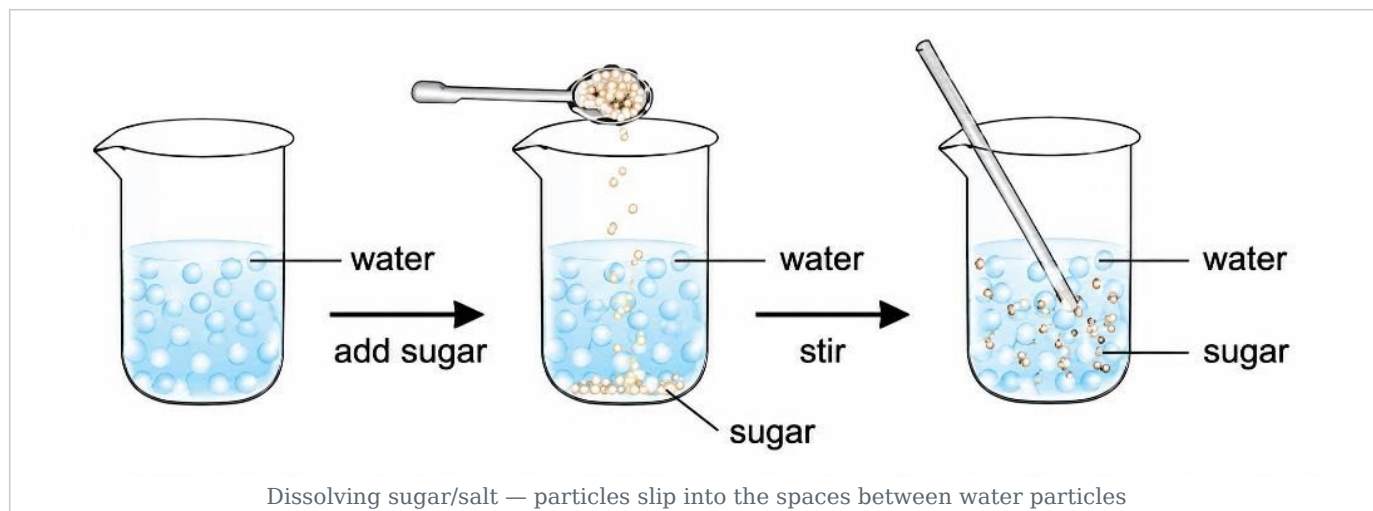
- The particles of matter are very small, and can be broken into further particles.
- Example: on dilution of a colourful solution we can still see the colour — meaning millions of particles are present, which just divide themselves on dilution.
- Which of these is matter — happiness, air, sandwich, thoughts, juice, eraser? → Air, sandwich, juice and eraser (they have mass, occupy space and can be broken into further particles).

1.2 Characteristics of Particles of Matter

Particles of matter have three characteristics: (1) they have spaces between them, (2) they are moving all the time, (3) they attract each other.

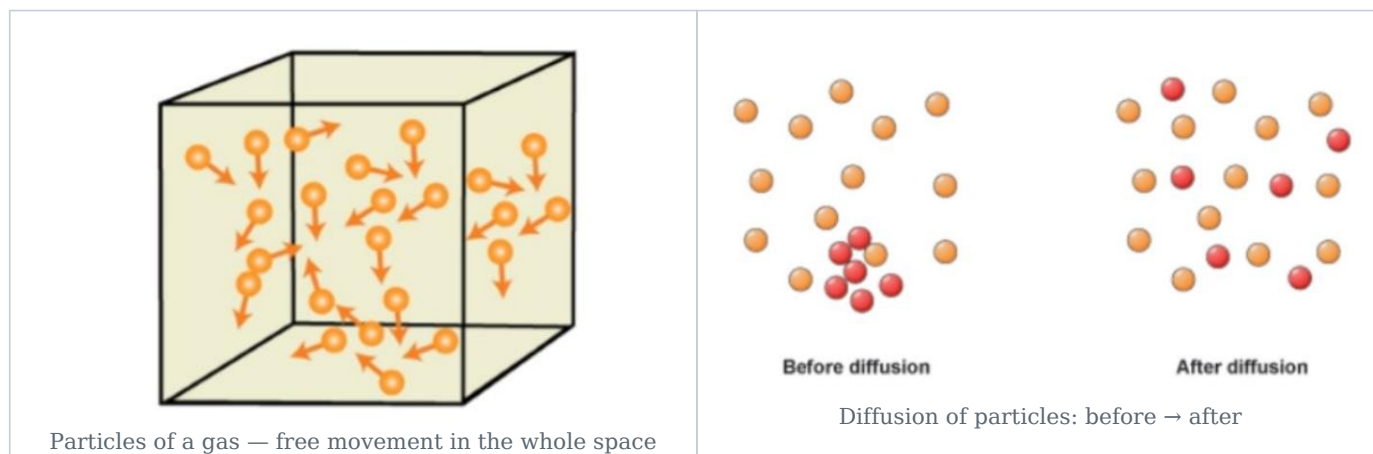
(a) Particles have spaces between them

- Salt gets dissolved in water because its particles have spaces between them — the salt particles get in between the spaces of water particles and a mixture is formed. These particles cannot be seen with the naked eye.



(b) Particles are continuously moving

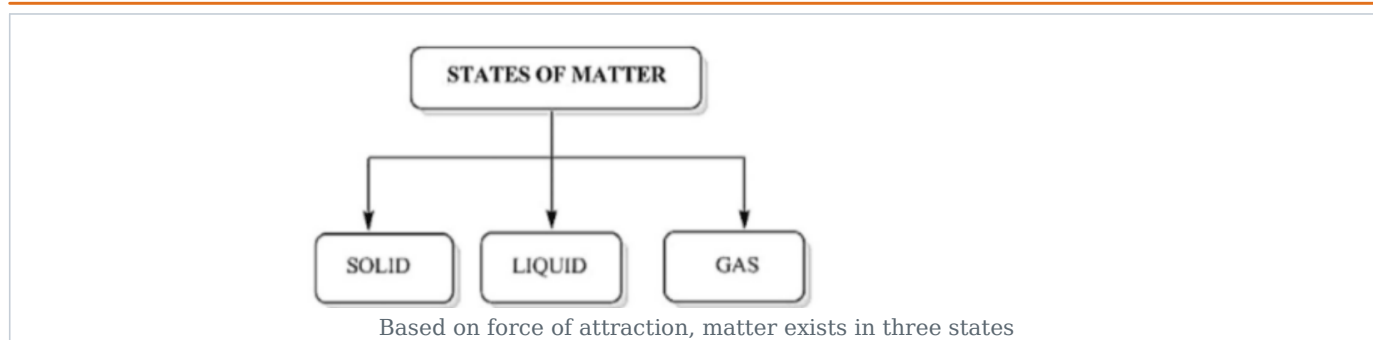
- Particles of matter are in motion all the time; hence they possess kinetic energy (energy due to motion).
- Particles intermix on their own: salt in water, various gases in air, ink in water.
- Diffusion — the process of mixing two different types of particles. Diffusion becomes faster on heating, because kinetic energy of particles increases on heating.



(c) Particles attract each other

- Particles are held together by a force of attraction, which varies in different forms of matter.
- Solids have the highest force of attraction (we cannot move our hands through a solid). Gases have the least (we can move our hands easily in air). Liquids are intermediate — we can move our hands through water but not as freely as in air.
- Increasing order of force of attraction: Gas < Liquid < Solid

1.3 States of Matter



The Solid State

- Solids have a specific shape, distinct boundaries and a fixed volume.

- Least kinetic energy; particles arranged in order; force of attraction is maximum; very little space between particles → cannot be compressed.
- Rubber band, sponge and salt are all solids — a rubber band and sponge change shape only when force is applied, and each grain of salt has its own definite shape.
- e.g. sugar, iron, gold, wood.

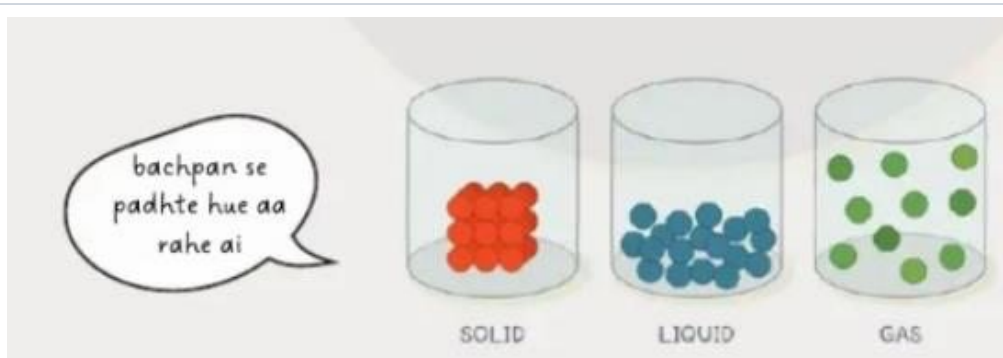
The Liquid State

- Liquids have a fixed volume but no fixed shape — they take the shape of the container (also called fluids as they can flow/pour). e.g. water, milk, oil, mercury, alcohol.
- Force of attraction is less than in solids → space between particles → they can flow easily; cannot be compressed.
- Particles have high kinetic energy and keep moving; they are not fixed.
- Can other matter diffuse into liquids? Yes — solids, liquids and gases can (there is space between liquid particles):
 - Solid into liquid — mixing sugar in tea;
 - Liquid into liquid — mixing ink in water;
 - Gas into liquid — presence of O_2 and CO_2 in water.

The Gaseous State

- Gases have neither fixed volume nor fixed shape; they completely fill the container. e.g. hydrogen, oxygen.
- Least (almost no) force of attraction → large spaces between particles → free movement in any direction; can be easily compressed and put into a small container, unlike solids and liquids.
- Different gases diffuse into each other easily; kinetic energy is the maximum.

Property	Solid	Liquid	Gas
Shape	Definite	Indefinite (shape of container)	Indefinite
Volume	Definite	Definite	Indefinite
Force of attraction	Maximum	Lesser than solid	Negligible
Compressibility	Cannot be compressed	Cannot be compressed	Can be compressed
Kinetic energy of particles	Minimum	More than solid	Maximum
Movement of particles	Only vibrate at fixed positions	Slide over one another	Move freely
Density	Highest	Lower than solid	Lowest
Flow	Cannot flow	Flow	Flow



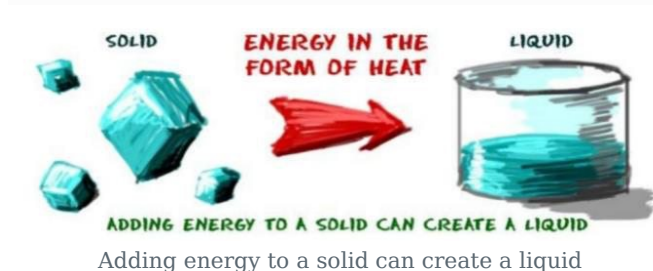
Arrangement of particles in solid, liquid and gas

1.4 Can Matter Change its State?

Water exists in three states — ice (solid), water (liquid), water vapour (gas) — showing that matter can change states by changing temperature or pressure.

Effect of change of temperature (heating)

- Solids: on heating, kinetic energy increases, force of attraction decreases; particles start vibrating and changing positions; slowly they become free and the solid converts into liquid.
- Melting point — temperature at which a solid melts to become a liquid at atmospheric pressure (m.p. of ice = 273.16 K).
- Fusion — the process of melting a solid into liquid.
- At melting point the temperature stops rising: the heat is used by particles to diminish the force of attraction — the energy stays "hidden" in the substance.
- Latent heat — the heat energy used to break the force of attraction between particles.
- Latent heat of fusion — heat energy required to change 1 kg of solid into liquid at its melting point at atmospheric pressure.
- Atmospheric pressure — pressure exerted by the weight of the atmosphere.



Adding energy to a solid can create a liquid

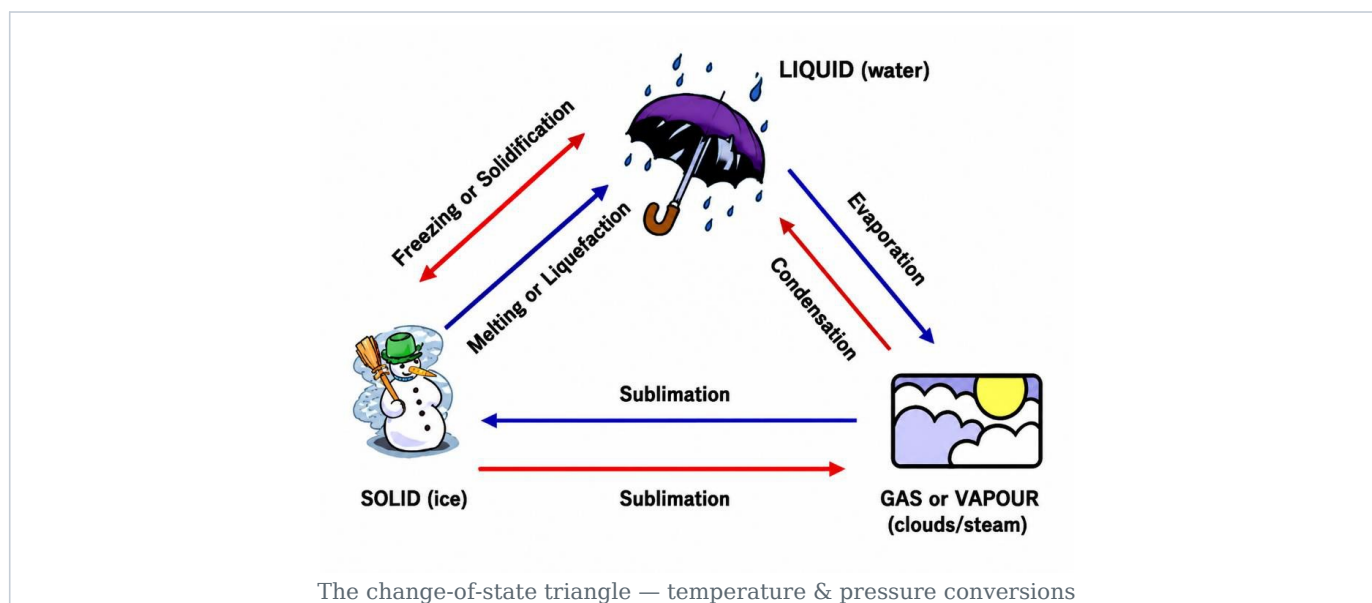


Energy from a stove heats liquid water and creates steam

- Liquids: kinetic energy increases → force of attraction decreases → particles move freely; a point comes when particles overcome the forces completely and the liquid changes into gas.
- Boiling point — temperature at which a liquid starts boiling at atmospheric pressure (b.p. of water = 373 K).
- Latent heat of vaporisation — heat energy required to change 1 kg of liquid into gas at its boiling point at atmospheric pressure.

Effect of decreasing temperature

- Gases → liquids: kinetic energy decreases and gas turns into liquid — Condensation / Liquefaction. Example: formation of clouds due to condensation of water vapour from Earth.
- Liquids → solids: kinetic energy decreases and liquid turns into solid (formation of ice).
- Sublimation — direct change of a solid into gas (and vice-versa). Example: camphor directly evaporates into air without becoming liquid.

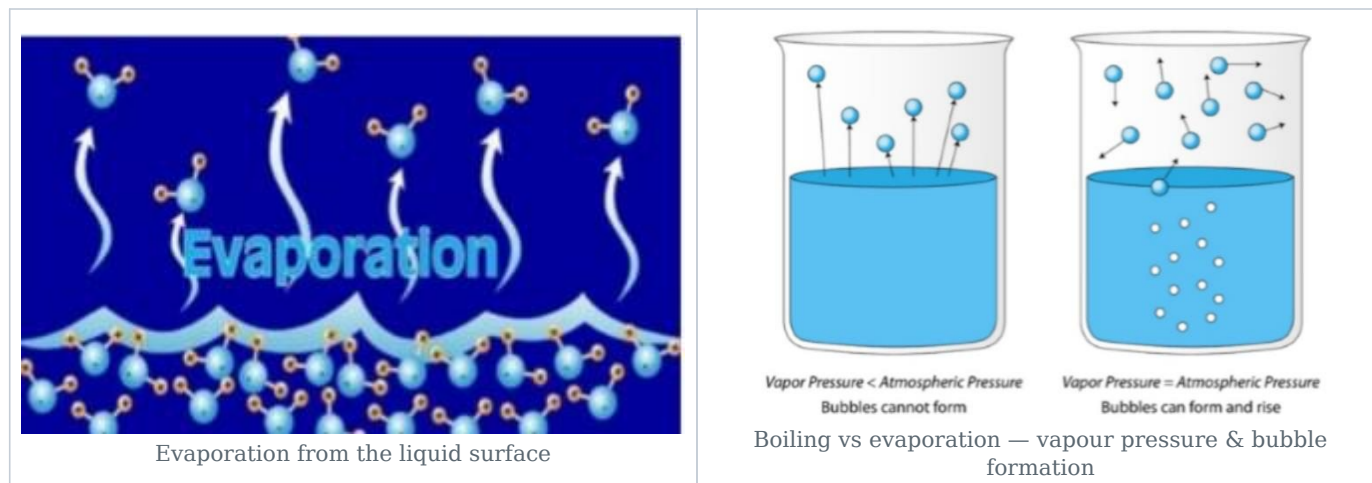


Effect of change of pressure

- Applying pressure brings particles closer, increasing the force of attraction.
- When we compress and cool a gas, it changes into a liquid.
- Dry ice — solid CO_2 ; it can directly turn into gas when pressure is decreased to 1 atmosphere.

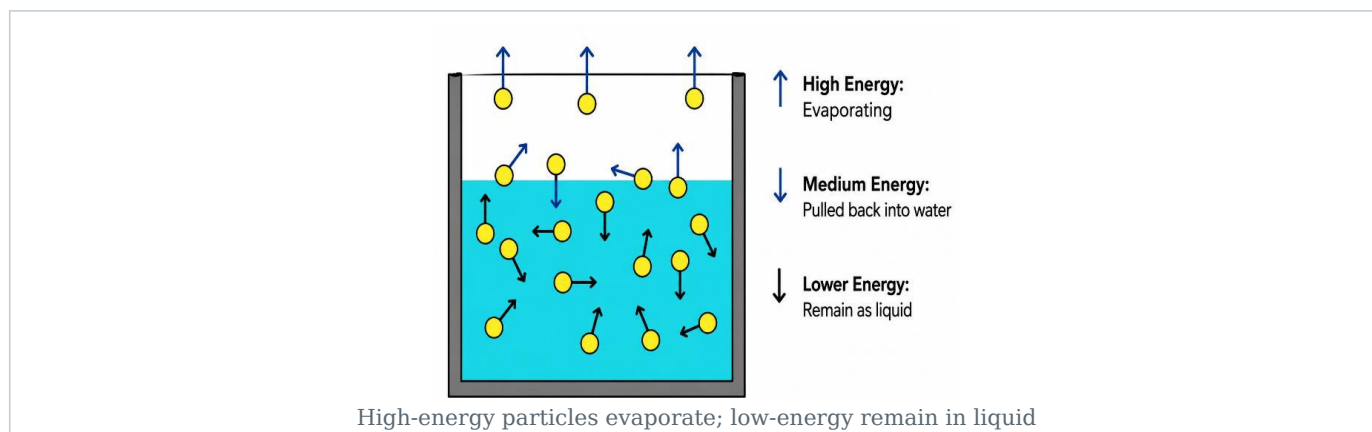
1.5 Evaporation

- Particles of liquid have extra kinetic energy; the most energetic particles overcome the forces of attraction and convert into vapour without any external force.
- Evaporation — change of a liquid into vapours at any temperature below its boiling point. (Evaporation $\neq$ boiling.)



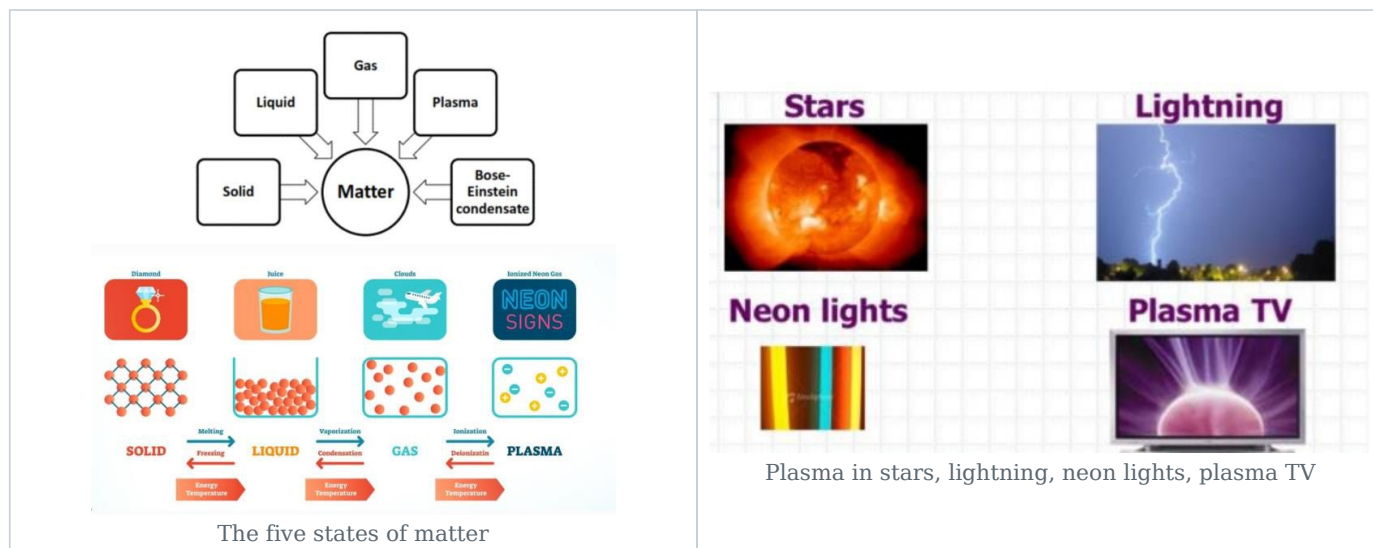
How evaporation causes cooling

- Evaporation uses the energy of liquid particles; particles absorb energy from the surroundings to compensate the loss $\rightarrow$ surroundings get cooled.
- Examples: palms feel cool when acetone (nail-paint remover) is put on them; people sprinkle water on roofs/ground on sunny days; we can sip hot tea faster from a saucer than a cup.
- Why cotton clothes in summer? Sweat evaporates taking energy from the body; cotton absorbs sweat and exposes it to the atmosphere $\rightarrow$ easy evaporation $\rightarrow$ keeps us cool.
- Why do water droplets appear on the outside of a glass of ice-cold water? Water vapours of air condense on the cold walls (gas $\rightarrow$ liquid droplets).



1.6 The Five States of Matter

- Besides solid, liquid and gas, scientists have discovered two more states: Plasma and Bose-Einstein Condensate (BEC).
- Plasma — particles are super-excited and super-energetic; the state consists of ionised gases. Fluorescent tubes and neon bulbs contain plasma: on passing electricity the gases get ionised and glow. The Sun and stars glow because of plasma.
- Bose-Einstein Condensate — fifth state, discovered by Albert Einstein based on studies by Indian scientist Satyendra Nath Bose; formed by condensing gases of extremely low densities to much lower temperatures.



1.7 Important Measurement Units

Quantity	SI / common unit
Mass	kg (kilogram)
Volume	m ³ (cubic metre); common unit — litre (L)
Temperature	kelvin (K); 0 °C = 273.16 K ≈ 273 K; K = °C + 273
Pressure	pascal (Pa); for gases — atmosphere (atm); 1 atm = 1.01 × 10 ⁵ Pa
Normal atmospheric pressure	1 atm (at sea level)

1.8 Fundamental Gas Laws

- Boyle's Law (P-V): at constant temperature, volume of a gas is inversely proportional to its pressure ($P \uparrow \rightarrow V \downarrow$).
- Charles's Law (V-T): at constant pressure, volume is directly proportional to the Kelvin temperature ($T \uparrow \rightarrow V \uparrow$).
- Gay-Lussac's Law (P-T): at constant volume, pressure is directly proportional to the Kelvin temperature.
- Avogadro's Law (V-n): at constant temperature and pressure, volume is directly proportional to the amount of gas (moles).

Combined gas law: $P_1V_1 / T_1 = P_2V_2 / T_2$ · Ideal gas equation: $PV = nRT$

- P = pressure, V = volume, n = number of moles, R = universal gas constant, T = temperature (kelvin).
- Ideal gas — a theoretical gas that follows these laws perfectly (particles have no volume, no attraction). Real gases deviate from ideal behaviour at high pressure and low temperature (finite volume, intermolecular forces).

**Gas Laws
Cheat sheet!**

	Boyle's Law	Charles's Law	Gay Lussac's Law	Avagadro's Law
Relationship between the physical properties	Pressure - Volume	Volume - Temperature	Pressure - Temperature	Volume - Amount
Constant	Temperature and mass	Pressure and mass	Volume	Temperature and Pressure
Mathematical representation	$V \propto \frac{1}{P}$	$V_t = V_0 \left[1 + \frac{t}{273.15} \right]$ $V_0 = \text{Volume of given mass of gas at } 0^\circ\text{C}$ $V_t = \text{Volume of given mass of gas at } t^\circ\text{C}$	$P \propto T$	$V \propto T$
Quantitative relationship	$\frac{P_1}{P_2} = \frac{V_2}{V_1}$	$\frac{V_1}{T_1} = \frac{V_2}{T_2}$	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$	$M = k_4 \frac{m}{V} = k_4 d$ $M = \text{Molar mass of the gas}$ $d = \text{Density}$

Gas laws at a glance (original cheat-sheet)

✂ RAPID REVISION

- Matter = mass + space; it is particulate. Force of attraction: Gas < Liquid < Solid.
- Melting of solid = fusion; hidden heat = latent heat (of fusion / of vaporisation).
- Sublimation: solid $\rightleftharpoons$ gas directly (camphor, dry ice — solid CO_2).
- Evaporation occurs below boiling point and causes cooling; rate $\uparrow$ with surface area, temperature, wind speed; $\downarrow$ with humidity.
- Five states of matter: solid, liquid, gas, plasma (ionised gas — sun, neon signs), BEC (Einstein + S.N. Bose).
- $K = ^\circ\text{C} + 273$; 1 atm = 1.01×10^5 Pa; $PV = nRT$.

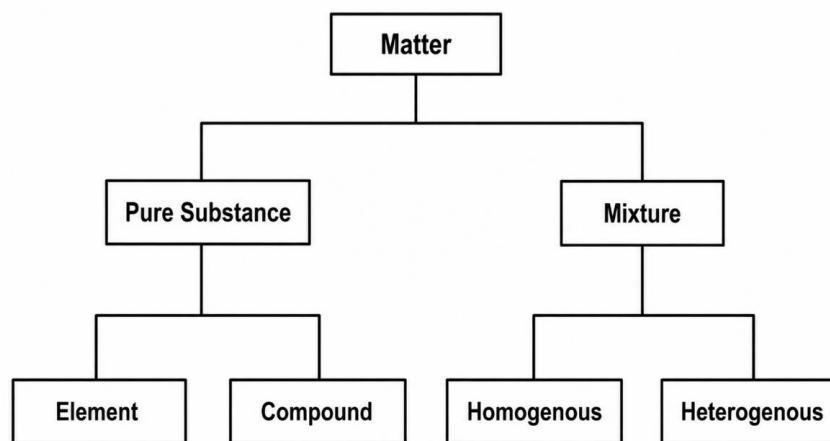
UNIT 02

Classification of Matter

Source: Basic Chemistry Part-II

2.1 Classification of Matter

At the bulk (macroscopic) level matter is classified into (a) Mixtures and (b) Pure substances:



Classification of matter

Phase — a definition

- Any distinct portion of matter that is uniform throughout in composition and properties is called a phase.

2.2 Pure Substances

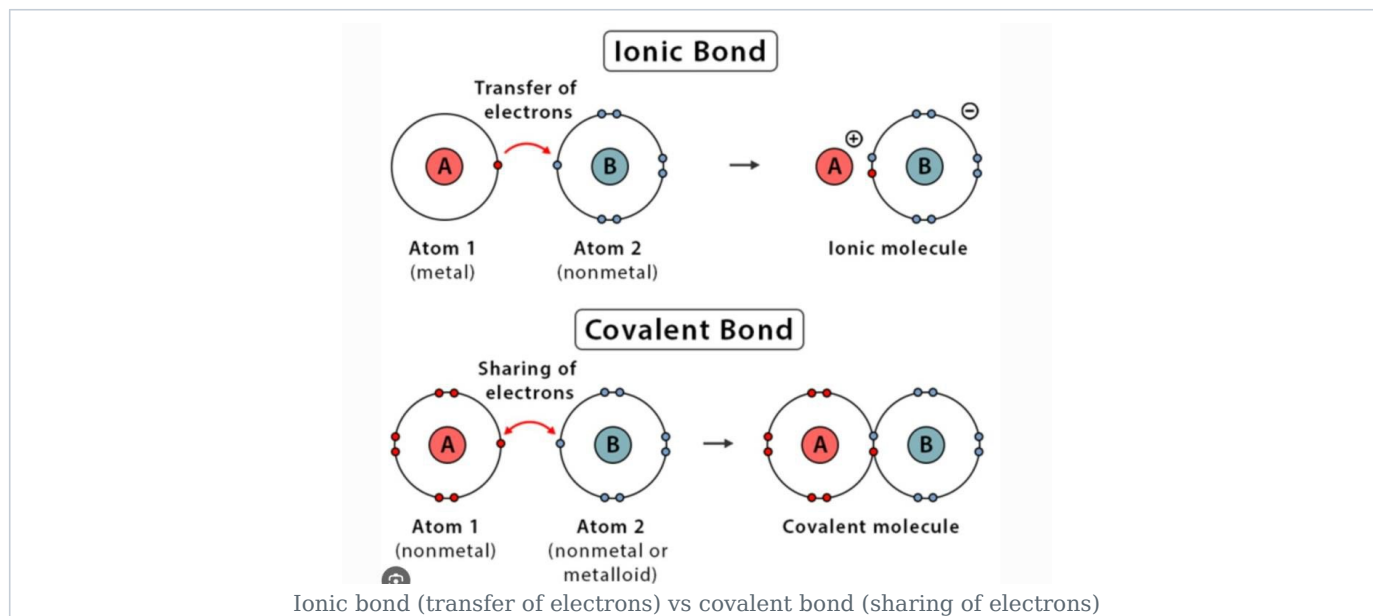
A material containing only one type of particle is a pure substance. Pure substances are further divided into elements and compounds.

Element

- An element is a pure substance that cannot be broken down into simpler substances by chemical means; its atoms have the same number of protons.
- Elements are split into three classes on the basis of physical and chemical properties: (1) Metals, (2) Non-metals, (3) Metalloids. (→ details in Unit 8.)

Compound

- A compound is a substance composed of two or more different elements chemically bonded together in a fixed ratio; it can be broken into elements only by chemical reactions.



Organic vs Inorganic compounds

Basis	Organic compounds	Inorganic compounds
Composition	Always contain carbon (C) and hydrogen (H), often with O, N, S or P	Typically lack C-H bonds
Bonding	Covalent bonds	Often ionic bonds
Physical properties	Gases, liquids or low-melting solids; generally insoluble in water; flammable	High-melting solids; often soluble in water
Source	Mostly from living organisms (carbon-based molecules)	From non-living sources like minerals
Examples	Glucose, methane, proteins, sugar, DNA, plastics, lipids	NaCl, HCl, water, salts, metals

- Exceptions: some carbon substances are classified as inorganic because they behave like minerals — CO_2 , CO, carbonates (CaCO_3), cyanides.
- Note: organic compounds exclude oxides of carbon, metallic carbonates, metal cyanides, metal carbides etc.

Ionic vs Covalent compounds

Basis	Ionic compounds	Covalent compounds
Bonding mechanism	Electron transfer (electrostatic attraction)	Sharing of valence electrons (directional shared pairs)
Elements involved	Usually a metal + a non-metal	Two or more non-metals
Physical state & m.p.	Hard, brittle crystalline solids; high melting/boiling points	Gases, liquids or soft solids; low melting/boiling points
Conductivity	Conduct when molten or dissolved in water (not as solids)	Generally poor conductors
Solubility	Often soluble in water	Tend to be insoluble in water (polar covalents can be exceptions)
Structure	Crystal lattices	Discrete molecules or network solids
Examples	NaCl, CaO, $(\text{NH}_4)_3\text{P}$	H_2O , CO_2 , CH_4 , SF_6

Physical & chemical properties of compounds

- Physical properties (measured without changing chemical identity): melting & boiling points (defined, distinct), density (mass per unit volume), solubility, electrical conductivity (esp. ionic compounds in liquid/aqueous state), appearance (colour, odour, physical state).
- Chemical properties (ability to transform into new substances): reactivity (e.g. with oxygen/acids), flammability, toxicity, heat of combustion, corrosion resistance.

2.3 Mixtures

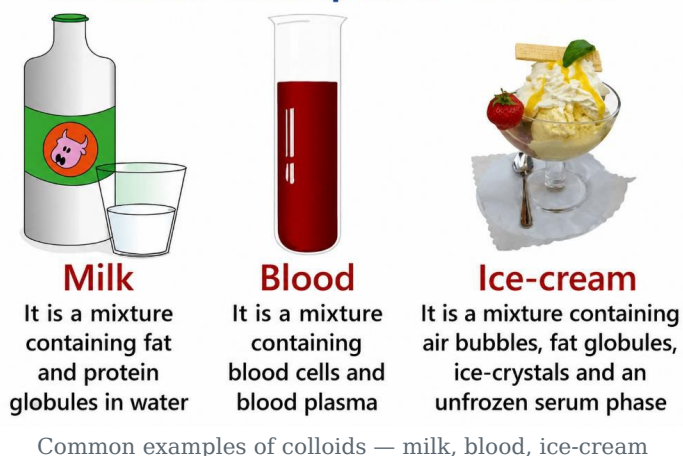
A mixture contains two or more substances in any ratio. Primarily of two types — heterogeneous and homogeneous.

Basis	Homogeneous mixture	Heterogeneous mixture
Composition	Uniform throughout (even at molecular level)	Non-uniform; varies throughout
Phases / appearance	Single phase	Two or more distinct phases
Visibility of components	Cannot be seen by naked eye	Can be seen by naked eye
Separation	Difficult — needs distillation etc.	Easy — physical means like filtration
Also called	Solutions	Suspensions or colloids
Examples	Air, saline solution, sugar solution, vodka, brass	Oil & water, salt + sugar, trail mix, pizza, soil, salad

2.4 Colloids

- Colloids are heterogeneous mixtures where microscopic insoluble particles (1 nm – 1000 nm, i.e. 10^{-7} to 10^{-4} cm) of one substance (dispersed phase — solute-like, smaller amount) are scattered throughout another (dispersion medium — solvent-like, larger amount).
- They appear homogeneous but are not; they are stable (particles do not settle).
- Filterability: particles pass through ordinary filter paper but can be separated by semipermeable membranes or centrifuges.
- Tyndall effect: they scatter light, making a light beam passing through them visible (e.g. fog, milk).
- Brownian movement: continuous zig-zag motion of colloidal particles due to bombardment by molecules of the dispersion medium.

Common Examples of Colloids



Classification of colloids (dispersed phase in dispersion medium)

Type	Dispersed / Medium	Examples
Sol	Solid in Liquid	Paint, cell fluids, blood
Solid sol	Solid in Solid	Coloured gemstones, alloys
Emulsion	Liquid in Liquid	Milk, hair cream, mayonnaise
Foam	Gas in Liquid	Whipped cream, soap lather
Solid foam	Gas in Solid	Pumice stone, foam rubber
Aerosol	Liquid in Gas	Fog, mist, hairspray
Solid aerosol	Solid in Gas	Smoke, dust
Gel	Liquid in Solid	Jelly, butter, cheese

- Applications: many medicines are colloidal for better absorption (e.g. argyrol); water purification (alum coagulation); paint, ink, rubber and food industries; blood is a colloid — specialised colloidal solutions are used in medical treatments.

2.5 Separation of Mixtures

Methods use physical properties — size, solubility, boiling point, magnetism — to isolate components without changing their chemical identities:

Method	Principle / use
Filtration	Separates insoluble solids from liquids (sand from water) using a filter.
Evaporation	Separates soluble solid from liquid (salt from water) by evaporating the liquid.
Distillation	Separates liquids with different boiling points (alcohol + water) by vaporising & condensing.
Fractional distillation	Liquids with closer boiling points (components of crude oil, petroleum refining).
Magnetic separation	Magnet separates magnetic materials (iron filings) from non-magnetic ones.
Chromatography	Separates components by different rates of movement across a stationary phase (dyes).
Centrifugation	Spins mixture at high speed; density-based separation (blood components).
Sublimation	Separates sublimable solids (iodine, ammonium chloride) from non-sublimable solids.
Crystallisation	Purifies solids by dissolving in hot solvent and cooling to form crystals.

✂ RAPID REVISION

- Pure substance = one type of particle (element or compound); compound = elements in fixed ratio.
- Organic → covalent, low m.p., flammable (glucose, CH₄); inorganic → ionic, high m.p., water-soluble (NaCl) — but CO₂, CO, carbonates, cyanides are inorganic despite carbon.
- Homogeneous = one phase (air, brass); heterogeneous = many phases (oil-water); colloids 1–1000 nm, show Tyndall effect + Brownian movement.
- Colloid types: sol (paint), gel (jelly), emulsion (milk), foam (soap lather), aerosol (fog/smoke).
- 9 separation methods — filtration, evaporation, (fractional) distillation, magnetic separation, chromatography, centrifugation, sublimation, crystallisation.

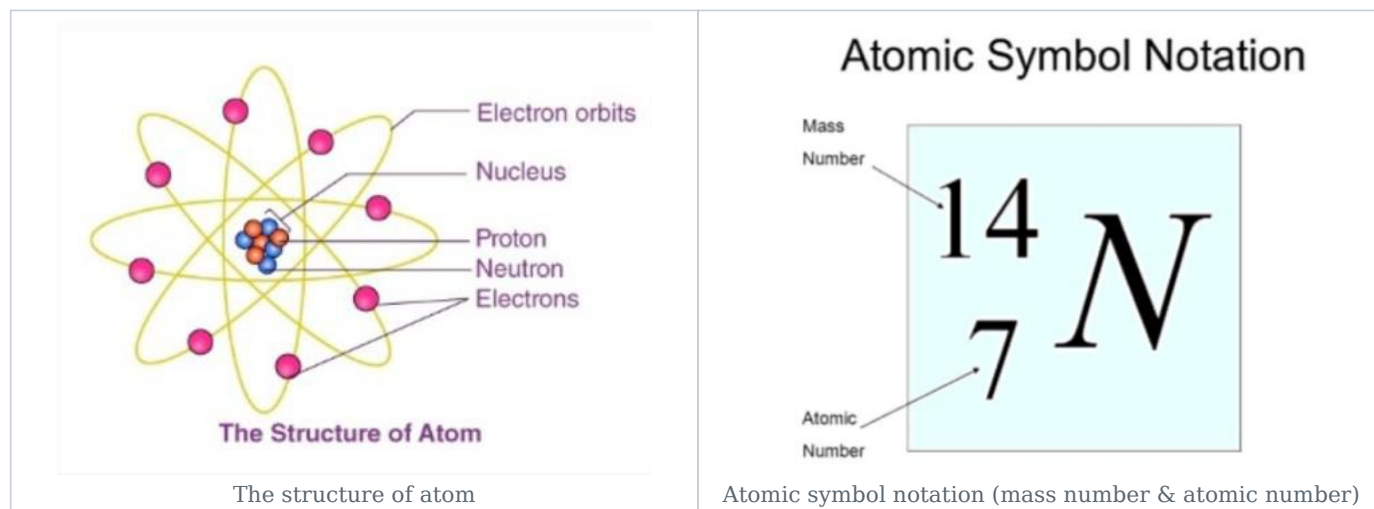
UNIT 03

Atomic Structure

Source: Atomic Structure lecture

3.1 What is Atomic Structure?

- The atomic structure of an element refers to the constitution of its nucleus and the arrangement of electrons around it. Matter is made of protons, electrons and neutrons.
- Protons and neutrons make up the nucleus; electrons surround it. The atomic number = total number of protons in the nucleus.



Laws of chemical combination

- Law of Conservation of Mass (Antoine Lavoisier) — matter can neither be created nor destroyed in a chemical reaction or physical change; total mass of reactants = total mass of products (atoms are only rearranged). Basis of balancing equations and stoichiometry. Limitation: holds only for low-energy (chemical) reactions — in nuclear reactions mass converts to energy.
- Law of Definite Proportions / Constant Composition (Joseph Proust, 1799) — a pure compound always contains the same elements combined in the same fixed proportion by mass, regardless of source or method of preparation. Limitations: not applicable to isotope-containing compounds (e.g. $^{35}\text{Cl}_2$ vs $^{37}\text{Cl}_2$) or non-stoichiometric compounds.
- Law of Multiple Proportions (John Dalton) — if two elements form more than one compound, the masses of one element combining with a fixed mass of the other are in a ratio of small whole numbers. E.g. in CO and CO_2 , fixed 12 g C combines with 16 g and 32 g O $\rightarrow 1 : 2$. Limitations: non-stoichiometric compounds, polymers/oligomers, isotopes.

3.2 Dalton's Atomic Theory

- John Dalton suggested that all matter is made up of atoms, which are indivisible and indestructible; atoms of an element are exactly the same, but atoms of different elements differ in size and mass.

Postulates

- Every matter is made up of atoms; atoms are indivisible.
- Specific elements have only one type of atom; each atom has its own constant mass that varies from element to element.
- Atoms undergo rearrangement during a chemical reaction; atoms can neither be created nor destroyed (only transformed).
- The theory successfully explained the laws of chemical reactions — conservation of mass, constant properties, multiple proportions and reciprocal proportions.

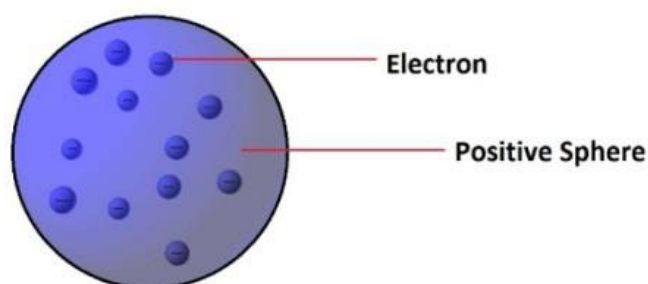
Demerits

- Could not explain the existence of isotopes.
- Said nothing about the structure of the atom; later, particles discovered inside the atom proved atoms are divisible.

3.3 Thomson's Model (Plum-pudding)

- Postulate: the atom is a sphere of positive charge in which electrons are embedded.
- Limitation: could not satisfactorily explain the results of Rutherford's scattering experiment.

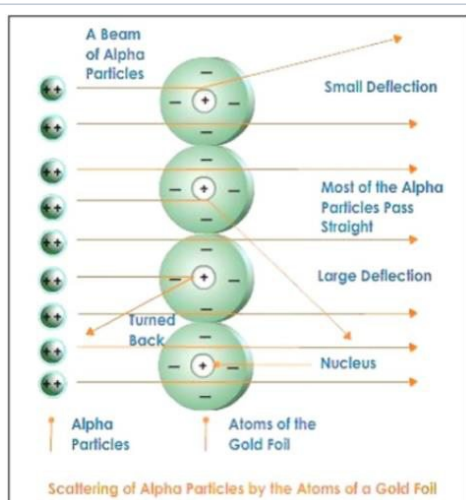
Thomson's Atomic Model (Plum – pudding model):-



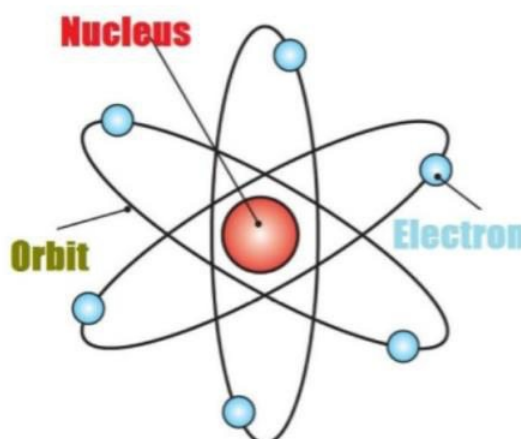
Thomson's atomic model — electrons embedded in a positive sphere

3.4 Rutherford's Model (Nuclear Model)

- Rutherford passed alpha rays through a thin gold foil, expecting the gold atoms to deflect the α -particles.



Scattering of α -particles by gold foil



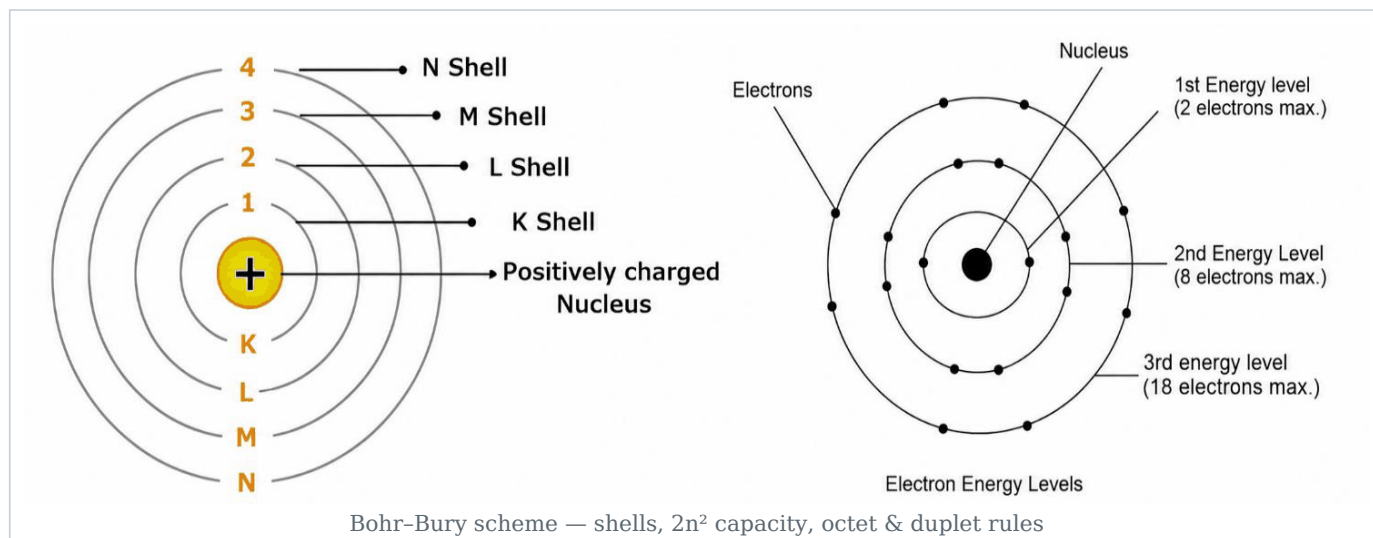
Rutherford's nuclear atom — orbits around nucleus

Observation	Inference
Most high-speed α -particles moved straight through the gold foil	Atom contains a lot of empty space
Some particles got diverted by small angles	Positive charges do not occupy much space
Only 1 out of 12000 particles bounced back	Positive charge is concentrated in a small area at the centre

- Nuclear model: atoms contain a lot of unoccupied space; a heavily positively charged nucleus lies at the centre; electrons revolve around it in circular paths called orbits; almost all the mass of the atom is in the nucleus (nucleus is much smaller than the atom).
- Drawback: could not explain atomic stability — by Maxwell's theory a charged particle moving in a circle radiates energy; the electron should lose speed and fall into the nucleus.
- Nucleons: protons and neutrons are collectively called nucleons.
- Neutron: discovered by J. Chadwick — carries no charge. (Electron — J.J. Thomson, cathode-ray tube; Proton — E. Rutherford, gold-foil experiment.)

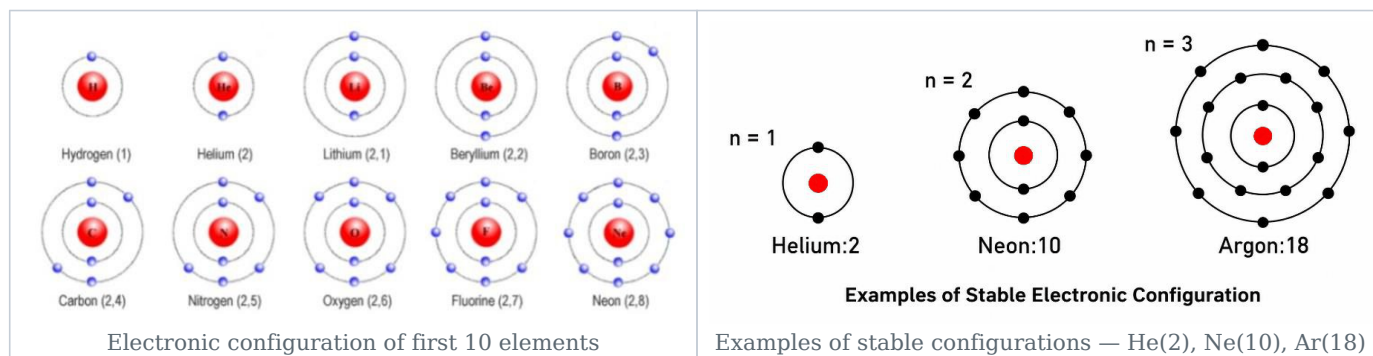
3.5 Bohr's Model of the Atom

- Electrons revolve around the nucleus only in special separate paths (orbits) of fixed energy — energy levels, denoted K, L, M, N... or $n = 1, 2, 3...$
- Electrons do not emit energy while moving in these special orbits.



Distribution of electrons in shells (Bohr-Bury rules)

- If orbit number = n , maximum electrons in the orbit = $2n^2$ ($K = 2, L = 8, M = 18, N = 32$).
- Orbits are filled from inside to outside — first $n = 1$, then $n = 2$, and so on; until inner shells are filled completely, outer shells cannot contain electrons.
- Octet rule: the outermost shell cannot accommodate more than 8 electrons even if it has capacity — 8 electrons in the outermost shell make the atom very stable.



Valency

- Valence electrons — electrons in the outermost orbit (valence shell) of an atom; they determine the combining capacity (valency).
- Atoms react to achieve a stable electronic configuration ($2 e^-$ in K, or $8 e^-$ in the outermost shell). Noble gases (He, Ne, Ar) are inert because they already have it.
- If the outermost shell has ≤ 4 electrons, valency = number of outer electrons (e.g. Ag has 1 $\rightarrow$ valency 1). If it is close to full, valency = $8 - \text{outer electrons}$ (oxygen has 6 $\rightarrow$ valency $8 - 6 = 2$).
- (Note: helium actually shares its 2 electrons — its valency is 0 in compounds.)

Atomic number & mass number

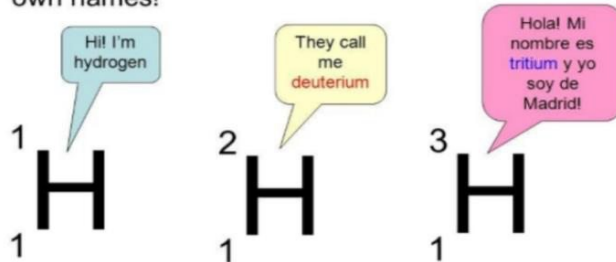
Atomic number (Z) = number of protons (atoms of different elements always have different Z) · Mass number (A) = protons + neutrons

3.6 Isotopes, Isobars & Other Species

- Isotopes — atoms of the same element with the same atomic number but different mass numbers. They are pure substances, have similar chemical nature but distinct physical characteristics.

Isotopes of Hydrogen

The three isotopes of Hydrogen even have their own names!



Isotopes of hydrogen — protium, deuterium, tritium

- Uses of isotopes: fuel of nuclear reactor — isotope of uranium; treatment of cancer — isotope of cobalt; treatment of goitre — isotope of iodine.
- Isobars — atoms of different elements with the same mass number but different atomic numbers (e.g. $^{32}_{15}\text{P}$ and $^{32}_{16}\text{S}$).

Isobars	Z	A	e ⁻	p	n
$^{40}_{18}\text{Ar}$	18	40	18	18	22 (40–18)
$^{40}_{19}\text{K}$	19	40	19	19	21 (40–19)
$^{40}_{20}\text{Ca}$	20	40	20	20	20 (40–20)

Term	Definition / example
Atomic number (Z)	Number of protons (or electrons) of a neutral atom.
Mass number (A)	Total number of protons and neutrons (nucleons) in an atom.
Nucleons	Protons + neutrons present in the nucleus.
Isotopes	Same Z, different A — $^1\text{H}_1$, $^2\text{H}_1$, $^3\text{H}_1$.
Isobars	Same A, different Z — $^{32}_{15}\text{P}$, $^{32}_{16}\text{S}$.
Isotones	Same number of neutrons — $^{14}_6\text{C}$, $^{15}_7\text{N}$, $^{16}_8\text{O}$.
Isoelectronic	Same number of electrons — N_2 , CO , CN^- .
Nuclear isomers	Same Z and A but different radioactive properties — uranium-X ($t_{1/2}$ 1.4 min) & uranium-Z ($t_{1/2}$ 6.7 h).
Isosters	Molecules with same number of atoms and same number of electrons — N_2 and CO .

3.7 Sub-atomic Particles — Comparison

Name	Electron (e ⁻)	Proton (p)	Neutron (n)
Approximate relative mass	1/1836	1	1
Mass in kg	9.109×10^{-31}	1.673×10^{-27}	1.675×10^{-27}
Mass in amu	5.485×10^{-4}	1.007	1.008
Charge (coulomb)	1.602×10^{-19}	1.602×10^{-19}	0
Actual charge (e.s.u.)	4.8×10^{-10}	4.8×10^{-10}	0

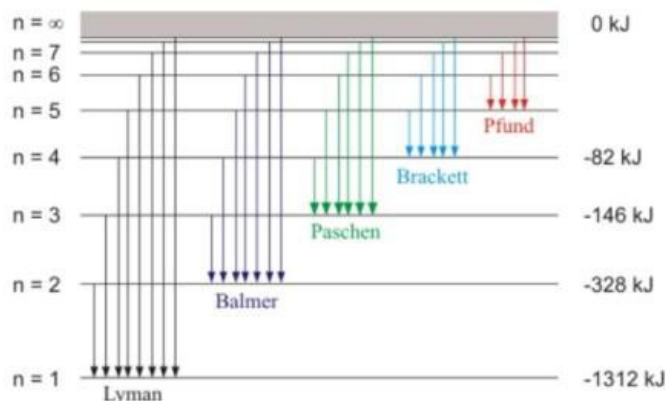
- Atoms consist of protons and electrons in balanced proportion. Protons sit in the interior, electrons in the exterior — so electrons can be removed from an atom.

3.8 Atomic Spectrum of Hydrogen

- When an electron jumps between energy levels, radiation is absorbed/emitted; R_H = Rydberg constant (108978 cm^{-1}); n_1 , n_2 take integral values:

n_1	n_2	Spectral series	Spectral region
1	2, 3, 4...	Lyman	UV

n_1	n_2	Spectral series	Spectral region
2	3, 4, 5...	Balmer	Visible
3	4, 5, 6...	Paschen	IR
4	5, 6, 7...	Brackett	IR
5	6, 7, 8...	Pfund	IR



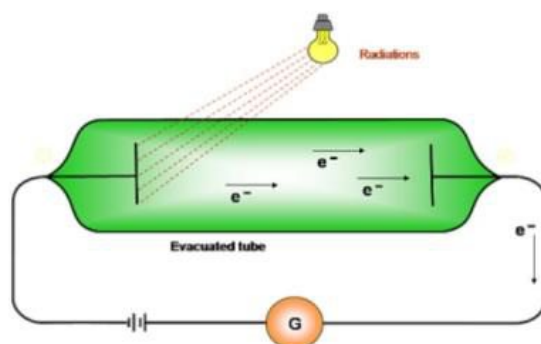
Hydrogen spectrum — series of lines in UV, visible and IR regions

3.9 Photoelectric Effect

- Ejection of electrons takes place from a metal surface when light of suitable frequency falls on it.
- Minimum frequency required for ejection = threshold frequency (ν_0).
- Kinetic energy of ejected electrons $\propto$ frequency of radiation; number of electrons ejected per second depends on intensity.

$$h\nu - h\nu_0 = \frac{1}{2} m_e v^2$$

Photoelectric effect:



Photoelectric effect — evacuated tube experiment

3.10 Planck's Quantum Theory

- Substances radiate or absorb energy discontinuously in the form of energy packets.
- The smallest packet of energy is a quantum (in light — photon).
- $E = h\nu$ — energy of a quantum $\propto$ frequency of radiation; h = Planck's constant = 6.626×10^{-34} J·s (or 6.626×10^{-27} erg·s).
- A body can radiate/absorb energy only in whole-number multiples of quanta: $h\nu, 2h\nu, 3h\nu \dots nh\nu$.

3.11 Bohr's Model — Quantitative Results

- Electrons revolve only in those orbits whose angular momentum (mvr) is an integral multiple of $h/2\pi$.
- Electron absorbs EMR energy when it jumps from a lower (ground) to higher (excited) level and vice-versa; $\Delta E = E_2 - E_1 = h\nu$.

- Energy of stationary state of H atom: $E_n = -R_H(1/n^2)$. For H-like species (He^+ , Li^{2+} , atomic number Z):
 Radius of n -th orbit $r_n = 52.9 \times n^2/Z \text{ pm}$ · $E_n = -2.18 \times 10^{-18} (Z^2/n^2) \text{ J} = -13.6 (Z^2/n^2) \text{ eV} = -313.6 (Z^2/n^2) \text{ kcal/mol}$ · $v = 2.18 \times 10^8 (Z/n) \text{ cm s}^{-1}$

Limitations of Bohr's theory

- Failed to explain spectra of atoms having more than one electron.
- Failed to account for splitting of spectral lines when the source is placed in a strong magnetic/electric field.
- Ignored the dual nature of particles and the uncertainty principle.

3.12 de Broglie Equation & Uncertainty Principle

$$\lambda = h/mv = h/p \quad \Delta x \cdot \Delta p \geq h/4\pi m$$

- Every moving particle has a wave character (de Broglie).
- It is impossible to determine simultaneously the exact position and exact momentum of an electron (Heisenberg).

3.13 Quantum Numbers

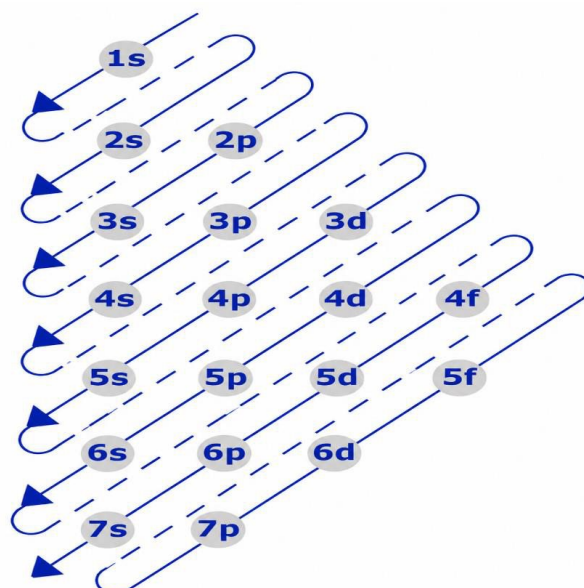
- Principal quantum number (n): main shell in which the electron resides; approximate distance from the nucleus; maximum electrons in a shell = $2n^2$.
- Azimuthal / angular momentum quantum number (l): number of subshells in the main shell — denoted s, p, d, f ; tells the shape of the subshell; for given n , $l = 0$ to $(n-1)$.
- Magnetic quantum number (m): preferred orientations of the electron in a subshell; for given l , $m = -l \dots 0 \dots +l$ ($2l+1$ values).
- Spin quantum number (s): direction of spin; only two values, $-\frac{1}{2}$ and $+\frac{1}{2}$.

Value of l	0	1	2	3
Notation of subshell	s	p	d	f
Number of orbitals ($2l+1$)	1	3	5	7

3.14 Rules for Filling of Electrons in Orbitals

Aufbau principle

- Electrons are added one by one to orbitals in order of increasing energy, starting with the lowest-energy orbital.
- In a neutral isolated atom, the lower the value of $(n + l)$ for an orbital, the lower its energy.
- Increasing order of orbital energy: $1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d$...



Aufbau diagonal rule — order of filling of orbitals

Pauli's exclusion principle

- An orbital can contain a maximum of two electrons, and these two must be of opposite spin.

Hund's rule of maximum multiplicity

- Electron pairing in p, d and f orbitals cannot occur until each orbital of a given subshell contains one electron (is singly occupied).

Exceptional configurations — stability of half-filled & fully-filled orbitals

- Cu (29 e⁻): expected $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^9$ — but shifting one electron from 4s to 3d makes the distribution symmetrical and more stable → actual configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$.
- Similarly Cr achieves a half-filled d^5 . Fully-filled and half-filled subshells are extra stable.

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- Discovery: electron — J.J. Thomson (cathode rays); proton — Rutherford (gold foil); neutron — Chadwick.
- Z = protons; $A = p + n$; isotopes same Z ; isobars same A ; isotones same neutrons; isoelectronic same e⁻.
- Shell capacity $2n^2$ ($K=2, L=8, M=18, N=32$); outermost shell ≤ 8 (octet rule).
- Spectral series: Lyman (UV) → Balmer (visible) → Paschen, Brackett, Pfund (IR).
- Photoelectric effect: $KE \propto \text{frequency}$; electron count $\propto \text{intensity}$; $h\nu - h\nu_0 = \frac{1}{2}mv^2$.
- $E_n = -13.6 \frac{Z^2}{n^2} \text{ eV}$; $r_n = 52.9 \frac{n^2}{Z} \text{ pm}$; $\lambda = h/mv$; $\Delta x \cdot \Delta p \geq h/4\pi m$.
- Aufbau ($n+l$ rule) → Pauli (max 2 e⁻, opposite spin) → Hund (pair only after single occupancy); Cu = $3d^{10}4s^1$, Cr = $3d^54s^1$.

UNIT 04

Periodic Classification & Trends

Sources: Periodic Table lecture + revision sections of Atomic Structure & Acids-Bases

4.1 Introduction

- Matter exists as elements, compounds and mixtures. Elements contain atoms of only one type (Na, Mg, Au...). 118 elements are known, all with different properties; to make their study easy, elements are divided into groups such that elements of the same group have similar properties.

4.2 Dobereiner's Triads (1829)

- 30 elements had been discovered by 1829. Dobereiner, a German scientist, grouped some elements in threes — triads — with similar chemical properties.
- The atomic mass of the middle element $\approx$ average of the atomic masses of the other two.

Triad	Elements	Atomic masses	Arithmetic mean
Triad-1	Li, Na, K	7, 23, 39	$(7+39)/2 = 23$
Triad-2	Cl, Br, I	35.5, 80, 126.5	$(35.5+126.5)/2 = 81$
Triad-3	Ca, Sr, Ba	40, 87, 137	$(40+137)/2 = 88.5 \approx 87$

Limitations

- Only three triads were recognised from the elements known at that time — only 9 elements (out of 30) could be arranged in triad form.

4.3 Newland's Law of Octaves (1866)

- By 1866, 56 elements were known. Newland, an English scientist, arranged them in increasing order of atomic masses and found that every 8th element repeats properties (8th = 1st, 9th = 2nd ...) — like octaves in music (sa, re, ga, ma...).
- Example: properties of lithium (Li) and sodium (Na) were found to be the same.

Limitations

- Applicable to only 17 elements, up to calcium (lighter elements only).
- Properties of newly discovered elements did not fit the law.
- At places two elements were put in one slot, in columns of unlike elements with very different properties.
- Placing of iron (Fe) far away from cobalt (Co) and nickel (Ni) — which have similar properties — could not be explained.

4.4 Mendeleev's Periodic Table (1869)

- Dmitry Mendeleev, a Russian chemist, arranged the 63 known elements in increasing order of atomic mass and found that elements with similar properties occur at regular intervals; he placed similar elements in a group — leaving certain empty boxes.
- Mendeleev's Periodic Law: "The properties of the elements are a periodic function of their atomic masses."
- Structure: horizontal rows = periods; vertical columns = groups (I to VIII). Groups I-VII subdivided into A and B; group VIII has no subgroups. Properties show regular gradation left→right in a period; elements of a group have similar properties.

Group	I	II	III	IV	V	VI	VII	VIII
Periods	A	B	A	B	A	B	A	B
Oxides:	R ₂ O	R ₂ O	R ₂ O ₃	R ₂ O	R ₂ O ₃	R ₂ O ₅	R ₂ O ₃	RO ₅
Hydride	RH	RH ₅	RH ₅	RH ₅	RH ₅	RH ₅	RH	
1.	H 1.008							
2.	Li 6.939	Be 2.312	B 10.81	C 12.011	N 14.007	O 15.999	F 18.999	
3.	Na 22.99	Mg 22.99	Al 24.31	Si 28.09	P 30.504	S 32.06	Cl 35.453	
4. First Series	K 39.12	Ca 40.00	Sc 44.96	Ti 47.59	V 50.94	Cr 50.28	Mn 54.54	Fe 55.85 Co 56.93 Ni 58.71
Second Series								
5. First Series	Ba 85.47	Sr 87.62	Y 88.31	Zr 51.22	Nb 59.51	Mo 55.54	Te 99	Ra 931.07 Rk 102.51 Pd 106.4
Second Series								
6. First Series	Cs 132.30	Be 137.34	La 138.91	Hf 178.40	Ta 108.55	W 183.85		Ra 190.2 Eh 152.2 Pd 155.09
Second Series								

Mendeleev's periodic table

Mendeleev's periodic table (with predicted elements)

Merits

- Predicted properties of elements from their positions; made the study of elements systematic — knowing one element of a group lets us predict the others.
- Gaps left for undiscovered elements helped find them — Gallium (Ga), Scandium (Sc), Germanium (Ge) were discovered this way.
- When inert gases were discovered they fit into a new group without disturbing the existing order.
- Corrected doubtful atomic masses of certain elements using expected positions and properties.

Limitations

- Isotopes (same element, different mass) violated the increasing-atomic-mass arrangement — unexplainable by Mendeleev.
- Position of hydrogen — properties resemble both group I and group VII, yet it was placed in group I without explanation.
- Wrong (decreasing) order of atomic masses of some pairs could not be explained — e.g. Co-Ni and Te-I.

4.5 Modern Periodic Table

- In 1913, Moseley showed that atomic number is the fundamental property of an element; Neil Bohr then constructed the Modern Periodic Table using atomic number.
- Modern Periodic Law: "The properties of elements are a periodic function of their atomic number."
- Basic concept: most properties of an element depend on the number of valence electrons; elements with the same number of valence electrons are grouped together.

Metals										Metalloids	Non-metals				Unknown chemical properties			
Alkali metals	Alkaline earth metals	Lanthanides	Actinides	Transition metals	Poor metals	Moderately active nonmetals	Highly active nonmetals	Noble gases										
Group names*																		
1 H	2	1 Hydrogen & the alkali metals 2 Alkaline earth metals 11 Coinage metals (Cu, Ag & Au) 12 Volatile metals 13 Boron Group										14 Carbon Group 15 Pnictogens 16 Chalcogens 17 Halogens 18 Noble gases	13 5 13 31 49 81 113	14 6 14 32 50 82 114	15 7 15 33 51 83 115	16 8 16 34 52 84 116	17 9 17 35 53 85 117	18 2 10 Ne 18 Ar 54 Xe 86 Rn 118 Uuo
3 Li	4 Be	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr	
11 Na	12 Mg	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe	
19 K	20 Ca	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe	
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe	
55 Cs	56† Ba	71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn	
87 Fr	88† Ra	103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Uut	114 Fl	115 Uup	116 Lv	117 Uus	118 Uuo	
Lanthanides	†57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	* Groups 3–10 are named after their first members i.e. Group 3 is the Scandium Group			
Actinides	†89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No				
The modern periodic table — 7 periods, 18 groups, s/p/d/f blocks																		

The modern periodic table — 7 periods, 18 groups, s/p/d/f blocks

Structure

- Periods: 7 horizontal rows; number of shells increases down a group (within a period all elements have the same number of shells).
- Groups: 18 vertical columns; elements of a group share the same number of valence electrons.
- Blocks: s, p, d, f — based on the orbital into which the last electron enters. (s-block: groups 1–2; p-block: 13–18; d-block: 3–12, transition elements; f-block: inner transition — lanthanides & actinides, placed at the bottom.)
- 1st period = shortest (2 elements: H, He); 2nd & 3rd = short periods (8 each); 4th & 5th = long periods (18 each); 6th = very long (32 elements); 7th is incomplete.
- Group 1 (extreme left) = alkali metals (Li, Na, K, Rb, Cs, Fr); group 18 (extreme right) = noble gases (He, Ne, Ar, Kr, Xe, Rn).
- In the middle lie metalloids / semi-metals — showing properties of both metals and non-metals.
- Helium exception: group-18 elements have 8 valence electrons except helium (2), but He is correctly placed in group 18 — it has a stable (duplet) configuration and similar inert behaviour.

Position of an element & electronic configuration

- An element's position corresponds to the quantum numbers of its last-filled orbital.
- In periods: the n of the outermost (valence) shell corresponds to the period number; each period fills the next principal energy level ($n = 1, 2, \dots$); number of elements in a period = twice the number of orbitals available in that level.
- In groups: elements of a group have similar valence-shell configuration (e.g. group 1 $\rightarrow ns^1$), giving analogous properties.

Period	Elements	Type
1st	2 (H, He)	Shortest
2nd	8 (Li $\rightarrow$ Ne)	Short
3rd	8 (Na $\rightarrow$ Ar)	Short
4th	18 (K $\rightarrow$ Kr)	Long
5th	18 (Rb $\rightarrow$ Xe)	Long
6th	32 (Cs $\rightarrow$ Rn)	Very long

Period	Elements	Type
7th	Incomplete (Fr onwards)	—

Key group characteristics

- Group 1 (Alkali metals): highly reactive, 1 valence electron, form +1 ions.
- Group 17 (Halogens): highly reactive non-metals, 7 valence electrons, form –1 ions.
- Group 18 (Noble gases): inert/unreactive, full valence shell.

4.6 Atomic Number, Electron Distribution & Valency (summary)

- Atomic number (Z) = total protons in the nucleus; atoms of different elements always have different Z; hydrogen has Z = 1.
- Electron distribution rules: $2n^2$ capacity; fill from inside out; ≤ 8 electrons in the outermost shell (octet rule).
- Valency: valence electrons in the outermost orbit. Across a period valency increases $1 \rightarrow 4$ then falls to 0; down a group it stays the same.
- Stable configuration = $2 e^-$ in K (if K is outermost) or $8 e^-$ in the outermost shell. Atoms react to attain it; Ag (1 outer e^-) donates one $\rightarrow$ valency 1.

4.7 Periodic Trends in Properties

Property	Across a period (left $\rightarrow$ right)	Down a group (top $\rightarrow$ bottom)
Valence electrons / shell	Valence e^- increase (1 $\rightarrow$ 8); valence shell constant	Valence e^- constant; valence shell increases
Valency	Increases till group 14, then decreases to 18	Remains constant
Atomic size / radius	Decreases — nuclear charge $\uparrow$, outer electrons pulled closer	Increases — number of shells increases
Metallic character	Decreases — effective nuclear charge $\uparrow$, tendency to lose $e^- \downarrow$	Increases — effective nuclear charge $\downarrow$, easier to lose e^-
Non-metallic character	Increases — tendency to gain $e^- \uparrow$	Decreases — tendency to gain $e^- \downarrow$
Electropositivity	Decreases (ionization energy $\uparrow$)	Increases (ionization energy $\downarrow$)
Ionization enthalpy	Increases (stronger nuclear hold)	Decreases (bigger size, shielding)
Electronegativity	Increases (increasing nuclear charge)	Decreases
Nature of oxides	Acidic nature increases	Acidic nature decreases
Chemical reactivity	Metals: decreases; Non-metals: increases	Metals: increases; Non-metals: decreases

Atomic radius in detail

- Atomic radius = distance from nucleus to the outermost electron; decreases left $\rightarrow$ right in a period, increases down a group. It helps define the covalent radius (non-metals) and metallic radius (metals); a third type is the van der Waals radius.
- Influencing factors: number of shells (more shells $\rightarrow$ larger radius); effective nuclear charge Z_{eff} (higher $\rightarrow$ smaller radius); screening effect (stronger shielding by filled orbitals $\rightarrow$ larger radius).

Ionic radius

- Distance between the nucleus centre and outermost electron of an ion. Decreases across a period, increases down a group. Anions are larger than the parent atom; cations are typically smaller.

Ionization energy

- Minimum energy needed to remove an electron from an isolated gaseous atom in its ground state. Increases across a period; decreases down a group. It measures the tendency of an atom to give up an electron — the lower the IE, the easier a neutral atom becomes a positive ion.

- Factors: atomic size, effective nuclear charge (Z_{eff}), screening effect, penetration effect, electronic configuration.

Electron gain enthalpy

- Enthalpy change when an electron is added to a neutral gaseous atom forming a gaseous anion. Negative value = energy released = greater stability; positive = energy absorbed = decreased stability. Increases left→right; decreases down a group.
- Factors: atomic size, effective nuclear charge, screening effect, electronic configuration.

Electronegativity

- An atom's ability within a molecule to pull the shared pair of electrons towards itself — not a fixed property; varies with the bonding partner. Helps predict the nature of the bond formed.
- Factors: atomic size, effective nuclear charge (Z_{eff}), magnitude of positive charge on the atom.

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- Dobereiner triads — mean mass of ends $\approx$ middle (Li-Na-K, Cl-Br-I, Ca-Sr-Ba).
- Newlands — every 8th element repeats (only up to Ca); Fe away from Co/Ni — a flaw.
- Mendeleev — atomic mass basis; predicted Ga, Sc, Ge; corrected atomic masses; failed for isotopes & H position.
- Moseley/Bohr — modern law: properties are a periodic function of atomic number; 7 periods, 18 groups, 4 blocks.
- Across period: size ↓, IE ↑, EN ↑, metallic ↓, oxide acidity ↑. Down group: all reverse.
- IE factors — size, Z_{eff} , screening, penetration, configuration. Anion > atom > cation in size.

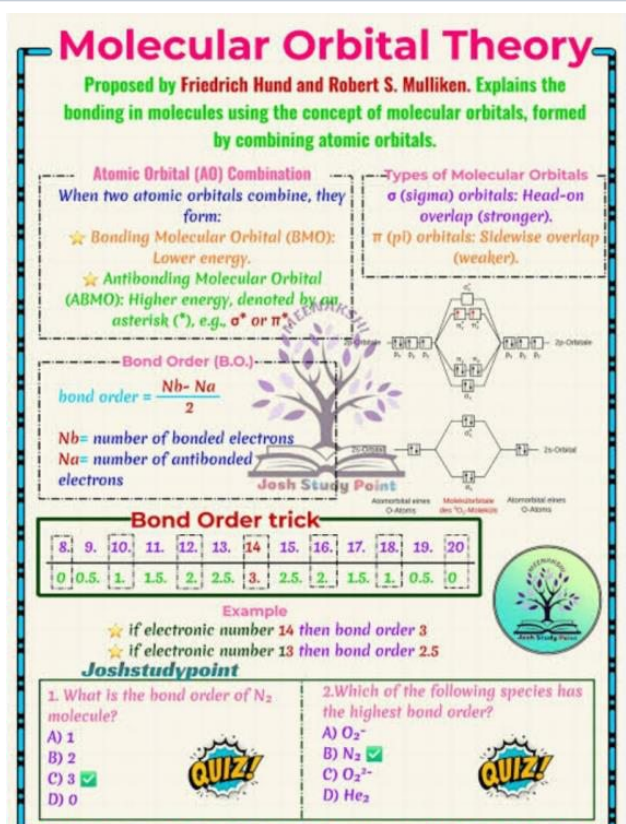
UNIT 05

Chemical Bonding & Molecular Structure

Sources: Chemical Bonding + Chemical Bond & Hydrogen Bond lectures

5.1 Chemical Bonding — Introduction

- Chemical bond — the attractive force that holds atoms together in a molecule to increase stability and minimise energy.
- Cause of combination: atoms combine to achieve a stable electronic configuration — usually a full octet (8 valence electrons) in the outermost shell.
- Strong bonds (ionic, covalent, metallic) involve transfer/sharing of electrons within compounds. Weak bonds (hydrogen, van der Waals) are intermolecular forces, crucial for molecular structure, solubility and state of matter.



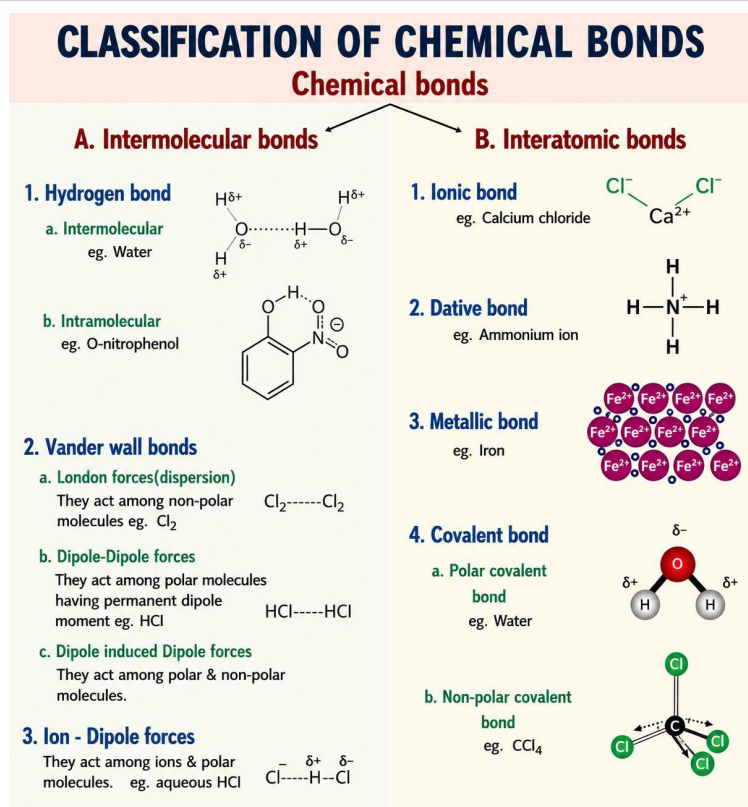
Chemical bonding — complete overview chart (original mind-map)

5.2 Types of Chemical Bonds

Bond type	Mechanism (action)	Key feature / example
Ionic (electrovalent)	Transfer — complete transfer of valence e^- from metal to non-metal; electrostatic attraction between ions	High m.p./b.p., water-soluble, conduct when molten/dissolved. e.g. NaCl
Covalent	Sharing of electron pairs between non-metals	Polar (unequal sharing, H_2O) or non-polar (equal, O_2 / H_2); multiple (double/triple) bonds possible
Metallic	Valence electrons delocalised — “sea of electrons” around positive metal ions	Electrical & thermal conductivity, malleability, ductility. e.g. Fe, Cu
Coordinate (dative)	Both shared electrons come from the same atom (donor-acceptor, one-sided sharing)	e.g. NH_4^+ , $BF_3 \cdot NH_3$

Bond type	Mechanism (action)	Key feature / example
Hydrogen (weak)	Dipole-dipole attraction — H (δ^+) bonded to N, O or F attracts an electronegative atom of another molecule	Stronger than van der Waals; DNA structure, properties of water
Van der Waals (very weak)	Temporary/induced dipoles from instantaneous uneven electron distribution (London dispersion forces)	Weakest attraction; noble gases, CH ₄

- Relative strength: Ionic $\approx$ Covalent > Metallic > Hydrogen bond > Van der Waals force.
- Key differences: ionic involves transfer (metal + non-metal), covalent involves sharing (non-metals); ionic/metallic bonds conduct electricity, covalent typically do not; ionic/covalent/metallic bonds bind atoms permanently while hydrogen/van der Waals are mere intermolecular forces.



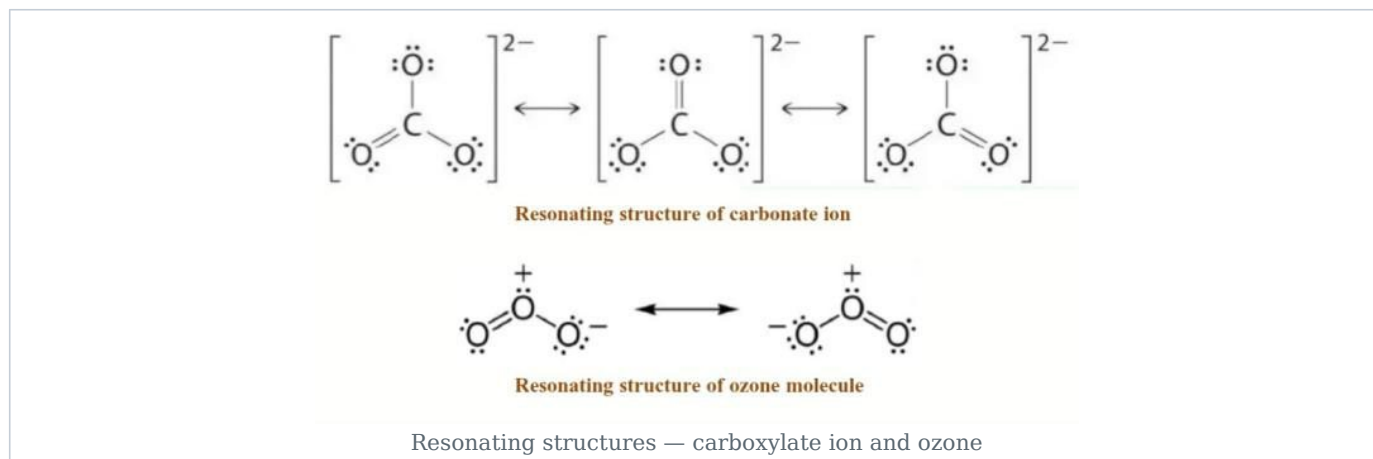
Classification of chemical bonds — intermolecular & intramolecular

5.3 Bond Parameters

- Bond length — equilibrium distance between the nuclei of two bonded atoms; measured by spectroscopy, X-ray & electron diffraction. Single bond > double bond > triple bond.
- Bond angle — angle between the orbitals containing bonding electron pairs around the central atom (in degrees); indicates distribution of orbitals and helps determine shape.
- Bond enthalpy — energy required to break one mole of a particular bond in the gaseous state: $A-B(g) \rightarrow A(g) + B(g)$, in kJ mol^{-1} . For polyatomic molecules (two O-H bonds in H₂O) the two breaks need different energies, so we use the mean bond enthalpy.
- Bond order — number of bonds between two atoms; bond strength $\propto$ bond order (triple > double > single).

5.4 Resonance

- When a molecule/ion cannot be described by a single structure, it is represented by several resonating (canonical) structures; none describes all properties completely, but each contributes to the true structure — the resonance hybrid.
- Resonance leads to extra stability of the molecule/ion.
- Resonance energy = energy of actual molecule — energy of the most stable resonating structure.



5.5 Polarity of Bonds & Dipole Moment

- Difference in electronegativity between two bonded atoms creates partial charges → polar covalent bonds / polar molecules.
- Dipole moment (μ) = charge on the bonded atoms $\times$ distance between them; a vector quantity directed from the positive to the negative end; unit: debye (D), $1 \text{ D} = 3.33564 \times 10^{-30} \text{ C}\cdot\text{m}$.

Applications of dipole moment

- Predicts whether a molecule is polar or non-polar.
- Compares relative polarities — greater μ , greater polarity.
- Determines shape: CO_2 has polar $\text{C}=\text{O}$ bonds but $\mu = 0 \rightarrow$ linear; SO_2 has $\mu = 1.60 \text{ D} \rightarrow$ bent.
- Helps calculate percentage ionic character.

Fajan's rules (polarisation)

- Polarisation = covalent character developed in an ionic bond. The ion distorting the other's electron cloud has polarising power; the ion getting distorted has polarisability. Higher either → more covalent character.
 - Small cation — smaller size → greater polarising power.
 - Large anion — greater size → higher polarisability.
 - Higher charge on either ion.
 - A medium of low polarising power (keeps ions apart) prevents polarisation.
 - Q. Why is LiCl covalent while NaCl is ionic? — Li^+ is very small → high polarising power → distorts Cl^- strongly → covalent character.

5.6 VSEPR Theory

- Valence Shell Electron Pair Repulsion: molecular shape is decided by minimising repulsion between electron pairs (bond pairs & lone pairs) around the central atom.
- Repulsion order: lone pair-lone pair $>$ lone pair-bond pair $>$ bond pair-bond pair.

VSEPR Theory

Definition

● **VSEPR** (Valence Shell Electron Pair Repulsion) theory states that electron pairs around a central atom repel each other and arrange themselves to be as far apart as possible, determining the molecule's 3D shape.

● This minimizes electron repulsion, resulting in a **more stable molecule** with a specific **geometry** based on the number of bonding and non-bonding (lone) electron pairs.

Lone pair - lone pair electrons Repulsion

Lone pair - Bond pair electrons Repulsion

Bond pair - Bond pair electrons Repulsion

Strong Repulsion

Weaker Repulsion

Hybridisation

sp
sp³
sp³d
sp³d²

VSEPR theory — definition & repulsion order

Number of Electron Dense Areas	Electron-Pair Geometry	Molecular Geometry				
		No Lone Pairs	1 lone Pair	2 lone Pairs	3 lone Pairs	4 lone Pairs
2	Linear SP	180° Linear				
3	Trigonal planar SP ²	120° Trigonal planar	120° Bent			
4	Tetrahedral SP ³	109.5° Tetrahedral	Trigonal pyramidal	Bent		
5	Trigonal bipyramidal SP ³ d	120° Trigonal bipyramidal	90° Sawhorse	T-shaped	Linear	
6	Octahedral SP ³ d ²	180° Octahedral	90° Square pyramidal	Square planar	T-shaped	Linear

Geometries for 2-7 electron-pair systems

5.7 Hybridisation

- Mixing of atomic orbitals (s, p...) to form new equivalent hybrid orbitals (sp, sp², sp³ ...) that decide geometry.
- sp → linear; sp² → trigonal planar; sp³ → tetrahedral; sp³d → trigonal bipyramidal; sp³d² → octahedral.

HYBRIDIZATION

WHAT IS HYBRIDIZATION? Chemist

The **mixing of atomic orbitals** (s, p, d) of similar energy to form new, **degenerate orbitals** called **hybrid orbitals** with **definite shape** and **orientation**.

- Explains the equivalent bonds in molecules.
- Leads to greater stability and definite geometry.

TYPES OF HYBRIDIZATION

- sp Hybridization**
 - Intermixing of 1 s and 1 p orbital.
 - Forms 2 linear hybrid orbitals.
 - Linear Geometry** and **180° Bond Angle**

Examples: BeCl₂, C₂H₂ (Ethyne)
- sp² Hybridization**
 - Intermixing of 1 s and 2 p orbitals.
 - Forms 3 trigonal planar hybrid orbitals.
 - Trigonal Planar geometry** and **120° Bond Angle**

Examples: BF₃, C₂H₄ (Ethene)
- sp³ Hybridization**
 - Intermixing of 1 s and 3 p orbitals.
 - Forms 4 tetrahedral hybrid orbitals.
 - Tetrahedral Geometry** and **~109.5° Bond Angle**

Examples: CH₄ (Methane), NH₃ (Ammonia), H₂O (Water)
(Varying angles due to lone pairs might be added subtly)

HYBRIDIZATION SUMMARY

Type	Orbitals Mixed	Geometry	Bond Angle	Examples
sp	1 s + p	Linear	180°	BeCl ₂ , C ₂ H ₂
sp ²	1 s + 2 p	Trigonal planar	120°	BF ₃ , C ₂ H ₄
sp ³	1 s + 3 p	Tetrahedral	~109.5°	CH ₄ , NH ₃ , H ₂ O

IMPORTANT NOTES:

- Hybrid orbitals form stronger **sigma bonds**.
- VSEPR geometry aligns with hybridization.
- Only valence orbitals participate.
- Proper spacing and alignment throughout.
- (Varying angles due to lone pairs might be added subtly)

Chemist

Hybridisation — summary infographic

5.8 Molecular Orbital Theory (MOT)

- Atomic orbitals combine to form molecular orbitals spread over the entire molecule; bonding (σ, π) and antibonding (σ*, π*) orbitals result.
- Bond order = $(N_b - N_a)/2$, where N_b , N_a = electrons in bonding and antibonding MOs. Higher bond order → greater stability.

Energy order of molecular orbitals

- For 1-14 electron systems (e.g. N₂): $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \pi 2p_x = \pi 2p_y < \sigma 2p_z < \pi^* 2p_x = \pi^* 2p_y < \sigma^* 2p_z$
- For 15-20 electron systems (e.g. O₂, F₂): $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \pi 2p_x = \pi 2p_y < \pi^* 2p_x = \pi^* 2p_y < \sigma^* 2p_z$

- Key feature: for 1–14 e^- systems $\pi 2p_x = \pi 2p_y$ lie lower than $\sigma 2p_z$. Each σ/π orbital holds 2 e^- ; π_x, π_y are degenerate (4 total).
- Orbitals are filled per the Aufbau principle, Hund's rule and Pauli's exclusion principle.

Number of electron pairs around central atom		Full description of the molecule					
BONDING (B)	LONE (L)	Example	Bond angles (°)	Geometry of Electron Pairs	Geometry of Atoms	3D Shape	Type
2	0	BeCl ₂	180	Linear	Linear		AB ₂
3	0	BF ₃	120	Trigonal planar	Trigonal Planar		AB ₃
2	1	SO ₂	Slightly less than 120	Trigonal planar	Bent or V Shaped		AB ₂ E
4	0	CH ₄	109.5	Tetrahedral	Tetrahedral		AB ₄
3	1	NH ₃	107.5	Tetrahedral	Trigonal Pyramidal		AB ₃ E
2	2	H ₂ O	104.5	Tetrahedral	Bent or V Shaped		AB ₂ E ₂
5	0	PCl ₅	120 in plane, 90 perpendicular to plane	Trigonal bipyramidal	Trigonal Bipyramidal		AB ₅
4	1	SF ₄	Complex	Trigonal bipyramidal	Seesaw		AB ₄ E
3	2	ClF ₃	Approx. 90	Trigonal bipyramidal	T-Shaped		AB ₃ E ₂
2	3	XeF ₂	180	Trigonal bipyramidal	Linear		AB ₂ E ₃
6	0	SF ₆	90	Octahedral	Octahedral		AB ₆
5	1	BrF ₅	Approx. 90	Octahedral	Square Pyramidal		AB ₅ E
4	2	XeF ₄	90	Octahedral	Square Planar		AB ₄ E ₂

Full description of molecular orbitals

Electron Groups	Bonding Groups	Lone Pairs	Electron Geometry	Molecular Geometry	Approximate Bond Angles	Example
2	2	0	Linear	Linear	180°	$\text{:S}=\text{C}=\text{S:}$
3	3	0	Trigonal planar	Trigonal planar	120°	$\text{:Cl}-\text{B}-\text{Cl:}$
3	2	1	Trigonal planar	Bent	<120°	$\text{:O}=\text{S}=\text{O:}$
4	4	0	Tetrahedral	Tetrahedral	109.5°	$\text{H}-\text{C}-\text{H}$
4	3	1	Tetrahedral	Trigonal pyramidal	<109.5°	$\text{H}-\text{P}-\text{H}$
4	2	2	Tetrahedral	Bent	<<109.5°	$\text{H}-\text{S}-\text{H}$
5	5	0	Trigonal bipyramidal	Trigonal bipyramidal	120° (equatorial) 90° (axial)	$\text{:Cl}-\text{P}-\text{Cl:}$
5	4	1	Trigonal bipyramidal	Seesaw	<120° (equatorial) <90° (axial)	$\text{:F}-\text{S}-\text{F:}$
5	3	2	Trigonal bipyramidal	T-shaped	<90°	$\text{:F}-\text{Cl}-\text{F:}$
5	2	3	Trigonal bipyramidal	Linear	180°	$\text{:F}-\text{Xe}-\text{F:}$
6	6	0	Octahedral	Octahedral	90°	$\text{:F}-\text{S}-\text{F:}$
6	5	1	Octahedral	Square pyramidal	<90°	$\text{:F}-\text{Br}-\text{F:}$
6	4	2	Octahedral	Square planar	90°	$\text{:F}-\text{Xe}-\text{F:}$

Energy level diagrams of MOs

Worked examples

- Ne₂ (20 e^-): all bonding and antibonding levels filled $\rightarrow N_b = N_a \rightarrow$ bond order = $(10-10)/2 = 0 \rightarrow$ molecule does not exist.
- O₂ (16 e^-): $\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* 2p_x^1 \pi^* 2p_y^1 \rightarrow$ bond order = $(10-6)/2 = 2$; two unpaired electrons $\rightarrow$ paramagnetic.
- N₂ (14 e^-): bond order = $(10-4)/2 = 3$ (diamagnetic); C₂ (12 e^-) bond order 2; N₂⁺ (13 e^-) bond order 2.5.

✂ RAPID REVISION

- Ionic = transfer, covalent = share, metallic = sea of e^- , coordinate = one-sided share, H-bond & vdw = weak intermolecular.
- Strength: ionic/covalent > metallic > H-bond > van der Waals.
- Bond length: single > double > triple; bond strength $\propto$ bond order.
- VSEPR repulsion: lp-lp > lp-bp > bp-bp; sp/sp²/sp³ $\rightarrow$ linear/trigonal/tetrahedral.
- MO bond order = $(N_b - N_a)/2$: N₂ = 3, O₂ = 2 (paramagnetic), Ne₂ = 0; for $\leq 14 e^-$, $\pi 2p$ lies below $\sigma 2p$.
- Dipole moment: CO₂ $\mu = 0$ (linear), SO₂ $\mu = 1.6$ D (bent); LiCl covalent (Fajan's rules), NaCl ionic.

UNIT 06

Chemical Reactions

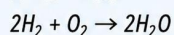
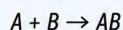
Source: Chemical Reaction lecture

6.1 Classification of Chemical Reactions

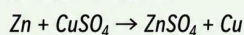
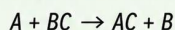
Chemical reactions are classified by how atoms rearrange. The five primary types are combination, decomposition, single displacement, double displacement and combustion/redox reactions — they explain everyday phenomena like rusting, digestion and photosynthesis.

Reaction type	Pattern	Example
Combination (synthesis)	$A + B \rightarrow AB$	$C + O_2 \rightarrow CO_2$; $2H_2 + O_2 \rightarrow 2H_2O$
Decomposition	$AB \rightarrow A + B$	$CaCO_3 \xrightarrow{\text{heat}} CaO + CO_2$; $2KClO_3 \rightarrow 2KCl + 3O_2$
Single displacement	$A + BC \rightarrow AC + B$	$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$; $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
Double displacement	$AB + CD \rightarrow AD + CB$	$AgNO_3 + NaCl \rightarrow AgCl \downarrow + NaNO_3$; $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 \downarrow + 2NaCl$
Redox (oxidation-reduction)	Electron transfer (oxidation + reduction)	$Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$
Combustion	$\text{Fuel} + O_2 \rightarrow CO_2 + H_2O + \text{energy}$	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
Neutralisation	$\text{Acid} + \text{Base} \rightarrow \text{Salt} + \text{Water}$	$HCl + NaOH \rightarrow NaCl + H_2O$
Precipitation	$\text{Aqueous} + \text{aqueous} \rightarrow \text{insoluble solid}$	$2NaOH + CuSO_4 \rightarrow Cu(OH)_2 \downarrow + Na_2SO_4$

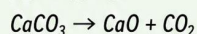
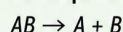
1. Combination Reaction



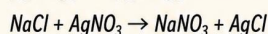
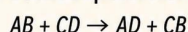
3. Displacement Reaction



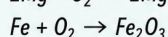
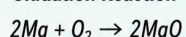
2. Decomposition Reaction



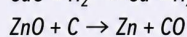
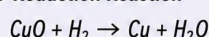
4. Double Displacement Reaction



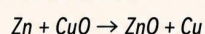
5. Oxidation Reaction



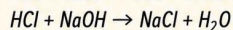
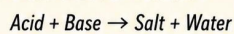
6. Reduction Reaction



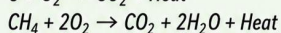
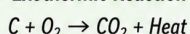
7. Redox Reaction



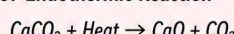
8. Neutralization Reaction



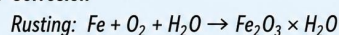
9. Exothermic Reaction



10. Endothermic Reaction



11. Corrosion



12. Rancidity

Oxidation of oils / fats causing bad smell and taste.

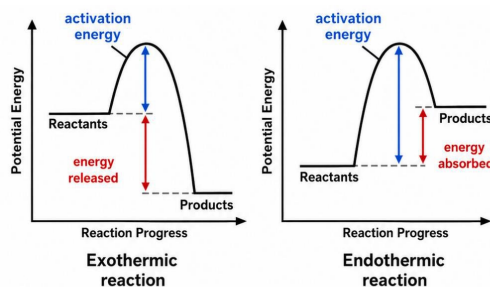


Types of chemical reactions — poster summary

- Neutralisation is a double displacement in which an acid and a base form salt and water.
- Precipitation reaction — a double displacement producing an insoluble solid (precipitate).

6.2 Exothermic vs Endothermic Reactions

Basis	Exothermic	Endothermic
Energy change	Releases energy (heat, light, sound) to surroundings	Absorbs energy from surroundings
Temperature	Surroundings become hot	Surroundings become cold
Enthalpy	Negative ($\Delta H < 0$)	Positive ($\Delta H > 0$)
Bonding	Typically forms stronger bonds	Breaks bonds (energy absorbed)
Examples	Combustion of fuels, rusting of iron, nuclear fission, mixing acids with bases, freezing of water	Photosynthesis, melting of ice, evaporation, thermal decomposition, dissolving NH_4NO_3 in water



EXOTHERMIC VS ENDOTHERMIC REACTIONS	
Exothermic reaction	Endothermic reaction
Energy is lost from the system to the surroundings	Energy is gained from the surroundings by the system
Temperature of the surroundings, reaction mixture, solution increases (become hotter)	Temperature of the surroundings, reaction mixture, solution decreases (become colder)
reactants $\rightarrow$ products + energy	reactants + energy $\rightarrow$ product
ΔH is Negative	ΔH is Positive
If the energy lost is greater than the energy taken in	If the energy taken in is greater than the energy lost
ΔS (Entropy change) is +ve	ΔS (Entropy change) is -ve
products have less energy than the reactants	products have more energy than the reactants
Energy needed to break the bonds in reactants is less than the energy released in making new bonds in the products	Energy needed to break the bonds in reactants is greater than the energy released in making new bonds in the products
Spontaneous reaction	Non spontaneous reaction
Examples of exothermic reactions: Respiration	Examples of endothermic reactions: Photosynthesis, Digestion

Exothermic vs endothermic — energy profiles & comparison

6.3 Effects of Oxidation in Everyday Life

Corrosion

- When a metal is attacked by substances around it (moisture, acids etc.), it is said to corrode.
- Iron gets coated with a reddish-brown powder — rusting of iron; the black coating on silver and the green coating on copper are other examples of corrosion.

Rancidity

- When fats and oils are oxidised, they become rancid — smell and taste change.
- Prevention: antioxidants are added to fatty/oily foods; airtight containers slow oxidation; chip packets are flushed with nitrogen gas.

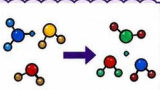
6.4 Reactions in Daily Life

- Rusting of iron — redox reaction;
- Cooking food — decomposition;
- Respiration — redox + decomposition;
- Baking cake with baking soda — decomposition + displacement;
- Vinegar + baking soda — double displacement + gas evolution;
- Burning candle — combustion.

TYPES OF CHEMICAL REACTIONS

What is a Chemical Reaction? InfoPic Notes

A chemical reaction is a process in which one or more substances (REACTANTS) are converted into new substances (PRODUCTS) with different chemical properties due to the REARRANGEMENT of atoms.



1. COMBINATION (Synthesis) Reaction

➡ Two or more substances combine to form one single product.

General form: $A + B \rightarrow AB$

Example: $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$

★ Often exothermic (releases heat)

2. DECOMPOSITION Reaction

➡ A single compound breaks down into two or more simpler substances.

General form: $AB \rightarrow A + B$

Example: $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$

★ Usually requires heat, light, or electricity

3. DISPLACEMENT (Single Replacement) Reaction

➡ A more reactive element displaces a less reactive element from its compound.

General form: $A + BC \rightarrow AC + B$

Example: $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$

★ Occurs based on REACTIVITY SERIES ↑

4. DOUBLE DISPLACEMENT (Metathesis) Reaction

➡ Exchange of ions between two compounds to form new compounds.

General form: $AB + CD \rightarrow AD + CB$

Example: $\text{Na}_2\text{SO}_4 + \text{BaCl}_2 \rightarrow \text{BaSO}_4 \downarrow + 2\text{NaCl}$

★ Often forms a precipitate, gas, or water

5. PRECIPITATION Reaction

➡ Formation of an insoluble solid (PRECIPITATE) from two aqueous solutions.

Example: $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} \downarrow + \text{NaNO}_3$

★ Common in analytical chemistry

6. NEUTRALIZATION Reaction

➡ Reaction between an acid and a base to form salt and water.

General form: $\text{Acid} + \text{Base} \rightarrow \text{Salt} + \text{Water}$

Example: $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$

7. REDOX (Oxidation-Reduction) Reaction

➡ Reactions involving transfer of electrons.

Oxidation:
LOSS of electrons

→

Reduction:
GAIN of electrons

Example: $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$

★ Oxidation and reduction always occur together

8. EXOTHERMIC Reaction

★ Temperature of surroundings INCREASES

9. ENDOTHERMIC Reaction

➡ Reactions that absorb heat energy.

Example: Photosynthesis

★ Temperature of surroundings DECREASES

Core reaction types with everyday examples (original poster)

➤ RAPID REVISION

- Five core types: combination, decomposition, single & double displacement, redox/combustion.
- Neutralisation = acid + base → salt + water; precipitation gives insoluble solid.
- Exothermic $\Delta H < 0$ (feels hot); endothermic $\Delta H > 0$ (feels cold).
- Corrosion — rust (Fe), black Ag_2S on silver, green patina on copper; rancidity prevented by antioxidants/ N_2 flushing.

UNIT 07

Acids, Bases & Salts

Source: Acids, Bases and Salts lecture

7.1 Acids

Definitions

- Arrhenius: compounds which when dissolved in water increase the concentration of H^+ (hydronium, H_3O^+) ions.
- Brønsted-Lowry: any hydrogen-containing substance which can donate a proton to another substance.
- Lewis: substances capable of accepting a pair of electrons — e.g. $AlCl_3$, BF_3 .

Examples

- Strong acids: H_2SO_4 (sulphuric), HCl (hydrochloric), HNO_3 (nitric), $HClO_3$ (chloric), HBr (hydrobromic).
- Weak acids: HF (hydrofluoric), H_2SO_3 (sulphurous), H_2CO_3 (carbonic), $HCOOH$ (formic/methanoic), CH_3COOH (acetic/ethanoic) etc. — all organic acids are weak.

Physical properties

- Sour in taste; sticky in touch; good conductors of electricity.
- Strong acids are corrosive to skin and cause painful burns.

Action with indicators

Indicator	Acids	Bases
Blue litmus	Turns red	No change (red litmus → blue for bases)
Methyl orange	Turns pink	Turns yellow
Phenolphthalein	Remains colourless	Turns pink
Turmeric	No change	Turns deep red
Red cabbage	Turns pink	Turns dark blue
Hibiscus	No change	Turns blue
Olfactory (onion, vanilla, clove oil)	No change in smell	Smell changes in basic medium

Classification of acids

- By source: organic acids (from natural sources) and inorganic acids (H_2SO_4 , HCl , HNO_3 , HBr ...). Hydracids — acids without oxygen.
- By concentration: dilute acid (more water) and concentrated acid (less/no water).
- By H^+ released: strong acids (highly increase H^+ in water — HCl , H_2SO_4 , HNO_3) and weak acids (small amount of H^+ — all organic acids, H_2CO_3 ...).
- By basicity (number of hydronium ions produced by one molecule in water): monobasic — 1 H^+ (HCl , HBr , HF); dibasic — 2 H^+ (H_2SO_4 , H_2SO_3 , H_2CO_3); tribasic — 3 H^+ (H_3PO_4 , citric acid).

Chemical properties of acids

- With metals: Acid + Metal → Salt + $H_2 \uparrow$ · $2HCl + 2Na \rightarrow 2NaCl + H_2 \uparrow$; $H_2SO_4 + Mg \rightarrow MgSO_4 + H_2 \uparrow$; $2HNO_3 + 2K \rightarrow 2KNO_3 + H_2 \uparrow$; $HCl + Au \rightarrow$ no reaction.
- With metal oxides: Acid + Metal oxide → Salt + H_2O · $3H_2SO_4 + Al_2O_3 \rightarrow Al_2(SO_4)_3 + 3H_2O$; $2HCl + K_2O \rightarrow 2KCl + H_2O$; $H_2CO_3 + MgO \rightarrow MgCO_3 + H_2O$; $2HCl + Ag_2O \rightarrow 2AgCl + H_2O$.
- With bases (neutralisation): Acid + Base → Salt + Water · $HCl + KOH \rightarrow KCl + H_2O$; $H_2SO_4 + Mg(OH)_2 \rightarrow MgSO_4 + 2H_2O$; $2HNO_3 + Ca(OH)_2 \rightarrow Ca(NO_3)_2 + 2H_2O$.
- With metal carbonates/bicarbonates: Carbonate + Acid → Salt + H_2O + $CO_2 \uparrow$ · $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$; $Ca(HCO_3)_2 + 2HCl \rightarrow CaCl_2 + 2H_2O + 2CO_2$.
- Use: Baking soda + tartaric acid → Baking powder.

Acid rain

- When rainwater mixes with pollutant gases (non-metal oxides) it forms dilute acid — acid rain. Non-metal oxide + $\text{H}_2\text{O} \rightarrow \text{acid}$ ($\text{SO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3/\text{H}_2\text{SO}_4$; $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3$).
- Effects: skin problems in humans/animals; harms plants; damages monuments; soil degradation; harms aquatic life.

7.2 Bases

Definitions

- Arrhenius: compounds which when dissolved in water increase the concentration of OH^- (hydroxide) ions.
- Brønsted-Lowry: any substance that can accept a proton from another substance.

Examples & properties

- NaOH , KOH , NH_4OH , $\text{Ca}(\text{OH})_2$, $\text{Mg}(\text{OH})_2$ etc.
- Physical properties: bitter in taste, slippery in touch, good conductors of electricity; mild corrosive action on skin.

Classification of bases

- By strength: strong bases (highly increase OH^- — NaOH , KOH , $\text{Ca}(\text{OH})_2$, $\text{Mg}(\text{OH})_2$) and weak bases (do not produce a large number of OH^- ions — NH_4OH , CH_3NH_2).
- By acidity (number of hydroxide ions produced by one molecule in water): monoacidic — 1 OH^- (NaOH , LiOH , KOH); diacidic — 2 OH^- ($\text{Mg}(\text{OH})_2$, $\text{Ca}(\text{OH})_2$); triacidic — 3 OH^- ($\text{Al}(\text{OH})_3$, $\text{Fe}(\text{OH})_3$).

Chemical properties of bases

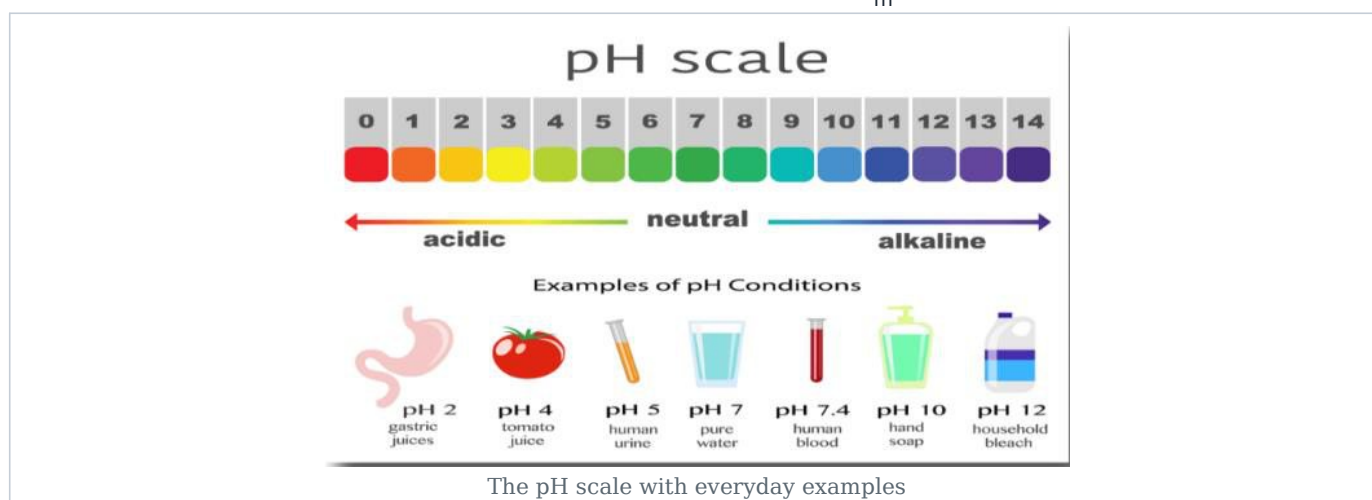
- With acids (neutralisation): $\text{Base} + \text{Acid} \rightarrow \text{Salt} + \text{Water}$ · $2\text{LiOH} + \text{H}_2\text{SO}_4 \rightarrow \text{Li}_2\text{SO}_4 + 2\text{H}_2\text{O}$; $\text{Al}(\text{OH})_3 + 3\text{HCl} \rightarrow \text{AlCl}_3 + 3\text{H}_2\text{O}$.
- With non-metallic oxides: $\text{Base} + \text{Non-metal oxide} \rightarrow \text{Salt} + \text{Water}$ · $2\text{NaOH} + \text{CO}_2 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$; $\text{Ca}(\text{OH})_2 + \text{SO}_2 \rightarrow \text{CaSO}_3 + \text{H}_2\text{O}$.

7.3 pH — Strength of Acidic or Basic Solutions

- pH stands for Potenz (potential) of Hydrogen — "potenz" is German for potential/capability.
- pH of a solution = negative logarithm (base 10) of the hydrogen-ion concentration (mol/L):

$$\text{pH} = -\log_{10}[\text{H}^+] \quad (1 \text{ mole} = 6.022 \times 10^{23} \text{ particles; SI unit of concentration mol/L})$$

- Logarithm recap: $2^3 = 8 \Rightarrow \log_2 8 = 3$; $\log_{10} 1000 = \log_{10} 10^3 = 3$; $\log_m m = 1$; $\log_{10} m^n = n \cdot \log m$.



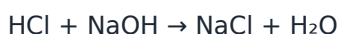
Importance of pH in everyday life

- Our body works within a narrow pH range of 7.0 to 7.8; plants and animals also survive in narrow ranges.
- Agriculture: every crop needs a particular pH — rice grows better in slightly acidic soil, sugarcane in neutral, citrus fruits in alkaline soil.

- Medicines: certain diseases are diagnosed on the basis of pH of blood and urine.
- Digestion: HCl in the stomach helps digestion; excess acid lowers pH → pain/irritation. Antacids like milk of magnesia $\text{Mg}(\text{OH})_2$ adjust the pH.
- Tooth decay: mouth bacteria degrade sweets; when pH falls to 5.5, tooth enamel (calcium phosphate — hardest substance in the body) corrodes. Saliva is slightly alkaline; toothpaste neutralises excess acid.
- Bee sting leaves acid in the body — applying baking soda (a base) gives relief.

7.4 Salts

- Compounds formed by the displacement of H^+ ions of an acid by a metallic (or ammonium) ion — e.g. NaCl, MgCl_2 , MgSO_4 , Na_2CO_3 .



Types of salts

Type	Definition	Examples
Neutral salt	Complete neutralisation of an acid by a base	NaCl, KCl, K_2SO_4
Acidic salt	Still contains replaceable H^+ ; increases H^+ in water ($\text{pH} < 7$); behave like acids	NaHSO_4 , NaH_2PO_4 (sodium dihydrogen phosphate), NaHSO_3 (sodium bisulphite)
Basic salt	Contains OH^- ; increases OH^- in water ($\text{pH} > 7$); behave like bases	$\text{Mg}(\text{OH})\text{Cl}$ (from $\text{Mg}(\text{OH})_2 + \text{HCl}$), $\text{Pb}(\text{OH})\text{Cl}$, $\text{Cu}(\text{OH})\text{Cl}$, $\text{Cu}(\text{OH})\text{NO}_3$
Double salt	Salt crystals containing two salts	Potash alum $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$; Mohr's salt $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$; dolomite $\text{CaCO}_3 \cdot \text{MgCO}_3$
Mixed salt	Contains more than one cation or anion	NaKCO_3 (sodium potassium carbonate); CaOCl_2 (bleaching powder — calcium oxychloride)

Properties of salts

- Salts are ionic compounds; they conduct electricity in aqueous or molten state; most salts are soluble in water.

Water of crystallisation & related terms

- Water of crystallisation — the number of water molecules present with one formula unit of a salt crystal (e.g. Glauber's salt $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$, Epsom salt $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, blue vitriol $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, gypsum $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$).
- Efflorescence — property of losing water of crystallisation fully/partially when exposed to air.
- Hygroscopy — property of absorbing moisture from surroundings without dissolving (silica gel, quicklime CaO , P_2O_5).
- Deliquescence — absorbing moisture, becoming moist and finally dissolving (caustic soda NaOH, caustic potash KOH, MgCl_2).
- Drying (desiccating) agents — absorb moisture from others: conc. H_2SO_4 , silica gel. Dehydrating agents — remove water molecules from other substances (e.g. blue vitriol on heating loses water).

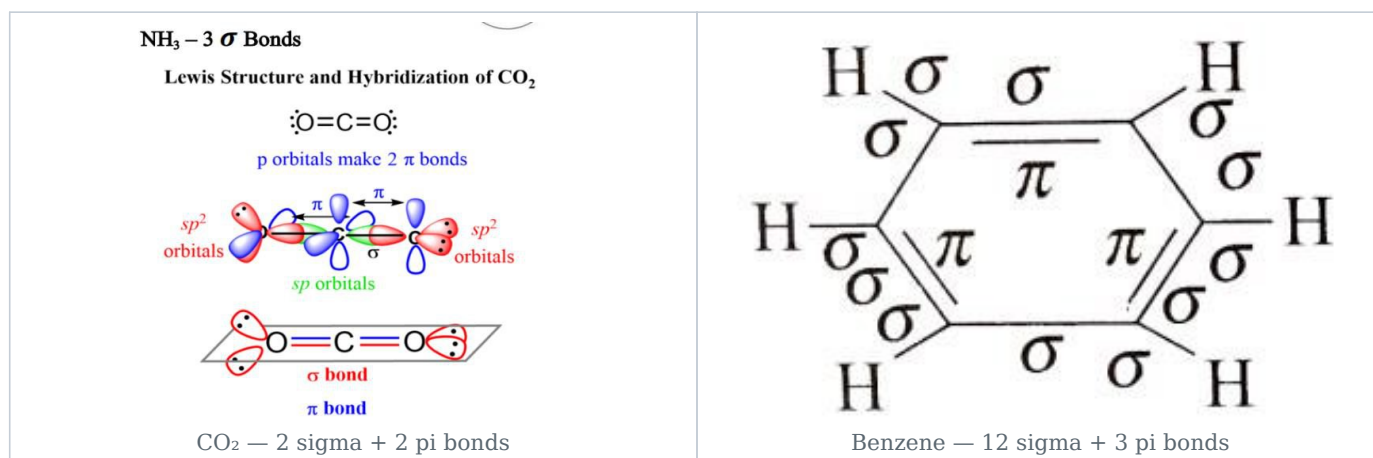
Uses of common salts

Salt	Formula	Use
Sodium chloride (common salt)	NaCl	Edible salt, food preservative, decing agent
Calcium carbonate	CaCO_3	Antacid, toothpaste abrasive, building material
Sodium bicarbonate (baking soda)	NaHCO_3	Baking soda, antacid, fire extinguisher
Potassium nitrate (saltpetre)	KNO_3	Fertiliser, food preservative, gunpowder ingredient
Calcium chloride	CaCl_2	Road de-icing, food additive, desiccant
Sodium carbonate (washing soda)	Na_2CO_3	Water softener, glass manufacture, pH regulator
Ammonium chloride	NH_4Cl	Electrolyte in dry cells, fertiliser, cough medicine
Magnesium sulphate (Epsom salt)	MgSO_4	Epsom salt baths, fertiliser, medicine

Salt	Formula	Use
Potassium chloride	KCl	Salt substitute, fertiliser, electrolyte in TV screens
Sodium hydroxide (caustic soda)	NaOH	Soap making, paper manufacturing

7.5 From Salts to Carbon — Bridge to Organic Chemistry

- Vital Force theory claimed all organic compounds can only come from natural sources and cannot be made by humans — proved wrong by Friedrich Wöhler in 1828 by converting ammonium cyanate (NH_4OCN) into urea (NH_2CONH_2).
- Basics about carbon: atomic number = 6, mass number = 12; 6 e^- , 6 protons, 6 neutrons; valence electrons = 4 $\rightarrow$ valency 4.
- Covalent compounds — formed by sharing of electrons; covalent bond — bond due to sharing, can be single, double or triple.
- Bond counts: CO_2 has 2 σ + 2 π bonds; N_2 has 1 σ + 2 π ; benzene has 12 σ + 3 π bonds.



✂ RAPID REVISION

- Acids give H^+ (Arrhenius), donate proton (Brønsted), accept e^- pair (Lewis — BF_3 , AlCl_3).
- Indicators: acid $\rightarrow$ blue litmus red, methyl orange pink; base $\rightarrow$ phenolphthalein pink, turmeric red.
- Basicity = H^+ per molecule (HCl mono, H_2SO_4 di, H_3PO_4 tri); acidity = OH^- per molecule.
- $\text{pH} = -\log[\text{H}^+]$; body 7.0–7.8; tooth decay below pH 5.5; antacid = $\text{Mg}(\text{OH})_2$.
- Acidic salt still has H^+ (NaHSO_4); basic salt has OH^- ($\text{Cu}(\text{OH})\text{NO}_3$); double salt — alum, Mohr's salt.
- Efflorescence = lose water of crystallisation; deliquescence = absorb + dissolve; hygroscopic = absorb only.
- Wöhler (1828): ammonium cyanate $\rightarrow$ urea killed the Vital Force theory.

UNIT 08

Metals, Non-metals & Metallurgy

Sources: Metal & Nonmetal lecture + element classification part of Basic Chemistry Part-II

8.1 Metals — Physical Properties

Property	Meaning
Malleability	Can be converted into thin sheets on beating
Ductility	Can be drawn into wires
Sonorosity	Produce a ringing (singing) sound when struck
Lustre	Shining surface
Conductivity	Good conductors of heat & electricity
Density	High density
Hardness / brittleness	Very hard and non-brittle
Melting & boiling points	High

Exceptions

- Brittle metals — beryllium, arsenic, zinc — fracture rather than deform under stress.
- Alkali metals (Na, K) are soft (cut with a knife) and have low density.
- Lead, bismuth and tungsten conduct less than most metals.
- Mercury is liquid at room temperature; gallium (Ga) and caesium (Cs) have very low melting points.
- Most metals are silver-grey — copper is reddish-brown, gold is yellow.
- Magnetic metals: iron, cobalt, nickel.

8.2 Metals — Chemical Properties**Reaction with oxygen**

- Metal + Oxygen → Metal oxide (basic). $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$; $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$. The product depends on reactivity — iron gives FeO/Fe₂O₃/Fe₃O₄ on heating with O₂.

Reaction with water

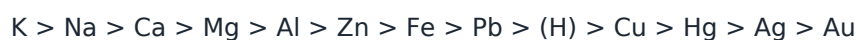
- Metal + H₂O → Metal hydroxide/oxide + H₂↑. Reactivity decides whether reaction is with cold water, hot water or steam.
- $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2\uparrow$ (vigorous); $\text{Mg} + 2\text{H}_2\text{O} (\text{hot}) \rightarrow \text{Mg}(\text{OH})_2 + \text{H}_2$; $\text{Fe} + \text{H}_2\text{O} (\text{steam}) \rightarrow \text{Fe}_3\text{O}_4 + \text{H}_2$.

Reaction with metal salts — displacement

- A metal displaces another metal from its salt solution only if it is more reactive: $\text{P} + \text{AB} \rightarrow \text{PB} + \text{A}$.
- $\text{Na} + \text{CuCl}_2 \rightarrow \text{NaCl} + \text{Cu}$ (reaction occurs); $\text{Cu} + \text{NaCl} \rightarrow$ no reaction; $\text{Cu} + \text{Ca}(\text{NO}_3)_2 \rightarrow$ no reaction.
- Double displacement: $\text{AB} + \text{PQ} \rightarrow \text{PB} + \text{AQ}$ (e.g. $\text{Na}_2\text{SO}_4 + \text{BaCl}_2 \rightarrow \text{BaSO}_4 + 2\text{NaCl}$).
- Reaction with non-metals: metals react with non-metals to form ionic compounds — $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$; $2\text{K} + \text{I}_2 \rightarrow 2\text{KI}$; $\text{Mg} + \text{H}_2 \rightarrow \text{MgH}_2$.

Reactivity series

Metals arranged in decreasing order of reactivity:



METALS REACTIVITY SERIES

Metal Name	Symbol	Reactivity
Cesium	Cs	HIGHLY REACTIVE Cs, Fr, Rb & K Violent Reaction with Cold Water
Francium	Fr	
Rubidium	Rb	
Potassium	K	
Sodium	Na	
Lithium	Li	Na, Li, Ba, Ra, Sr & Ca Moderate reaction with Cold water, Strong reaction with hot water or steam
Barium	Ba	
Radium	Ra	
Strontium	Sr	
Calcium	Ca	
Magnesium	Mg	MODERATELY REACTIVE Mg Very Slow Reaction with Cold Water, Strong reaction with hot water or steam
Aluminum	Al	
Titanium	Ti	
Manganese	Mn	
Zinc	Zn	
Chromium	Cr	
Iron	Fe	
Cadmium	Cd	
Cobalt	Co	
Nickel	Ni	
Tin	Sn	
Lead	Pb	
Hydrogen	H	Nonmetal (For Comparison)
Antimony	Sb	LEAST REACTIVE Sb, Bi, Cu, W, Hg, Ag, Au & Pt Do not react with cold water, hot water, steam or dilute acids
Bismuth	Bi	
Copper	Cu	
Tungsten	W	
Mercury	Hg	
Silver	Ag	
Gold	Au	
Platinum	Pt	

Metals reactivity series — original poster

8.3 Types of Metals

1. Alkali metals (Group 1, s-block): Li, Na, K, Rb, Cs, Fr

- Highly reactive — most reactive metal group; always form +1 cations ($\text{Na} \rightarrow \text{Na}^+ + \text{e}^-$).
- Form water-soluble bases: $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2 \uparrow$.
- Soft and light (low density, cut by knife); low m.p./b.p.; highly electropositive (one valence electron).
- Applications: lithium-ion batteries, sodium compounds in pharmaceuticals, potassium compounds in organic synthesis.

2. Alkaline earth metals (Group 2, s-block)

- Be, Mg, Ca, Sr, Ba, Ra (except Be, called alkaline earth metals — oxides/hydroxides are alkaline, found in earth's crust). Form +2 cations: $\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^-$; $\text{Ca} \rightarrow \text{Ca}^{2+} + 2\text{e}^-$. They too form water-soluble bases, but solubility is less than group 1: $\text{Mg} + 2\text{H}_2\text{O} \rightarrow \text{Mg(OH)}_2 + \text{H}_2$.
- Mnemonic from notes: "Be, Mg, Ca, Sr, Ba, Ra" — Be, Mg, Ca, Sr, Ba, Ra.

3. Transition metals (d-block, groups 3-12)

- Show variable valency; form colourful compounds (partially filled d orbitals).
- High m.p./b.p. (higher than alkali/alkaline earth); hard and strong; malleable and ductile.
- Widely used as catalysts — substances that change the rate of reaction without participating in it.
- Fe, Co, Ni are magnetic — used in magnetic storage and electromagnets.

4. Inner transition metals (f-block: lanthanides Ce-Lu, actinides Th-Lr)

- Placed below the main body of the periodic table; characterised by filling of f orbitals.
- Heavy metals with high m.p./b.p. and high atomic numbers; variable valency.
- Many are radioactive — useful in nuclear energy and medical imaging; U and Pu are among the densest natural elements.
- Most are rare in the earth's crust, hence valuable.

8.4 Alloys

- Alloy — homogeneous mixture of two or more metals (and sometimes non-metals). On their own most metals are less useful: pure iron is soft and rusts easily, but mixed with other metals it gains desirable properties. Some steels include non-metals too. Metals form compounds with non-metals by giving outer-shell electrons.

Alloy	Composition	Uses
Brass	80% Cu + 20% Zn	Utensils, cartridge cases, decorative items
Bronze	70-95% Cu + 5-30% Sn	Coins, machine & turbine blades, statues
Stainless steel	Fe + Cr (different proportions) + Ni (trace)	Kitchen utensils, surgical instruments
Mild steel	0.15-0.25% C + rest Fe	Nails, chains, ship plates, girders
Tool steel	Fe + larger % of W/Mo + trace V/Co	Machine tools, drills, saws
Nichrome	60% Ni + 25% Fe + 15% Cr	Heating elements
Alnico	Al + Ni + Co	Strong permanent magnets

- More alloys: german silver (Cu+Zn+Ni — silverware), gunmetal (Cu+Sn+Zn), type metal (Pb+Sb), duralumin (Al+Cu+Mg+Mn — aircraft), 22-carat gold (Au+Ag/Cu), magnalium (Al+Mg), permalloy (Ni+Fe), invar (Fe+Ni), stainless & tool steels, tungstens, solder (Pb+Sn).
- Remember: iodine is a non-metal but shiny; graphite (carbon) conducts and has a high m.p.; mercury is a liquid metal; bromine is a liquid non-metal (red, volatile).

8.5 Non-metals (19 elements)

- Elements having a tendency to gain or share electrons. 17 non-metals in the periodic table, of which 12 are gases; all others are solids except bromine — a red volatile liquid.

Properties

- Brittle — non-malleable, non-ductile; non-lustrous, non-sonorous.
- Bad conductors of heat & electricity (exception: graphite — a good conductor).
- Low density; low melting & boiling points; generally solid or gaseous (bromine alone is liquid).
- Electronegative — tendency to attract the shared pair of electrons in covalent compounds; non-metals are more electronegative than metals.
- With oxygen: non-metal + O₂ → acidic oxide — C + O₂ → CO₂ (CO₂ + H₂O → H₂CO₃); S + O₂ → SO₂ (SO₂ + H₂O → H₂SO₃).
- Chemical behaviour: accept or share electrons → form covalent compounds (e.g. N≡N).
- Uses: hydrogen — fuel, ammonia, hydrogenation; oxygen — respiration, welding; nitrogen — fertilisers, food preservation; chlorine — water purification, PVC, disinfectant; sulphur — sulphuric acid, vulcanisation; carbon — fuels, pencil leads, diamonds in jewellery.

8.6 Hydrogen & its Compounds

- Most abundant element in the universe; on Earth it occurs in combined form — water, organic matter, minerals.
- Shows similarity with alkali metals (H → H⁺ + e⁻) and halogens (H + e⁻ → H⁻); exists as diatomic H₂.
- Isotopes: protium (¹H), deuterium (²H), tritium (³H) — same Z = 1, different A.

	Protium	Deuterium	Tritium
no of e ⁻	1	1	1
no. of proton	1	1	1
no of neutron	0	1	2

Isotopes of hydrogen — p, d, t

Electrons are pulled towards chlorine.

Electronegativity: H(2.2) < Cl(3.6)

Polarity of H-Cl bond (H^{δ+}-Cl^{δ-})

Heavy water (D₂O)

- Contains deuterium instead of protium; discovered in 1932 by H.C. Urey and E.W. Washburn.

- Uses: nuclear moderator & coolant (slows neutrons in reactors); tracer in experiments (reaction/metabolism studies); production of deuterium compounds.

Soft & hard water

- Soft water — lathers easily with soap; hard water — does not lather (salts of Ca^{2+} and Mg^{2+}).
- Causes of hardness: bicarbonates, chlorides and sulphates of calcium and magnesium.
- Effects: soap wastage; scale formation in boilers; problems in dyeing/laundry.

Basis	Temporary hard water	Permanent hard water
Cause	Bicarbonates of Ca & Mg	Chlorides and sulphates of Ca & Mg
Removal	Boiling, Clark's method (adding Ca(OH)_2)	Cannot be removed by simple boiling; washing-soda treatment

Hydrogen peroxide (H_2O_2)

- Discovered 1818 by Thenard; pale blue liquid; m.p. -0.43°C ; b.p. 150.2°C ; density 1.47 g/cm^3 .
- Strong oxidising agent (can also act as a reducing agent); unstable — decomposes to water and oxygen.
- Uses: germicide/antiseptic; bleaching textiles, paper, hair, leather, oils, fats; manufacture of sodium perborate & sodium carbonate; milk/food preservation; oxygen source in submarines.

8.7 Sulphur (Non-metal)

- 0.03–0.1% of earth's crust. Main sources: gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$), Epsom salt ($\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$), barite (BaSO_4), galena (PbS), zinc blende (ZnS), copper pyrites (CuFeS_2); also H_2S from volcanoes; present in proteins of egg, onion, garlic, mustard, pulses.
- Sulphur dioxide (SO_2): colourless, pungent; liquefies at 263 K; highly water-soluble. Uses — petroleum refining, bleaching wool/silk, disinfectant & preservative, refrigerant.
- Sulphur hexafluoride (SF_6): colourless, odourless, non-toxic, octahedral; excellent electrical insulator.
- Sulphuric acid (H_2SO_4) — "King of chemicals": colourless, oily, hygroscopic; high b.p.; very strong acid. Uses — petroleum refining, fertiliser production, explosives/detergents/plastics, storage batteries, medicines (nitroglycerine).

8.8 Inert (Noble) Gases

- He, Ne, Ar, Kr, Xe, Rn — also called rare gases; very low reactivity due to completely filled outer shells; colourless, odourless, non-flammable.

Gas	Uses
Helium	Filling airships/observation balloons; O_2 mixture for deep-sea divers (avoids nitrogen narcosis); asthma treatment; inflating aeroplane tyres; inert atmosphere in melting/welding
Neon	Discharge/advertising tubes with characteristic colours; beacon lights for air navigation (fog-penetrating)
Argon	Filling electric bulbs & fluorescent lamps (prevents filament oxidation); with N_2 in gas-filled lamps; welding shield
Krypton	High-intensity airport runway lamps, photographic flashes
Xenon	Powerful lamps, camera flashes, medical anaesthesia
Radon	Cancer treatment (radiotherapy) — radioactive, used carefully

8.9 Metallurgy

Metallurgy — the branch of science dealing with the extraction of metals from ores and their purification.

Basic terms

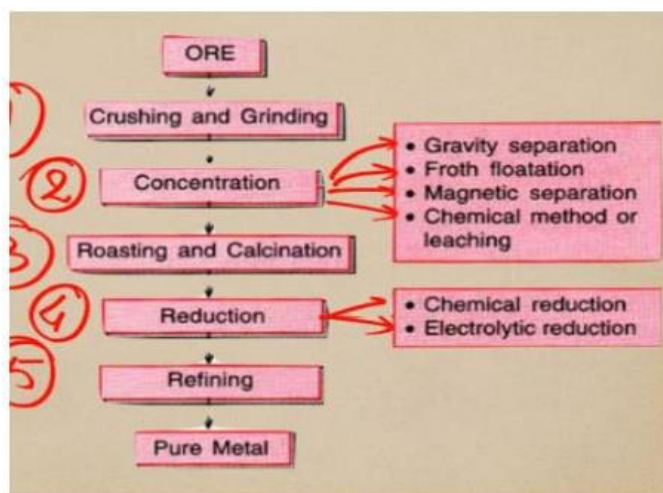
Term	Meaning
Minerals	Naturally occurring compounds of metals mixed with soil, sand, limestone, rocks etc.
Ore	Mineral from which a metal is extracted commercially at low cost with minimum effort

Term	Meaning
Gangue (matrix)	Earthly impurities — mud, silica, sand — present in the ore
Flux	Substance added in smelting to promote fluidity and remove impurities as slag
Slag	Product formed when flux reacts with gangue — e.g. CaO (flux) + SiO_2 (gangue) → CaSiO_3 (slag)

Common ores of metals

Metal	Ore	Formula	Metal	Ore	Formula
Aluminium	Bauxite	$\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$	Zinc	Zinc blende / Calamine	ZnS / ZnCO_3
Iron	Haematite	Fe_2O_3	Mercury	Cinnabar	HgS
Copper	Copper pyrites	CuFeS_2	Magnesium	Magnesite	MgCO_3
Lead	Galena	PbS	Manganese	Pyrolusite	MnO_2
Tin	Cassiterite (Tinstone)	SnO_2	Sodium	Common salt	NaCl
Calcium	Limestone	CaCO_3	Iron (other)	Magnetite / Siderite	Fe_3O_4 / FeCO_3

Steps involved in metallurgy



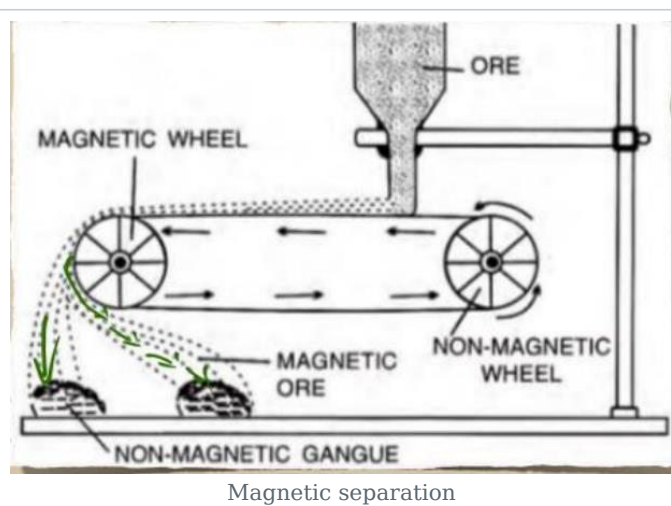
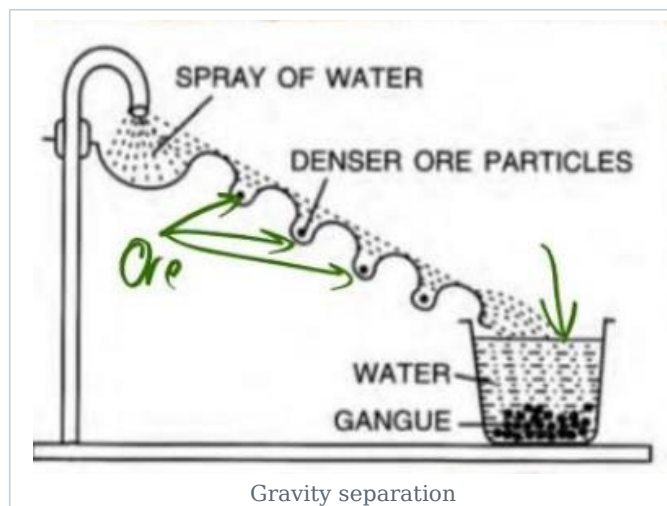
Flow of metallurgy — from ore to pure metal

1. Crushing & grinding

- Ores crushed to fine powders using big jaw crushers & ball mills. Pulverisation — crushing & grinding of ore to powdered form.

2. Dressing (concentration) of ores

- Removal of impurities (sand, soil, rock particles) from powdered ore. Methods:
 - Hydrolytic (gravity separation): needs density difference between ore and impurities — water washes lighter gangue away, denser ore particles settle.
 - Magnetic separation: for ores attracted by a magnet — magnetic wheel carries ore away from non-magnetic gangue.
 - Leaching (chemical): a suitable chemical dissolves the ore; insoluble impurities are filtered. E.g. bauxite + hot conc. NaOH → soluble sodium aluminate $\text{Na}[\text{Al}(\text{OH})_4]$; then CO_2 passed through the solution precipitates aluminium oxide (Al_2O_3).



3. Roasting & calcination

Roasting	Calcination
I. The ore is heated in the excess of air. II. Generally, sulphide ores are roasted, so SO_2 is given off $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$ (at 800-900 degree) III. Volatile impurities are removed as oxides (SO_2, P_2O_5, As_2O_3) and the ore becomes porous and more reactive.	I. The ore is heated in the absence of air II. Carbonate and hydrated ores are calcined and CO_2 or water vapours are given off $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$ III. Moisture and organic impurities are removed and the ore becomes porous and more reactive.

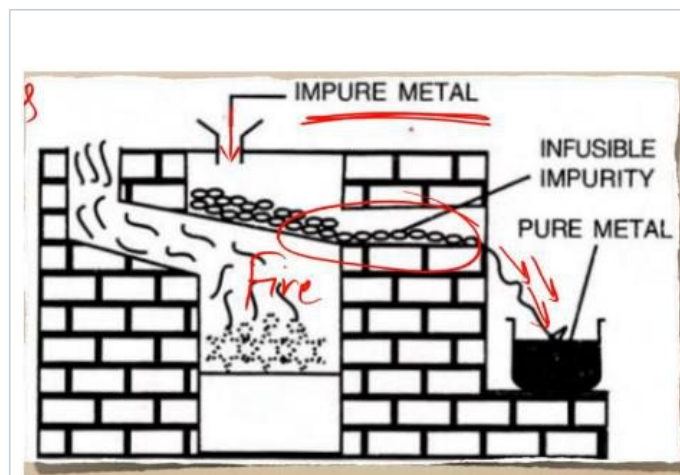
Roasting (with O_2 , sulphide ores) vs calcination (without air, carbonate ores)

- Roasting: ore heated in excess air — $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$; volatile impurities removed. For sulphide ores. Example — $\text{HgS} + \text{O}_2 \xrightarrow{\Delta} \text{HgO} + \text{SO}_2$, then $\text{HgO} \xrightarrow{\text{heat}} \text{Hg} + \text{O}_2$.
- Calcination: ore heated in absence/limited air — $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$; $2\text{ZnCO}_3 \rightarrow 2\text{ZnO} + 2\text{CO}_2$; hydrated oxides lose water; ore becomes porous. For carbonate ores.

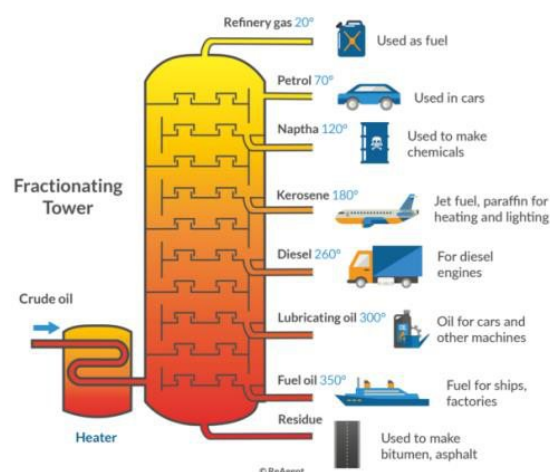
4. Reduction of metal oxides

- Highly reactive metals (K, Na, Ca, Mg, Al): their compounds cannot be reduced by carbon — extracted by electrolytic reduction of molten chloride/oxide ($\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$ at cathode; $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$ at anode).
- Mildly reactive (Zn, Fe, Pb): oxide + reducing agent (C, CO, H_2 or a more reactive metal) — $2\text{ZnO} + \text{C} \rightarrow 2\text{Zn} + \text{CO}_2$; $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$ (thermite uses Al).
- Least reactive (Hg, Ag, Au, Pt): simply heating the oxide gives the metal — $2\text{HgO} \rightarrow 2\text{Hg} + \text{O}_2$; $2\text{Ag}_2\text{O} \rightarrow 4\text{Ag} + \text{O}_2$.

5. Refining of metals

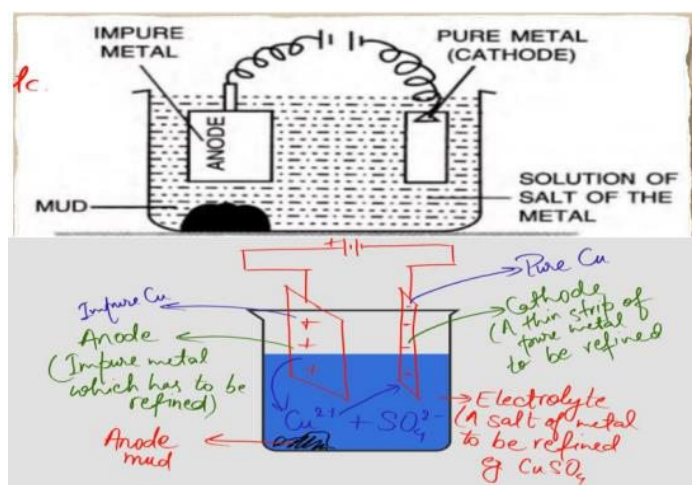


Liquation (Pb, Sn)



Fractional distillation of petroleum

- Distillation: for volatile metals (Hg, Zn) — impure metal heated above its b.p., vapour separated from non-volatile impurities and condensed as pure liquid metal.
- Liquation: for low-melting metals — lead and tin; metal flows down a sloping hearth, leaving impurities behind.
- Electrolytic refining: for Cu, Al, Zn etc. — impure metal = anode, pure metal = cathode, metal-salt solution = electrolyte; pure metal deposits on the cathode, mud/impurities settle as anode mud.



Electrolytic refining of copper

➤ RAPID REVISION

- Metal exceptions: Hg liquid; Ga/Cs low m.p.; Na/K soft; Cu red-brown, Au yellow; Fe-Co-Ni magnetic.
- Reactivity: $K > Na > Ca > Mg > Al > Zn > Fe > Pb > H > Cu > Hg > Ag > Au$.
- Group 1 $\rightarrow +1$, strongest bases in water; group 2 $\rightarrow +2$; d-block $\rightarrow$ variable valency, coloured compounds, catalysts; f-block $\rightarrow$ radioactive (actinides).
- D_2O — Urey 1932 — moderator + tracer. H_2O_2 — Thenard 1818 — oxidant, antiseptic, bleaching.
- H_2SO_4 = King of chemicals. Noble gas uses: He — divers/balloons; Ne — signs; Ar — bulbs; Kr — runway lamps; Xe — flashes; Rn — radiotherapy.
- Metallurgy: crushing $\rightarrow$ dressing (gravity/magnetic/leaching) $\rightarrow$ roasting or calcination $\rightarrow$ reduction (electrolysis / C-CO / heat) $\rightarrow$ refining (distillation-liquation-electrolytic).

UNIT 09

s, p, d & f Block Elements

Source: Important S,P,D,F Elements lecture

9.1 s-Block Elements (Groups 1 & 2)

- Group 1 — Alkali metals: Li, Na, K, Rb, Cs, Fr — named so because they form strongly alkaline hydroxides on reacting with water.
- Group 2 — Alkaline earth metals: Be, Mg, Ca, Sr, Ba, Ra (except Be) — oxides/hydroxides are alkaline and commonly found in the earth's crust.

s-block on the periodic table	<table border="1"> <thead> <tr> <th>Alkali Metals</th><th>Alkali Earth Metals</th></tr> </thead> <tbody> <tr> <td>Alkali metals and their salts give characteristic colour to Bunsen flame. The flame energy causes an excitation of the outermost electron which on dropping back to ground state emits absorbed energy as a visible light.</td><td>Due to small size of Be & Mg, outermost n electrons are tightly bounded. So not excited to higher level, hence they do not give flame test.</td></tr> <tr> <td>Li - Crimson red; Na - Golden yellow K - Violet; Rb - Red violet; Cs - Blue</td><td>Ca - Brick red; Sr - Crimson red; Ba - Apple green</td></tr> </tbody> </table> Flame test colours of s-block metals	Alkali Metals	Alkali Earth Metals	Alkali metals and their salts give characteristic colour to Bunsen flame. The flame energy causes an excitation of the outermost electron which on dropping back to ground state emits absorbed energy as a visible light.	Due to small size of Be & Mg, outermost n electrons are tightly bounded. So not excited to higher level, hence they do not give flame test.	Li - Crimson red; Na - Golden yellow K - Violet; Rb - Red violet; Cs - Blue	Ca - Brick red; Sr - Crimson red; Ba - Apple green
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General configuration & trends

Aspect	Alkali metals (Gr 1)	Alkaline earth metals (Gr 2)
Configuration	$[NG] ns^1$	$[NG] ns^2$
Ion formed	M^+ (monovalent)	M^{2+} (divalent)
Reactivity	Very high — kept under oil (Li in paraffin wax); reactivity $\propto$ ionisation potential	Less reactive than alkali metals; order: Be < Mg < Ca < Sr < Ba
Melting point	Decreases down: Li > Na > K > Rb > Cs	Irregular: Ca > Sr > Ba > Li-like trend; Be & Mg high
Flame colour	Li — crimson red; Na — golden yellow; K — violet/lilac; Rb — red-violet; Cs — blue	Ca — brick red; Sr — crimson; Ba — apple green

Physical Properties of s-block Elements	
Alkali Metals	Alkali Earth Metals
One electron in outermost shell & General formula- ns^1	Two electrons in outermost shell & General formula- ns^2
Francium is a radioactive element.	Radium is a radioactive element.
All are silvery white	All are silvery white.
Light soft, malleable and ductile metals with metallic lustre.	These metals are harder than alkali metals.
Atomic size	
Alkali Metals	Alkali Earth Metals
Largest in their respective period (except noble gas element)	Smaller than IA group elements, since extra charge on nucleus attracts the electron cloud.
Size increases from Li to Cs due to addition of an extra shell. $Li < Na < K < Rb < Cs$	Size increases gradually from Be to Ba $Be < Mg < Ca < Sr < Ba$
Note- In s-block elements Beryllium (Be) has the smallest size, while Caesium (Cs) has the largest size.	

Alkali metals vs alkaline earth metals — original comparison

Important compounds of s-block

- NaCl (sodium chloride): white crystalline solid, high m.p., highly water-soluble; table salt — seasoning & preservative; production of chemicals, food processing, water softening.
- NaOH (sodium hydroxide): white solid, strongly alkaline solution; pulp & paper, textiles, soaps, detergents, cleaners, biodiesel, pH regulation in water treatment.
- NaHCO₃ (sodium bicarbonate): white crystalline solid, mildly alkaline; baking soda (leavening agent), antacid, cleaning & deodorising.
- Na₂CO₃ (sodium carbonate): white, odourless, alkaline; soda ash / washing soda — glassmaking, detergents, chemicals, water treatment, pool pH regulator.
- Ca(OH)₂ (calcium hydroxide): white powder, low solubility, strong alkaline solution; slaked lime — mortar & plaster, wastewater treatment, agriculture pH, pickling, toothpaste & antacids.
- CaSO₄ (calcium sulphate): gypsum (CaSO₄·2H₂O) — plaster of Paris, plasterboard, cement; anhydrous — soil conditioner, brewing water conditioner.
- Plaster of Paris: white powder that hardens with water (made by heating gypsum to remove water); casts, moulds, sculptures, casting broken limbs, dental impressions.

9.2 p-Block Elements (Groups 13-18)

- General outer configuration $ns^2 np^{1-6}$ — the last electron enters a p-orbital. The block has metals, non-metals and metalloids → wide property variety. Multiple oxidation states (–3 to +5). Fluorine is the most reactive non-metal; Ne/Ar unreactive. Several show allotropy (carbon — diamond, graphite, fullerenes).

p block Elements

p-block on the periodic table

General electronic configurations and oxidation states for each group of the p-block elements—

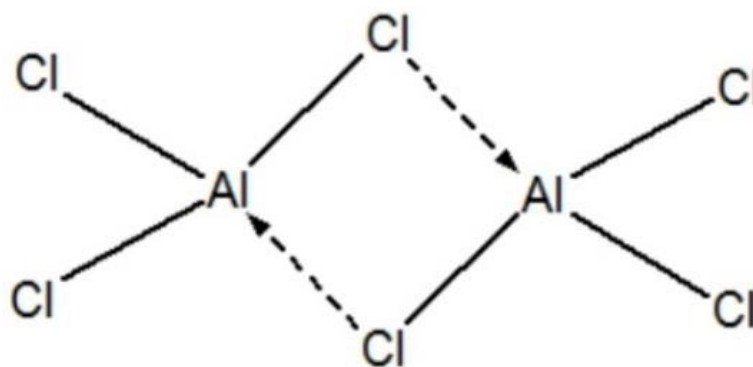
Group	General Electronic Configuration	First Member	Group Oxidation State	Other Oxidation States
13	$ns^2 np^1$	B (Boron)	+3	+1
14	$ns^2 np^2$	C (Carbon)	+4	+2, –4
15	$ns^2 np^3$	N (Nitrogen)	+5	+3, –3
16	$ns^2 np^4$	O (Oxygen)	+6	+4, +2, –2
17	$ns^2 np^5$	F (Fluorine)	+7	+5, +3, +1, –1
18	$ns^2 np^6$ (for He)	He (Helium)	+8 (theoretical)	+6, +4, +2

Configurations & oxidation states

Group 13 — Boron family (B, Al, Ga, In, Tl)

- M.P. decreases B → Ga then increases up to Tl; $IE_1 \ll IE_2 \ll IE_3$; metallic character increases B → Tl (B is a non-metal).
- Boron preparation: from boric acid — $B_2O_3 + 3Mg \rightarrow 2B + 3MgO$; from BCl₃ — $2BCl_3 + 3H_2 \xrightarrow{1270\text{ K}} 2B + 6HCl$ (or with Zn at 900 °C); by electrolysis of fused B₂O₃ + MgO + MgF₂ at 1100 °C; thermal decomposition — $B_2H_6 \xrightarrow{\Delta} 2B + 3H_2$.
- Orthoboric acid (H₃BO₃): from borax — $Na_2B_4O_7 + H_2SO_4 + 5H_2O \rightarrow Na_2SO_4 + 4H_3BO_3$; from colemanite — $Ca_2B_6O_{11} + 2SO_2 + 11H_2O \rightarrow 2Ca(HSO_3)_2 + 6H_3BO_3$. Weak monobasic acid: $B(OH)_3 + H_2O \rightleftharpoons H^+ + [B(OH)_4]^-$; with NaOH gives sodium metaborate — $H_3BO_3 + NaOH \rightleftharpoons NaBO_2 + 2H_2O$.
- Borax (Na₂B₄O₇·10H₂O): $4H_3BO_3 + Na_2CO_3 \rightarrow Na_2B_4O_7 + 6H_2O + CO_2$; aqueous solution is alkaline due to hydrolysis ($NaBO_2 + 2H_2O \rightarrow NaOH + H_3BO_3$).
- Aluminium compounds: alumina Al₂O₃ — $2Al(OH)_3 \xrightarrow{\Delta} Al_2O_3 + 3H_2O$; aluminium chloride AlCl₃ — white hygroscopic solid, sublimes at 183 °C, forms addition compounds (NH₃, PH₃, COCl₂), hydrolyses: $AlCl_3 + 3H_2O \rightarrow Al(OH)_3 + 3HCl$.

$3\text{H}_2\text{O} \rightarrow \text{Al}(\text{OH})_3 + 3\text{HCl}$; exists as dimer Al_2Cl_6 .



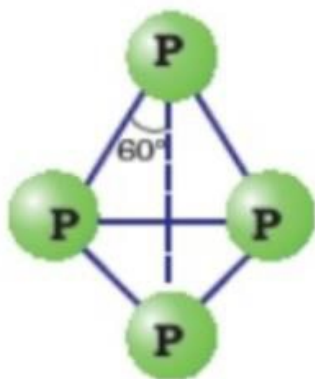
Dimeric structure of Al_2Cl_6

Group 14 — Carbon family (C, Si, Ge, Sn, Pb)

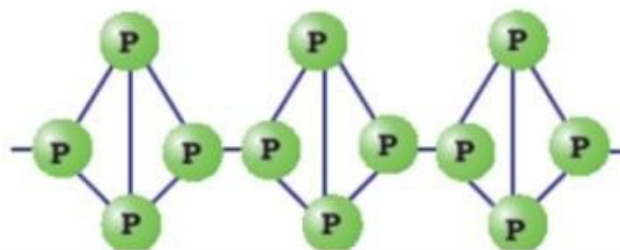
- IE decreases $\text{C} \rightarrow \text{Sn}$ then increases up to Pb; C & Si non-metals, Ge metalloid, Sn & Pb metals.
- Catenation: C and Si combine with their own atoms to form long-chain polymers.
- Carbon monoxide: tests — burns with a blue flame; turns filter paper soaked in platinum/palladium chloride pink or green.
- Carbon dioxide: $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$ (from carbonates); $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$ (combustion). Turns lime water milky — $\text{Ca}(\text{OH})_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$; excess CO_2 dissolves the milky — $\text{CaCO}_3 + \text{H}_2\text{O} + \text{CO}_2 \rightarrow \text{Ca}(\text{HCO}_3)_2$. Dry ice — solid CO_2 , sublimates directly.
- Carbides: salt-like/ionic (C_2^{2-} acetylide or C^{4-} methanide) — CaC_2 , Al_4C_3 , Be_2C ; covalent — SiC (carborundum), boron carbide; interstitial — transition metal lattices with C in interstices — WC, VC.
- Silicon compounds: sodium silicate — $\text{Na}_2\text{CO}_3 + \text{SiO}_2 \rightarrow \text{Na}_2\text{SiO}_3 + \text{CO}_2$ (fused with sand); silicones — Si-O-Si polymers from hydrolysis of alkyl/aryl chlorosilanes; silicates — salts of silicic acid H_4SiO_4 , built of tetrahedral SiO_4^{4-} units (sp^3 hybridised).

Group 15 — Nitrogen family (N, P, As, Sb, Bi)

- Atomic radius increases down the group (small increase As $\rightarrow$ Bi); oxidation states -3 to $+5$; stability of $+3$ increases down; IE decreases N $\rightarrow$ Bi.
- Oxy-acids of nitrogen: $\text{H}_2\text{N}_2\text{O}_2$ (hyponitrous), HNO_2 (nitrous), HNO_3 (nitric), HNO_4 (pernitric).
- Ammonia (NH_3): $\text{NH}_4\text{Cl} + \text{NaOH} \rightarrow \text{NH}_3 + \text{NaCl} + \text{H}_2\text{O}$; $\text{Mg}_3\text{N}_2 + 6\text{H}_2\text{O} \rightarrow 3\text{Mg}(\text{OH})_2 + 2\text{NH}_3$; Haber process — $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$. Basic (red litmus $\rightarrow$ blue); forms complexes — $\text{Cu}^{2+} + 4\text{NH}_3 \rightarrow [\text{Cu}(\text{NH}_3)_4]^{2+}$, $\text{Ag}^+ + 2\text{NH}_3 \rightarrow [\text{Ag}(\text{NH}_3)_2]^+$, $\text{Cd}^{2+} + 4\text{NH}_3 \rightarrow [\text{Cd}(\text{NH}_3)_4]^{2+}$; precipitates metal hydroxides — $\text{FeCl}_3 + 3\text{NH}_4\text{OH} \rightarrow \text{Fe}(\text{OH})_3 \downarrow$ (brown), $\text{AlCl}_3 + 3\text{NH}_4\text{OH} \rightarrow \text{Al}(\text{OH})_3 \downarrow$ (white), $\text{CrCl}_3 + 3\text{NH}_4\text{OH} \rightarrow \text{Cr}(\text{OH})_3 \downarrow$ (green).
- Phosphorus allotropes: white — translucent waxy solid, extremely reactive, poisonous, water-insoluble; red — from heating white P without air, does not burn spontaneously; black — further heating of red P.
- Phosphine (PH_3) and phosphorus halides (PCl_3 , PCl_5); oxy-acids of phosphorus: H_3PO_4 (orthophosphoric), H_3PO_3 (phosphorous), $\text{H}_4\text{P}_2\text{O}_7$ (pyrophosphoric), HPO_3 (metaphosphoric).



P_4 tetrahedron (white phosphorus)



Red phosphorus — polymeric chain

Group 16 — Oxygen family / chalcogens (O, S, Se, Te, Po)

- Electronic configuration $ns^2 np^4$; common oxidation states $-2, +4, +6$; electronegativity decreases $O \rightarrow Po$.
- Oxides: all form dioxides MO_2 — SO_2 is a gas, SeO_2 volatile solid, TeO_2/PoO_2 non-volatile crystalline solids. Ozone (O_3) — unstable, decomposes easily to O_2 ; strong oxidising agent (liberates nascent oxygen).
- Sulphur allotropes: rhombic (α) — bright yellow, density 2.07 g/cm^3 , S_8 puckered rings, insoluble in water; monoclinic (β) — stable above 369 K , dull yellow, soluble in CS_2 , slowly changes to rhombic; plastic — amorphous, from pouring molten sulphur into cold water, insoluble in water & CS_2 .
- Sulphuric acid (H_2SO_4): powerful dehydrating agent (strong affinity for water); chars sugar/wood/paper to black carbon mass; moderately strong oxidising agent; decomposes carbonates, bicarbonates, sulphides, sulphites, thiosulphates, nitrites at room temperature — $H_2SO_4 + Na_2CO_3 \rightarrow Na_2SO_4 + H_2O + CO_2$; hot conc. acid liberates acids from salts — $H_2SO_4 + CaCl_2 \rightarrow CaSO_4 + 2HCl$.

Sr. No.	Property	Oxygen	Sulfur	Selenium	Tellurium	Polonium
1.	Configuration	$[He]2s^2 2p^4$	$[Ne]3s^2 3p^4$	$[Ar]4s^2 4p^4$	$[Kr]5s^2 5p^4$	$[Xe]6s^2 6p^4$
2.	Common oxidation state	-2	-2, +4, +6	+4, +6	+4, +6	
3.	Atomic radius (pm)	66	104	116	143	167
4.	First ionization energy (kJ/mol)	1314	1000	941	869	812
5.	Electronegativity	3.5	2.5	2.4	2.1	2.0

Group 16 — oxygen vs sulphur vs selenium vs tellurium (original comparison)

Group 17 — Halogens (F, Cl, Br, I, At)

- $ns^2 np^5$; F and Cl used in water purification/disinfectants; iodine essential in medicine.
- Interhalogen compounds: XX' (ClF , BrF), XX'_3 (ClF_3 , BrF_3), XX'_5 (BrF_5 , IF_5), XX'_7 (IF_7) — formula depends on size ratio of central & peripheral halogen.
- Hydrogen halides: reducing power $HF < HCl < HBr < HI$; HBr/HI are oxidised by H_2SO_4 — $2HBr + H_2SO_4 \rightarrow 2H_2O + SO_2 + Br_2$; all react with $KMnO_4/K_2Cr_2O_7$ liberating halogen — $2KMnO_4 + 16HCl \rightarrow 2KCl + 2MnCl_2 + 8H_2O + 5Cl_2$.
- Dipole moment & bond strength & thermal stability: $HI < HBr < HCl < HF$; bond length: $HF < HCl < HBr < HI$; acid strength: $HF < HCl < HBr < HI$. Best-known pseudohalide ion: CN^- .
- Oxy-acids of chlorine: $HClO$ (hypochlorous) $< HClO_2$ (chlorous) $< HClO_3$ (chloric) $< HClO_4$ (perchloric) — acidity increases with oxidation number of the halogen.

Group 18 — Noble gases (He, Ne, Ar, Kr, Xe, Rn)

- Inert because of: stable electronic configuration (complete octet), high ionization energies, low electron affinity.
- Stable xenon compounds: XeO_3 (pyramidal), XeO_4 (tetrahedral), $XeOF_4$ (square pyramidal), XeO_2F_2 (distorted octahedral).
- Uses — see Unit 8 (He, Ne, Ar, Kr, Xe, Rn table).

Type XX'_1 ($n = 1$) (with linear shape)	Type XX'_3 ($n = 3$) (with T-shape)	XX'_5 ($n = 5$) (with square pyramidal shape)	XX'_7 ($n = 7$) with pentagonal bipyramidal shape)
ClF	ClF ₃	ClF ₅	
BrF BrCl	BrF ₃	BrF ₅	
ICl, IBr, IF	ICl ₃ , IF ₃	IF ₅	IF ₇

Interhalogen types

Molecule	Total electron pairs (BP + LP)	Hybridisation	Shape
XeF ₂	5	sp ³ d	Linear
XeF ₄	6	sp ³ d ²	Square planar
XeF ₆	7	sp ³ d ³	Distorted octahedral

Xenon oxyfluorides/oxides

Key trends across the p-block (down a group)

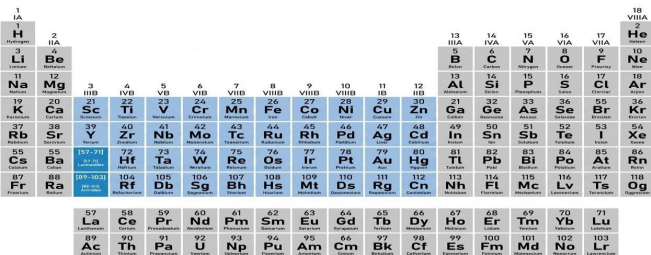
- Electronegativity ↓ — bigger atoms, more shielding ($F > I$).
- Atomic size ↑ — extra shells ($C < Pb$).
- Ionization energy ↓ — easier to remove outer electrons ($B > TI$).

Important p-block compounds — quick table

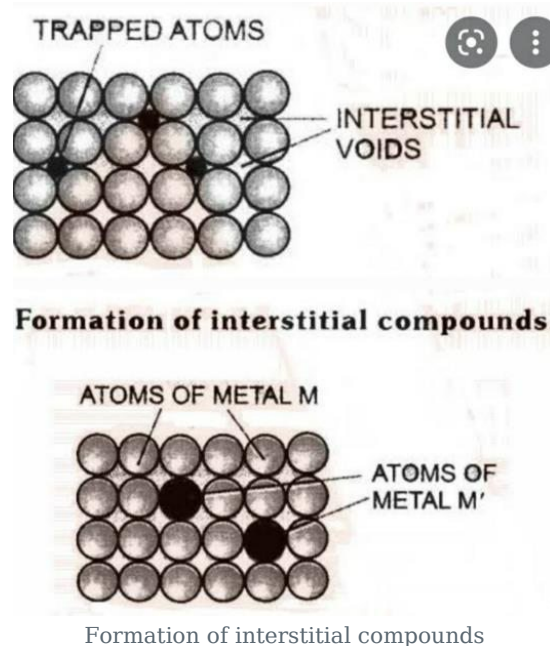
Compound	Group	Uses	Key property
Ammonia NH ₃	15	Fertilisers (NH ₄ NO ₃), cleaning agents, making HNO ₃	Colourless pungent gas; highly soluble; strongly basic
Silicon dioxide SiO ₂	14	Glass, cement, semiconductor	Hard crystalline (quartz); water-insoluble
Sulphuric acid H ₂ SO ₄	16	Phosphate fertilisers, battery acid, chemical synthesis	Strong, corrosive, high affinity for water
Phosphoric acid H ₃ PO ₄	15	Superphosphate fertilisers, soft-drink acidifier, rust removal	Weak, non-volatile; crystallises
Aluminium oxide Al ₂ O ₃	13	Making Al metal, abrasives, refractories	Very high m.p., hard, electrical insulator
Chlorine Cl ₂	17	Water disinfection, PVC, bleaching	Yellow-green gas, reactive, toxic
Hydrofluoric acid HF	17	Etching glass, cleaning metals, making F-compounds	Weak but highly corrosive
Boric acid H ₃ BO ₃	13	Antiseptic, insecticide, borosilicate glass	Weak acid, water-soluble, mild antiseptic
Dichloromethane CH ₂ Cl ₂	14	Paint remover, degreasing, pharma solvent	Volatile, sweet odour, non-flammable

9.3 d-Block Elements (Transition Metals)

- Four series aligned with filling of 3d, 4d, 5d, 6d orbitals:
- 3d: Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn · 4d: Y, Zr, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Cd · 5d: La, Hf, Ta, W, Re, Os, Ir, Pt, Au, Hg · 6d: incomplete.



d-block on the periodic table



Electronic configuration

- General: $(n-1)d^{1-10} ns^{1-2}$.
- Exceptions (energy gap between $(n-1)d$ and ns is small; half/full-filled d sets are extra stable): $3d$ — Cr, Cu; $4d$ — Nb, Mo, Tc, Ru, Rh, Pd, Ag; $5d$ — Pt, Au; $6d$ — Rg.

Atomic radii

- In a period: $Sc \rightarrow Mn$ radius decreases (nucleus-valence attraction beats repulsion); Fe, Co, Ni nearly equal; $Cu \rightarrow Zn$ increases.
- In groups: $4d$ radii $> 3d$; $4d \approx 5d$ (because of lanthanide contraction).

Magnetic properties

- Paramagnetism — unpaired d electrons (d^1 – d^9 configurations) attract towards magnetic field.
- Diamagnetism — all electrons paired; repelled by field.
- Ferromagnetism — strongest form (Fe, Co, Ni).
- Magnetic moment $\mu = \sqrt{n(n+2)}$ B.M., n = number of unpaired electrons.

Interstitial compounds & alloys

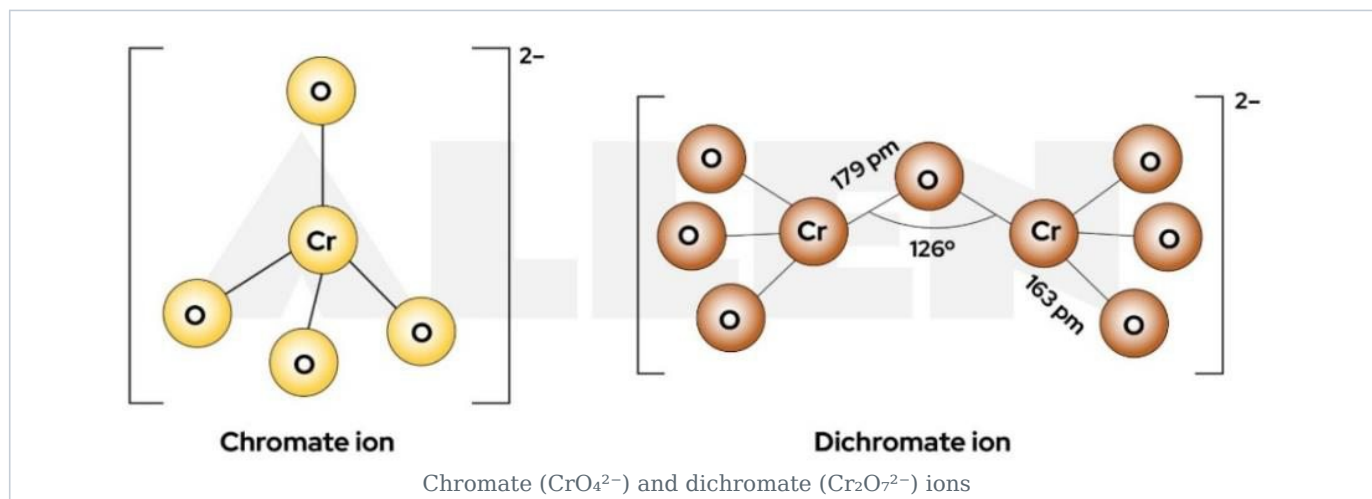
- Small atoms (C, N, H) occupy lattice gaps of the transition metal — TiC (hardness — cutting tools), Mn_4N (magnetic), Fe-H (hydrogen storage/fuel cells).
- Alloys: brass (Cu-Zn), alnico magnets (Al-Ni-Co), titanium alloys (Ti-Al/V — aerospace, implants). Transition metals form alloys readily due to similar radii.

Complexes, catalysis, density, oxidation states & colour

- Coordination complexes: central metal ion + ligands (molecules/ions donating lone pairs) via coordinate covalent bonds; coordination number 2–12; geometry — octahedral, square planar, tetrahedral. Examples: $[Fe(CN)_6]^{3-}$ hexacyanoferrate(III), $[Cu(NH_3)_4]^{2+}$ tetraamminecopper(II), $[PtCl_4]^{2-}$ tetrachloridoplatinate(II). Complexes are often coloured, catalytically active, and biologically important (cofactors).
- Catalysis: transition metals activate molecules, enable bond-making/breaking (hydrogenation, CO_2 conversion); vital in petroleum refining, chemical production, renewable energy.
- Density: high and tightly packed; $Sc < Ti < V < Cr < Mn < Fe < Co < Ni = Cu > Zn$ (Zn lower — large atomic volume); minimum in $3d$ — Sc; maximum — Ni/Cu; in groups $3d < 4d < 5d$; densest elements — Os (22.51 g/cm³), Ir (22.61 g/cm³).
- Variable oxidation states: +1 to +8 (d -orbital electrons available); $Ni(CO)_4$ and $Fe(CO)_5$ show zero oxidation state.

- ## Important d-block compounds

- Potassium dichromate ($K_2Cr_2O_7$): bright orange-red crystals; powerful oxidising agent; water-soluble (orange solution); stable normally, decomposes on heating or with reducing agents. Uses — qualitative analysis/titrations, (historical) photography, pigments & dyes, wood preservation (biocide). Highly toxic and carcinogenic — use restricted.
- Potassium permanganate ($KMnO_4$): dark purple crystals; strong oxidising agent. Uses — water treatment (impurity/odour removal), dilute antiseptic (wounds, dermatitis), organic synthesis/oxidiser, analytical reagent for reducing agents.



9.4 f-Block Elements (Inner Transition Metals)

1 H Hydrogen																	2 He Helium						
3 Li Lithium	4 Be Beryllium																	5 B Boron	6 C Carbon	7 N Nitrogen	8 O Oxygen	9 F Fluorine	10 Ne Neon
11 Na Sodium	12 Mg Magnesium																	13 Al Aluminum	14 Si Silicon	15 P Phosphorus	16 S Sulfur	17 Cl Chlorine	18 Ar Argon
19 K Potassium	20 Ca Calcium	21 Sc Scandium	22 Ti Titanium	23 V Vanadium	24 Cr Chromium	25 Mn Manganese	26 Fe Iron	27 Co Cobalt	28 Ni Nickel	29 Cu Copper	30 Zn Zinc	31 Ga Gallium	32 Ge Germanium	33 As Arsenic	34 Se Selenium	35 Br Bromine	36 Kr Krypton						
37 Rb Rubidium	38 Sr Strontium	39 Y Yttrium	40 Zr Zirconium	41 Nb Niobium	42 Mo Molybdenum	43 Tc Technetium	44 Ru Ruthenium	45 Rh Rhodium	46 Pd Palladium	47 Ag Silver	48 Cd Cadmium	49 In Indium	50 Sn Tin	51 Sb Antimony	52 Te Tellurium	53 I Iodine	54 Xe Xenon						
55 Cs Cesium	56 Ba Barium	57 La Lanthanum	72 Hf Hafnium	73 Ta Tantalum	74 W Tungsten	75 Re Rhenium	76 Os Osmium	77 Ir Iridium	78 Pt Platinum	79 Au Gold	80 Hg Mercury	81 Tl Thallium	82 Pb Lead	83 Bi Bismuth	84 Po Polonium	85 At Astatine	86 Rn Radon						
87 Fr Francium	88 Ra Radium	89 Ac Actinium	104 Rf Rutherfordium	105 Db Dubnium	106 Sg Seaborgium	107 Bh Bohrium	108 Hs Hassium	109 Mt Meitnerium	110 Ds Darmstadtium	111 Rg Roentgenium	112 Cn Copernicium	113 Nh Nihonium	114 Fl Flerovium	115 Mc Moscovium	116 Lv Livermorium	117 Ts Tennessine	118 Og Oganesson						
			58 Ce Cerium	59 Pr Praseodymium	60 Nd Neodymium	61 Pm Promethium	62 Sm Samarium	63 Eu Europium	64 Gd Gadolinium	65 Tb Terbium	66 Dy Dysprosium	67 Ho Holmium	68 Er Erbium	69 Tm Thulium	70 Yb Ytterbium	71 Lu Lutetium							
			90 Th Thorium	91 Pa Protactinium	92 U Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium							

Lanthanides & actinides on the periodic table

Lanthanides (4f) – “rare earths”

- Silvery-white, malleable, high electrical conductivity; high density; high m.p./b.p. (exceptions among lighter ones).
- Unique magnetic properties (para/ferromagnetic at low temperatures); relatively reactive but tarnish slowly (oxide layer).
- Variable oxidation states; +3 most common; share many properties due to similar configurations.

Actinides (5f)

- Mostly radioactive (α , β , γ decay); U and Pu notably dense; high m.p./b.p. (strong metallic bonding).
- Variable valency from partially filled 5f orbitals — highest unstable +7 (Np, Pu); highest stable +6 (U); allotropy common; unique low-temperature magnetism.

Aspect	Lanthanides	Actinides
Occurrence	More abundant, naturally found	Mostly synthetic/very scarce (U, Pu natural)
Radioactivity	Largely non-radioactive	All radioactive
Stability	Most +3 states stable	+3 to +7, unstable oxidation states common
Applications	Lighting, catalysts, magnets, electronics	Nuclear reactors, research, nuclear weapons

◀ RAPID REVISION

- s-block: alkali = ns^1 (+1), alkaline earth = ns^2 (+2); flame colours — Na golden, K violet, Ca brick-red, Sr crimson, Ba apple-green.
- PoP = gypsum heated (loses $1\frac{1}{2}$ H_2O); gypsum = $CaSO_4 \cdot 2H_2O$; borax = $Na_2B_4O_7 \cdot 10H_2O$.
- p-block: B family — B non-metal $\rightarrow$ Tl metal; CO_2 turns lime water milky; SiC = carborundum; ozone = strong oxidant.
- Sulphur allotropes: rhombic (S_8 , α), monoclinic (β , >369 K), plastic (amorphous).
- Halogen orders: acidity $HClO < HClO_2 < HClO_3 < HClO_4$; acid strength $HF < HCl < HBr < HI$; reducing power $HF < HCl < HBr < HI$.
- d-block: $(n-1)d^{1-10}ns^{1-2}$; exceptions Cr, Cu...; $\mu = \sqrt{n(n+2)}$ B.M.; densest — Os/Ir; chromate orange-red $K_2Cr_2O_7$, permanganate purple $KMnO_4$.
- Lanthanides: +3 common, 4f; actinides: radioactive, 5f; $4d \approx 5d$ radii (lanthanide contraction).

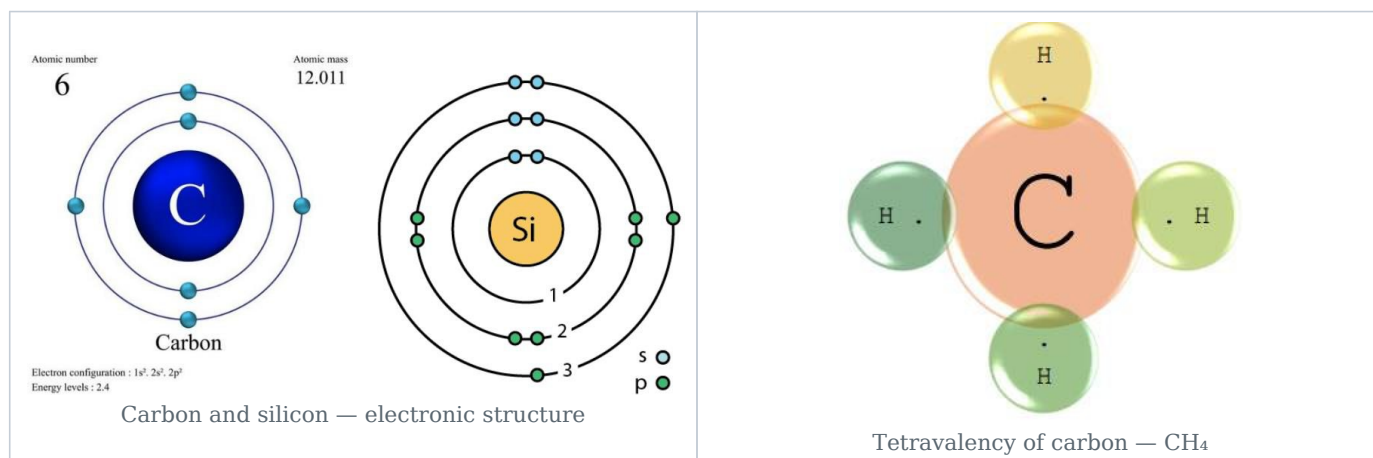
UNIT 10

Organic Chemistry & Chemistry in Everyday Life

Sources: Organic Chemistry lectures + organic part of Acids-Bases-Salts

10.1 Organic Compounds & the Uniqueness of Carbon

- Organic compounds — compounds containing carbon (oxides of carbon, metallic carbonates, metal cyanides/carbides are excluded).
- Vital Force theory (organic compounds only from living sources) was disproved by Wöhler (1828): ammonium cyanate → urea.
- Unique nature of carbon: tetravalency (4 covalent bonds per C atom) and catenation — self-linking ability to form straight, branched and cyclic chains. Elements showing catenation: C, Si, Ge, S, P, N (strongest in C).



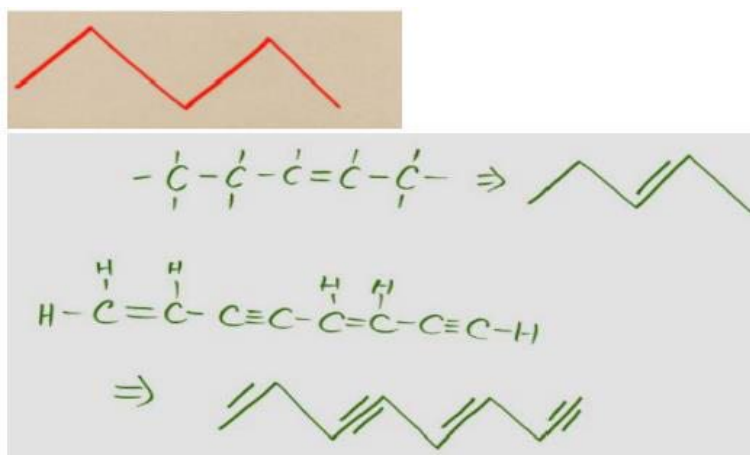
Allotropy & allotropes of carbon

- Allotropy — property of an element to exist in different physical forms. Coal, diamond & graphite are natural allotropes; O₂/O₃, white/red phosphorus, rhombic/monoclinic sulphur also show allotropy.
- Coal types: peat → lignite → bituminous → anthracite.
- Graphite — the only non-metal that conducts electricity: electrodes, lubricant (slippery), moderator in nuclear reactors, pencil lead.

10.2 Nomenclature & Structural Representation

Carbon count	1	2	3	4	5	6	7	8	9	10
Root word	Meth	Eth	Prop	But	Pent	Hex	Hept	Oct	Non	Dec

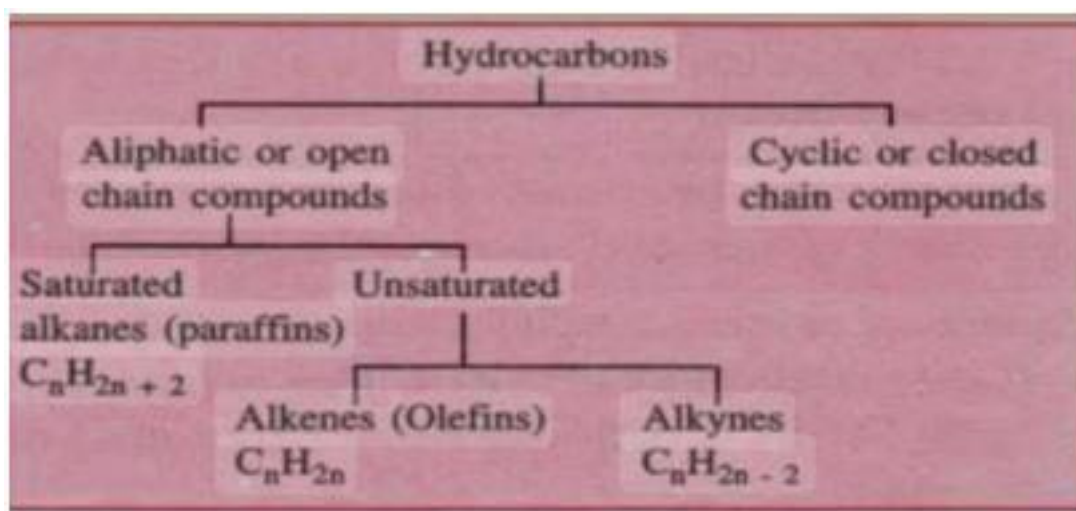
- A compound can be drawn as a complete structural formula, a condensed formula — pentane: C₅H₁₂ or CH₃CH₂CH₂CH₂CH₃ — or a bond-line (zig-zag) formula.



Complete, condensed and bond-line representations of pentane

10.3 Hydrocarbons & their Classification

- Hydrocarbons — compounds containing only carbon & hydrogen: CH_4 (methane), C_2H_4 (ethene), C_2H_2 (ethyne).



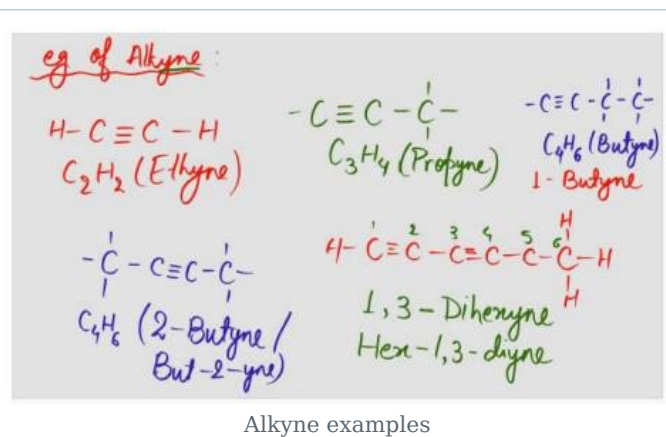
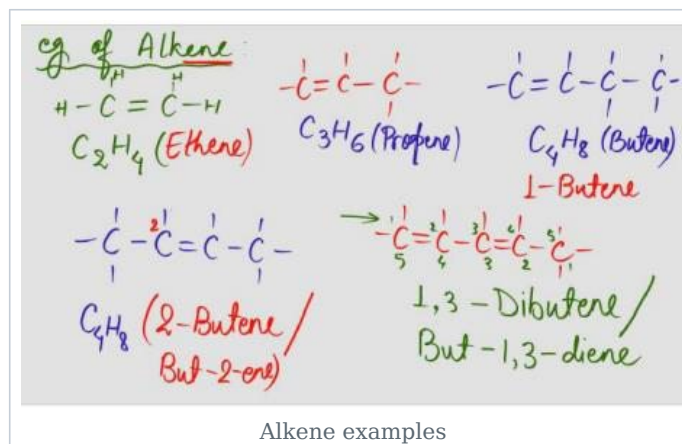
Classification of hydrocarbons

Saturated hydrocarbons — Alkanes ($\text{C}_n\text{H}_{2n+2}$)

- Only single covalent bonds; also called paraffins (less reactive); methane is the first member.
- Non-polar; weak van der Waals forces; first four members are gases, C_5 – C_{17} liquids, C_{18} onwards solids; low reactivity.

Unsaturated — Alkenes (C_nH_{2n}) & Alkynes ($\text{C}_n\text{H}_{2n-2}$)

- Alkenes — at least one $\text{C}=\text{C}$ double bond; ethene (ethylene) first member. Non-polar; first 3 members gases, C_5 – C_{18} liquids, $\text{C}_{19}+$ gases/solids; colourless & odourless (exception ethene — sweet smell); insoluble in water.
- Alkynes — at least one $\text{C}\equiv\text{C}$ triple bond; ethyne (acetylene) first member. Non-polar; first 3 gaseous, C_5 – C_{13} liquids, $\text{C}_{14}+$ solid; odourless & colourless (exception ethyne); insoluble in water.

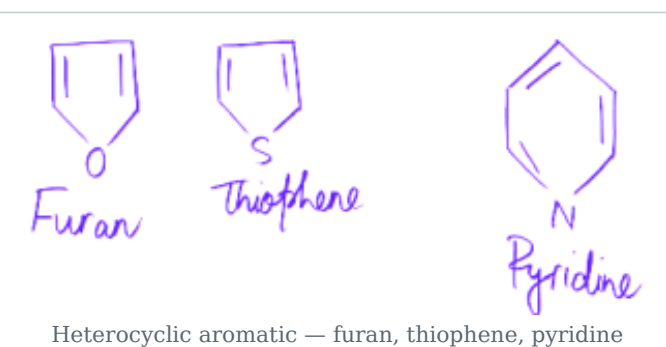
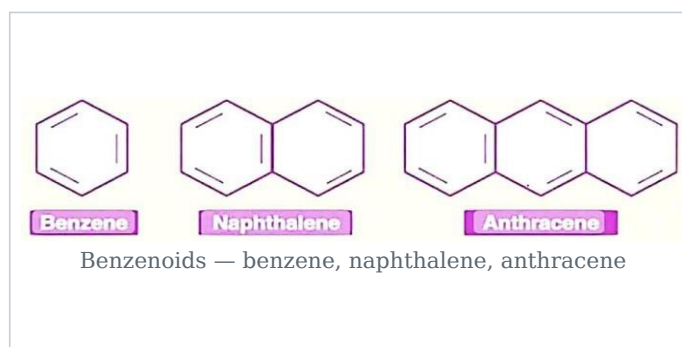


Homologous series

- A series of organic compounds with similar physical & chemical properties where each successive member differs by a CH_2 group — e.g. alkanes, alkenes, alkynes, alcohols, aldehydes.

10.4 Cyclic (Closed-chain) Compounds

- Rings of carbon atoms (≥ 3 carbons). Divided into alicyclic and aromatic compounds.
- Alicyclic (non-aromatic): single/double/triple bonds in any (non-alternating) fashion — cyclopropane, cyclobutane... Homocyclic/carbocyclic (only C in ring $\rightarrow$ cycloalkanes, cycloalkenes, cycloalkynes) or heterocyclic (ring contains other atoms — epoxides, 1,2-dioxolane, 1,4-dioxane).
- Aromatic (arenes): ring with alternating single & double bonds; usually pleasant smell. Sub-types:
 - Benzenoids — contain a benzene ring (benzene, naphthalene, anthracene);
 - Non-benzenoids — aromatic but no benzene ring (tropone, azulene);
 - Heterocyclic aromatic — ring contains N/O/S etc. (furan, thiophene, pyridine).



10.5 Functional Groups

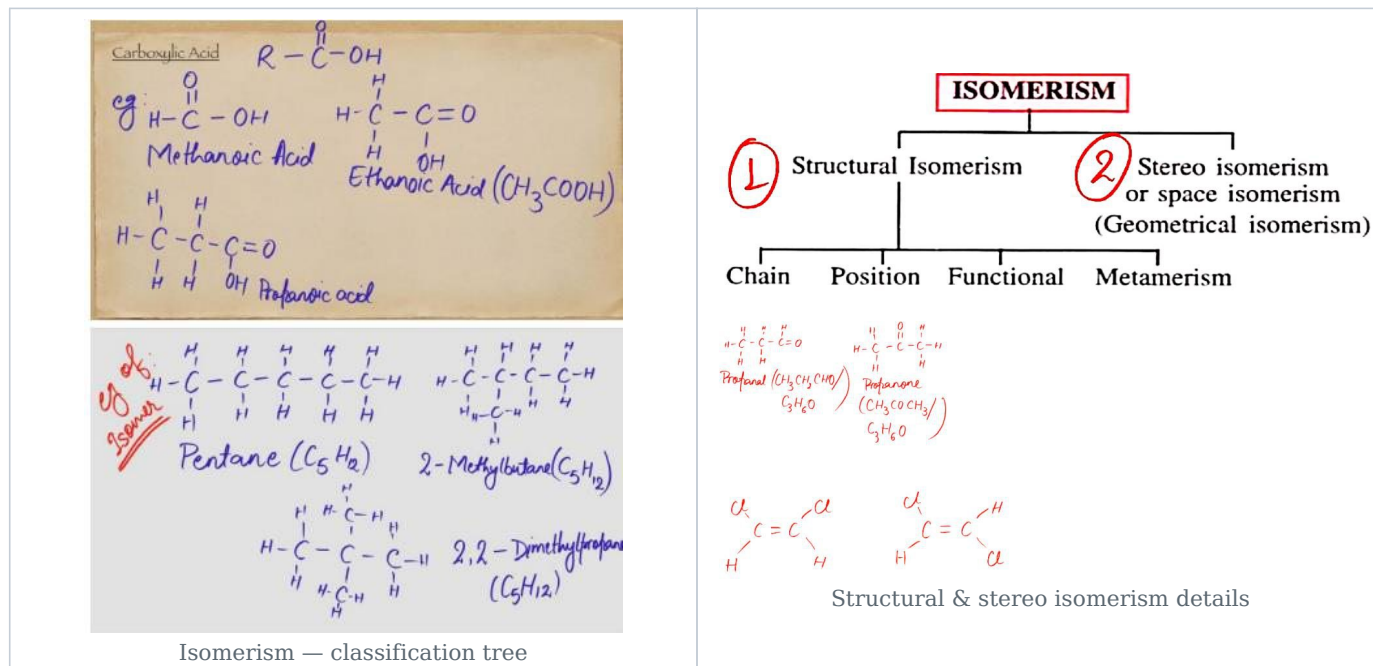
- An atom or group of atoms which, on being added to an organic compound, changes its properties — e.g. $\text{R}-\text{OH}$ (alcohol), aldehydes, ketones, carboxylic acids.

Functional group	Formula	Class	Suffix
Halide	$\text{R}-\text{X}$ ($\text{X} = \text{F}, \text{Cl}, \text{Br}, \text{I}$)	Haloalkane	-ane $\rightarrow$ halo
Hydroxyl	$\text{R}-\text{OH}$	Alcohol	-ol
Aldehyde	$\text{R}-\text{CHO}$	Aldehyde	-al
Carboxyl	$\text{R}-\text{COOH}$	Carboxylic acid	-oic acid
Keto	$\text{R}-\text{CO}-\text{R}'$	Ketone	-one
Ether	$\text{R}-\text{O}-\text{R}'$	Ether	-oxy

10.6 Isomerism

- Compounds with the same molecular formula but different arrangement of atoms show isomerism.

- Structural isomerism: chain (straight vs branched parent chain), position (same group, different position), functional (different functional groups), metamerism.
- Stereoisomerism — same formula and connectivity but different geometrical orientation (cis-trans / geometrical isomerism).



10.7 Chemical Properties of Carbon Compounds

Combustion

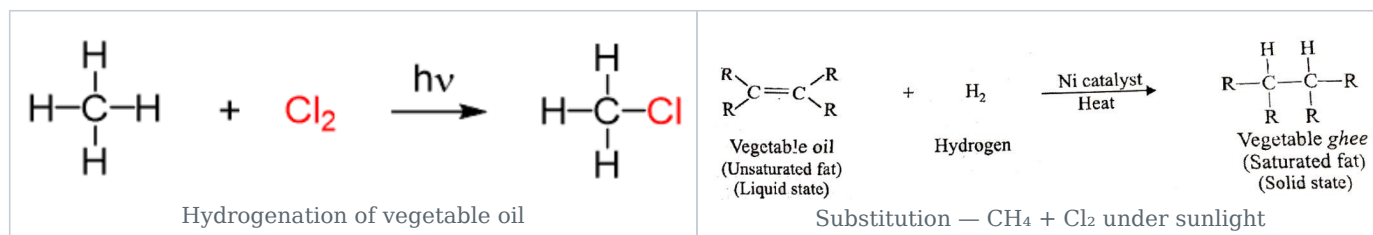


Oxidation

- Ethanol (CH_3CH_2OH) $\rightarrow$ ethanoic acid (CH_3COOH) with acidified $K_2Cr_2O_7$ or alkaline $KMnO_4$ (both strong oxidising agents).

Addition reaction (hydrogenation / reduction)

- Unsaturated compounds add H_2 in presence of Ni or Pd catalyst — used for hydrogenation of vegetable oil (liquid $\rightarrow$ ghee/vanaspati).



Substitution reaction

- One atom/group replaces another in a saturated compound — $CH_4 + Cl_2 \xrightarrow{h\nu} CH_3Cl + HCl$ (in presence of sunlight).

10.8 Important Carbon Compounds

Methane (CH_4) — 1st alkane

- Also called marsh gas; 80-85% of natural gas; colourless, odourless; produced in paddy fields.
- Preparation: $Al_4C_3 + H_2O \rightarrow Al(OH)_3 + CH_4 \uparrow$; $CH_3COONa + NaOH \xrightarrow{\Delta} Na_2CO_3 + CH_4$ (soda-lime decarboxylation).

Ethene (C_2H_4) — 1st alkene

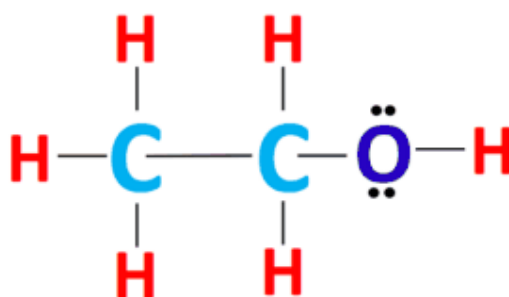
- Present in plants in small amounts as a hormone — responsible for ripening of fruits; colourless, sweet smell.
- Preparation: ethanol —conc. H_2SO_4 , 443 K $\rightarrow$ ethene + H_2O (conc. H_2SO_4 acts as a dehydrating agent).

Ethyne (C_2H_2) — 1st alkyne

- Structure $\text{H}-\text{C}\equiv\text{C}-\text{H}$; colourless gas with smell; lighter than air.
- Preparation: $\text{CaC}_2 + 2\text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2 + \text{C}_2\text{H}_2 \uparrow$ (from calcium carbide).

Ethanol ($\text{C}_2\text{H}_5\text{OH}$) — alcohol

- Liquid at room temperature; widely used as an organic solvent; consumption causes drunkenness — 100% (absolute) ethanol is lethal.
- Reactions: with Na — $2\text{C}_2\text{H}_5\text{OH} + 2\text{Na} \rightarrow 2\text{C}_2\text{H}_5\text{ONa}$ (sodium ethoxide) + H_2 ; neutral ($\text{pH} \approx 7.5$); hot conc. $\text{H}_2\text{SO}_4 \rightarrow$ ethene (dehydration).

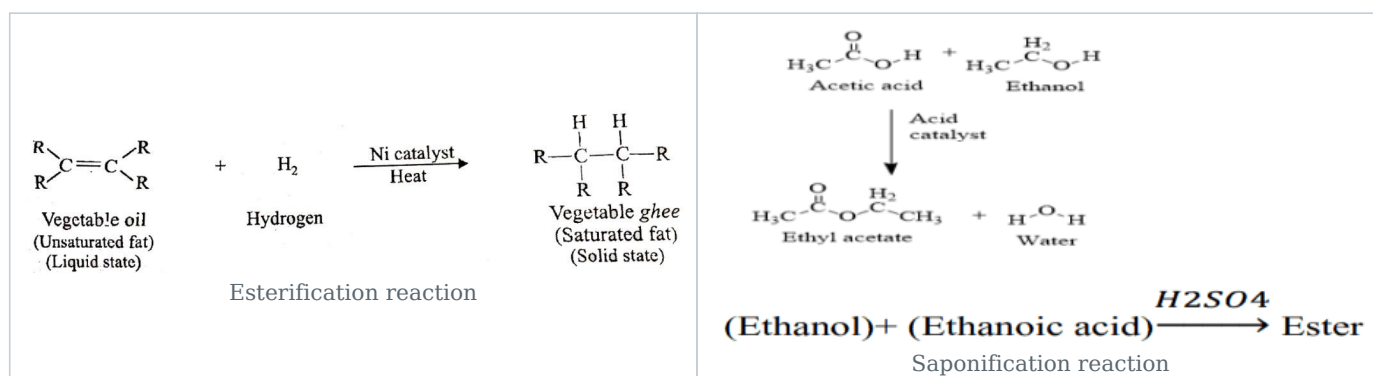


$\text{C}_2\text{H}_5\text{OH}$ Lewis structure

Lewis structure of ethanol ($\text{C}_2\text{H}_5\text{OH}$)

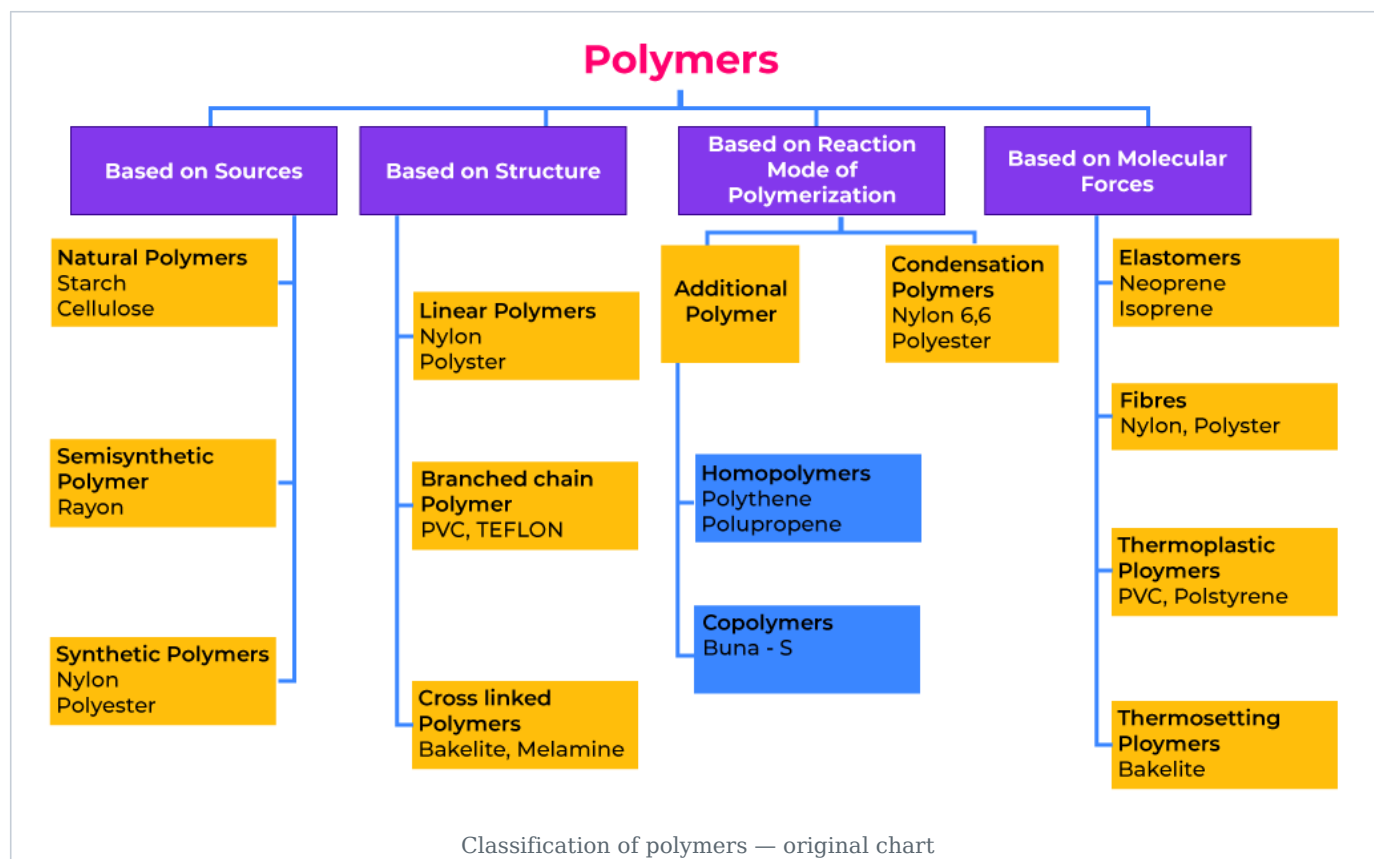
Ethanoic acid (CH_3COOH) — carboxylic acid

- Called glacial acetic acid — low melting point ($\approx 15-17^\circ\text{C}$, freezes like ice in cold climates); 5-8% solution = vinegar.
- Esterification: ethanol + ethanoic acid —conc. $\text{H}_2\text{SO}_4 \rightarrow$ ester (sweet-smelling) + water.
- Saponification (reverse of esterification): ester + base $\rightarrow$ alcohol + metallic salt of carboxylic acid (soap formation).
- With base: $\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa}$ (sodium ethanoate) + H_2O ; with carbonate: $2\text{CH}_3\text{COOH} + \text{K}_2\text{CO}_3 \rightarrow 2\text{CH}_3\text{COOK} + \text{H}_2\text{O} + \text{CO}_2$.



10.9 Polymers

- Polymer (Greek — “many units”) — chemical substance of high molecular mass formed by combining a large number of simple molecules (monomers). e.g. $n(\text{CH}_2=\text{CH}_2) \rightarrow \text{--[CH}_2-\text{CH}_2\text{]}_n$ (ethylene $\rightarrow$ polyethylene).
- Polymerisation — the process by which monomers combine into polymers: $n[\text{monomer}] \rightarrow \text{polymer}$.
- Polymers vs macromolecules: all polymers are macromolecules, but not vice-versa — chlorophyll ($\text{C}_{55}\text{H}_{72}\text{O}_5\text{N}_4\text{Mg}$) is a macromolecule without monomer units.



Classification

Basis	Types	Notes / examples
Source	Natural; Synthetic; Semisynthetic	Natural — starch (plant food reserve), cellulose (plant structure), proteins (animal building blocks), natural rubber (latex of <i>Hevea brasiliensis</i>); synthetic — polythene, PVC, nylon-66, teflon, orlon; semi — rayon (cellulose acetate), vulcanised rubber
Structure	Linear; Branched; Cross-linked (network)	Linear — close packing, high density/tensile strength (HDPE, nylon, polyester); cross-linked — hard, rigid, brittle (bakelite, glyptal, melamine-formaldehyde)
Mode of polymerisation	Addition; Condensation	Addition — unsaturated monomers, same empirical formula (polythene); condensation — with elimination of small molecules (nylon-6,6 from hexamethylenediamine + adipic acid)
Addition sub-types	Homopolymer; Copolymer	Homo — one monomer type (polythene); co — two or more monomers
Molecular forces	Elastomers; Fibres; Thermoplastics; Thermosetting	Elastomers — weakest forces, stretchable (rubber, buna-S/N); fibres — strong H-bonded, high tensile (nylon-66, terylene); thermoplastics — melt & remould (PE, polystyrene); thermosetting — cross-linked on heating, infusible (bakelite, melamine, urea-formaldehyde)

Types of polymerisation

- Chain-growth (addition): via reactive intermediates — free radical (benzoyl peroxide catalyst), cationic (Lewis acids BF_3 , AlCl_3 , SnCl_4 / H_2SO_4 — more stable carbocation = more reactive monomer), anionic (strong-base initiated).
- Step-growth (condensation): stepwise with elimination of small molecules (H_2O , NH_3 , HCl , ROH) — adipic acid + hexamethylenediamine; phenol + formaldehyde.

	Chain growth polymerisation	Step growth polymerisation
1.	In this type of polymerisation, repeated addition of monomer units to a growing polymer chain occurs.	Polymerisation occurs by a series of condensation reactions with the elimination of simple molecules such as H_2O , HCl , NH_3 , etc.
2.	The monomers involved are unsaturated compounds such as alkenes, alkadiene, etc. e.g., ethene, vinyl chloride, etc.	Monomers involved are saturated compound having two or more functional groups. e.g., terephthalic acid, ethylene glycol, etc.

Chain-growth vs step-growth

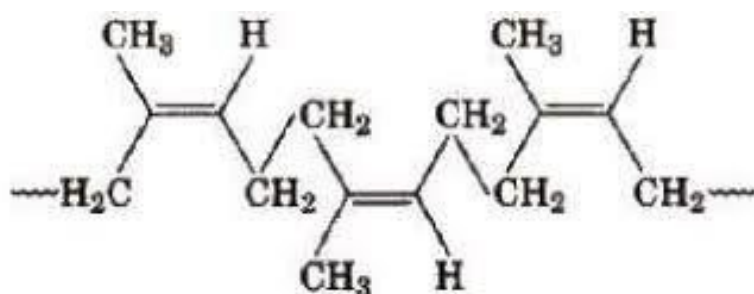
$n(CH_2=CH-CH=CH_2) + n(CH_2=CH-Ph)$
 butadiene styrene

$\downarrow$
 $\{CH_2-CH=CH-CH_2-CH_2-CH(Ph)\}_n$
 buna-S (butadiene styrene rubber)

Buna-S (butadiene-styrene rubber)

Important polymers

- LDPE — tough, flexible, transparent, inert, poor conductor; wire insulation, squeeze bottles, toys, flexible pipes. HDPE — denser, harder; containers, housewares, bottles.
- Polystyrene — thermoplastic of styrene; toys, radio/TV cabinets; foamed = thermocol.
- Polypropylene (PP) — Ziegler-Natta polymerisation of propylene.
- Nylon-66 — hexamethylenediamine + adipic acid (numbers = C atoms in each); very high tensile strength, abrasion-resistant; bristles, textiles, sheets. Nylon-6 — from caprolactam with water at high temperature.
- Bakelite — phenol + formaldehyde (acid/base catalyst, o/p-methylene bridges); combs, records, electrical switches; soft grades — laminating glue, varnishes, lacquers.
- Melamine-formaldehyde — very hard & tough; unbreakable crockery. Urea-formaldehyde resin — adhesives.
- Natural rubber — coiled cis-1,4-polyisoprene; van der Waals forces only; usable 10–50 °C, soft & sticky, attacked by oxidising agents/solvents.
- Vulcanisation: heating rubber with 3–5% sulphur (+ ZnO accelerator) at ≈ 110 °C for 20–30 min introduces –S–S– polysulphide crosslinks $\rightarrow$ limits chain motion $\rightarrow$ tough, elastic rubber.
- Neoprene — polymer of chloroprene (synthetic rubber); conveyor belts, gaskets, hoses. Buna-N — buta-1,3-diene + acrylonitrile copolymer; insulator; conveyor belts, printing rollers.
- PMMA (lucite/acrylite/plexiglass/perspex) — hard transparent polymer; unbreakable lights, aircraft windows, dentures. Glyptal — cross-linked thermosetting polyester (ethylene glycol + phthalic acid); paints & lacquers.
- Terylene (Dacron) — ethylene glycol + terephthalic acid at 420–460 K (zinc acetate + antimony trioxide); chemical/biological resistance; terycot/terylwool/terylsilk fabrics, magnetic tapes, conveyor belts.
- Biodegradable: PHB (3-hydroxybutanoic + 3-hydroxypentanoic acid copolymer); Nylon-2-Nylon-6 (glycine + amino caproic acid) — alternating polyamide, biodegradable.
- More: saran (vinylidene chloride — food wrapping); ABS rubber (acrylonitrile + butadiene + styrene); bubble gum — styrene-butadiene rubber; dynel (vinyl chloride + acrylonitrile — hair wigs); silk — natural polyamide fibre; thermocol — foamed polystyrene; superglue — methyl- α -cyanoacrylate (anionic polymerisation).



A section of the polymeric chain of natural rubber

A section of the polymeric chain of natural rubber (cis-1,4-polyisoprene)

10.10 Chemistry in Everyday Life

Medicines / drugs

- Drugs help treat diseases or reduce pain; treating disease with chemicals that destroy microorganisms without harming the body = chemotherapy.

Class	Action	Examples
Antiseptics	Prevent/destroy growth of harmful microorganisms (applied to living tissue)	Dettol (chloroxylenol + terpineol), Savlon, iodine, bithional (in soaps); 0.2% phenol is antiseptic
Disinfectants	Destroy microorganisms completely; toxic to tissues — floors, sanitary ware, clothes	1% phenol solution
Analgesics (painkillers)	Relieve pain	Narcotic — morphine (opium alkaloid, habit-forming); non-narcotic — aspirin, ibuprofen, naproxen
Antipyretics	Control fever	Aspirin, paracetamol
Antimalarials	Cure malaria fever	Quinine, chloroquine, primaquine
Tranquilizers	Act on CNS — calming (psychotherapeutic)	Sedatives/hypnotics — reserpine, barbiturates (luminal, seconal); antidepressants — benzedrine (amphetamine)
Anaesthetics	Local/general loss of sensation	General — nitrous oxide, chloroform, sodium pentothal; local — xylocaine (jelly), ethyl chloride (spray), procaine (injection)
Antibiotics	From microorganisms; stop growth of other microbes	Penicillin (pneumonia, bronchitis, sore throat); streptomycin (TB); chloromycetin (typhoid, meningitis); tetracycline (fever, dysentery, UTI)
Sulpha drugs	Antibacterial — sulfanilamide derivatives	Sulphathiazole (severe infections), sulpha guanidine (bacillary dysentery), sulpha pyridine (pneumonia), sulfadiazine (dysentery, UTI)

Rocket propellants

- Combination of (a) an explosive fuel and (b) an oxidiser that gives rockets the push to enter space.
- Requirements: ashless burning; large volume of gases per gram; fast combustion.
- Solid propellants: composite — polyurethane/polybutadiene binder + ammonium perchlorate oxidiser; double-base — nitrocellulose + nitroglycerin. Drawback — cannot be stopped once ignited (no start/stop control).
- Liquid propellants: mono — nitromethane, methyl nitrate; biliquid — fuel (kerosene, alcohol, hydrazine, MMH, liquid H₂) + oxidiser (liquid O₂, N₂O₄, nitrous acid). Higher thrust; start/stop controllable.
- Hybrid: solid fuel (acrylic rubber) + liquid oxidiser (nitrous oxide).
- Specific impulse (I_s) measures performance: $\propto \sqrt{T/M}$ — higher flame temperature T and lower average molecular mass M of exhaust gases give a better propellant.

Dyes

- Coloured substances applied (in solution/dispersion) to textiles, paper, leather, hair, plastics etc.
- Colour depends on light absorption — the reflected complementary colour is seen. Chromophores bear the colour; auxochromes (-OH, -NH₂, -Cl, -COOH) deepen it.
- By source: natural (alizarin, indigo) and synthetic (malachite green, orange-I/II, aniline yellow).
- By constitution: nitro (martius yellow), azo (aniline yellow, methyl orange, orange-I, congo red), triphenylmethane (malachite green, magenta), indigoid (indigo), anthraquinone (alizarin), phthalein (phenolphthalein).
- By application: acid dyes (azo-sulphonic salts — wool, silk, nylon; not cotton); basic dyes (amino-group salts — aniline yellow, malachite green); direct dyes (water-soluble, H-bonding fabrics — congo red, martius yellow); fibre-reactive dyes (irreversible attachment — triazine derivatives); insoluble azo dyes (60% of dyes; toxic — not for food); ingrain dyes (made in situ — para red); vat dyes (reduced to soluble, re-oxidised on fabric — indigo); mordant dyes (metal-ion mordant binds — alizarin, turkey red).

Chemicals in food

- Food additives improve qualities, appearance, taste, odour, nutritive value — preservatives, flavouring agents, sweeteners, dyes, antioxidants, fortifiers, emulsifiers, antifoaming agents.
- Antioxidants: prevent oxidation of fats in packaged food (chips, biscuits, cereals); also reduce free-radical ageing — BHT (butylated hydroxytoluene), BHA (butylated hydroxyanisole).
- Artificial sweeteners: saccharin (non-biodegradable, zero calories); aspartame (methyl ester of phenylalanine-aspartic dipeptide, soft drinks); alitame (more stable); sucralose (trichloro derivative of sucrose — no calories, no tooth decay); cyclamate (N-cyclohexyl sulfamate); L-glucose (sweet but not metabolised).
- Preservatives: protect against bacteria, yeasts, moulds — sodium benzoate (most common); sodium/potassium metabisulphite (jams, squashes, pickles — SO_2 forms sulphurous acid); salts of propionic & sorbic acid.
- Edible colours: synthetic azo dyes (tartrazine — harmful to children/asthma patients) vs natural safe colours (annatto, caramel, carotene, saffron); iron oxide (red), titanium dioxide (whiteness).

Soaps & detergents

- Soaps — sodium/potassium salts of higher fatty acids (lauric, myristic, palmitic, stearic, oleic, linoleic, linolenic). Biodegradable (microbes oxidise to CO_2) — no water pollution.
- Soap drawbacks: fail in hard water ($\text{Ca}^{2+}/\text{Mg}^{2+}$ form curdy precipitate); fail in acidic solutions (free fatty acids precipitate).
- Synthetic detergents — sodium salts of long-chain alkyl hydrogen sulphates or alkylbenzene sulphonates; work even in hard water (Ca/Mg salts soluble).
- Anionic detergents: sodium alkyl sulphates (from long-chain alcohols + conc. H_2SO_4 , then NaOH — biodegradable); alkylbenzene sulphonates (Friedel-Crafts alkylation + sulphonation + NaOH — e.g. sodium dodecylbenzenesulphonate). Effective in slightly acidic solutions.
- Cationic detergents: quaternary ammonium salts (chlorides/bromides/acetates) with long alkyl chain; expensive; germicidal — cetyltrimethylammonium chloride $[\text{CH}_3(\text{CH}_2)_{15}\text{N}^+(\text{CH}_3)_3]\text{Cl}^-$.
- Non-ionic detergents: high-molecular-mass esters — polyethylene glycol + stearic acid; or long-chain alcohols + ethylene oxide (base catalyst).
- Detergent pollution: branched chains resist bacterial breakdown (steric hindrance) → accumulation in water bodies; straight-chain detergents (sodium lauryl sulphate) are biodegradable.

Fuels & calorific value

- Fuel — any substance undergoing chemical/nuclear change to produce energy convertible into useful work.
- Calorific value — energy released per unit mass on complete combustion; SI unit J/kg.

Fuel	Calorific value (kJ/kg)	Fuel	Calorific value (kJ/kg)
Cow-dung cakes	6000–8000	Petrol	45000
Wood	17000–20000	Kerosene	45000
Coal	25000–33000	Diesel	45000
Methane	50000	CNG	52000

- Renewable sources — replenished by biogeochemical cycles (running water → rain, solar energy); sustainable development favours them. Non-renewable — cannot be regenerated (fossil fuels take millions of years) — use judiciously.
- Properties of an ideal fuel: easily attainable ignition temperature; high calorific value; low residue/harmless products (e.g. CNG burns clean); constant/moderate rate of combustion (no explosions); easy availability at low cost; easy storage & transport.
- Classification: by state — solid, liquid, gaseous fuels; by occurrence — natural (wood, coal, petroleum), biofuels (cow-dung cake), fossil fuels (coal — from plants; petroleum — from aquatic animals; millions of years old), artificial (water gas $\text{CO} + \text{H}_2$, producer gas — from hot coke with air/steam).
- Natural gas: found over petroleum deposits and around coal deposits (firedamp); mainly methane (+ higher hydrocarbons, traces of H_2S); high calorific value; easily liquefied — LNG.

Petroleum	Gasoline, Diesel, Kerosene
Coal	Coal tar
Biomass	Biodiesel obtained from Jatropha tree

Solid-liquid-gas fuel examples

Fuel	Composition	Sources
Water Gas	Carbon monoxide + Hydrogen	Reaction of red-hot coke with steam
Producer Gas	Carbon monoxide + Nitrogen	Reaction of red-hot coke with insufficient air
Coal Gas	Hydrogen + Methane + ethylene + Acetylene + CO + N ₂	Destructive distillation of coal
Natural gas	Methane (83%) + Ethane	Petroleum deposits
Liquified petroleum gas (LPG)	Butane (C ₄ H ₁₀) 95%	Fractional distillation of Petroleum
Compressed natural gas (CNG)	Methane (CH ₄) 95%	Petroleum deposits
Biogas or Gobar Gas	Methane (CH ₄) + Carbon dioxide (CO ₂) + Hydrogen (H ₂) + Nitrogen (N ₂)	Organic wastes

Composition & sources of gaseous fuels

Octane number & knocking

- Octane number — measure of a fuel's ability to resist knocking (premature combustion) in spark-ignition (petrol) engines; scale: iso-octane = 100, n-heptane = 0. Definition: percentage of iso-octane in an iso-octane/n-heptane mixture matching the fuel's antiknock quality.
- Knocking — uneven premature combustion → pressure waves → engine damage; higher-octane fuel resists it.
- Regular petrol 87-90; premium 91-100+; racing/aviation fuels > 100. High-compression (luxury/sports) engines need 95-100.
- India: moving toward 95 RON and E20 (20% ethanol blend — also raises octane); produces indigenous 97-octane reference fuel.
- RON vs MON: Research and Motor Octane Numbers; pumps in many countries show (RON+MON)/2 = Anti-Knock Index (AKI).
- Octane vs cetane: octane — petrol engines (spark-ignition); cetane number — diesel engines (compression-ignition; ignition delay).
- TEL (tetraethyl lead) was added to petrol as anti-knock agent (raises octane, smooth running at high compression) — but lead is neurotoxic/polluting → phased out globally (still in some aviation gasoline). Modern alternatives: ethanol and MTBE (methyl tertiary-butyl ether).

✂ RAPID REVISION

- Catenation C > Si; coal — peat → lignite → bituminous → anthracite; graphite conducts (only non-metal).
- Alkanes C_nH_{2n+2} (paraffins), alkenes C_nH_{2n}, alkynes C_nH_{2n-2}; homologues differ by CH₂.
- Aromatic = alternating double bonds; benzenoid/non-benzenoid/heterocyclic (furan, pyridine).
- Isomerism: chain, position, functional (structural); geometrical (stereo).
- Ethene ripens fruits; conc. H₂SO₄ dehydrates ethanol → ethene; vinegar = 5-8% CH₃COOH; glacial acetic acid m.p. 15-17 °C.
- Esterification (acid catalyst) ⇌ saponification (base); soap = Na/K salts of fatty acids.
- Polymers: LDPE/HDPE, nylon-66, bakelite, melamine (crocery); vulcanisation = S crosslinks at 110 °C with ZnO.
- Antiseptic 0.2% phenol vs disinfectant 1% phenol; sodium benzoate = common preservative; BHT/BHA = antioxidants.
- Rocket propellant = fuel + oxidiser; I_s ∝ √(T/M); solid — no start/stop; liquid — controllable.
- Octane number: iso-octane 100 / n-heptane 0; petrol 87-90; E20 India; TEL phased out → ethanol/MTBE.